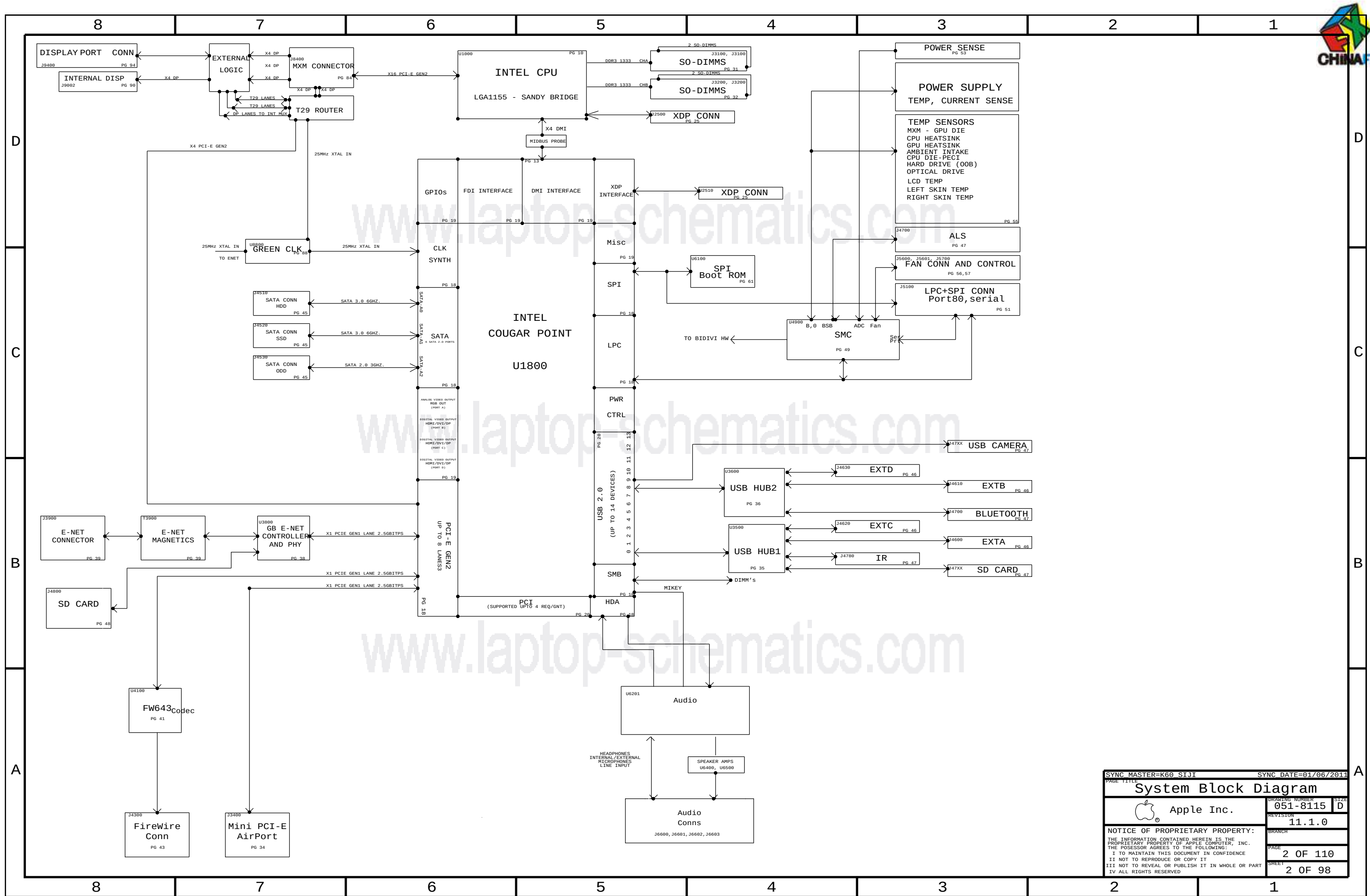
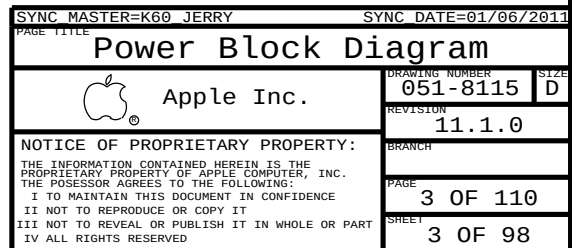




SYNC MASTER=K60 SIJI		SYNC DATE=01/06/2011	
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 Apple Inc.		051-8115	
		SIZE	
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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
085-0801	PCBA, MLB, DEV, K60	DEVELOPMENT, DEV_GROUP
639-1767	PCBA, MLB, K60, 2.5G, 4C, PRQ, P2_ODD	K60, 2P5GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P2, YES_DBG
639-1820	PCBA, MLB, K60, 2.7G, 4C, PRQ, P2_ODD	K60, 2P7GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P2, YES_DBG
639-1821	PCBA, MLB, K60, 2.8G, 4C, PRQ, P2_ODD	K60, 2P8GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P2, YES_DBG
639-2160	PCBA, MLB, K60, 2.5G, 4C, PRQ, P2_ODD, NO_DBG	K60, 2P5GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P2, NO_DBG
639-2159	PCBA, MLB, K60, 2.7G, 4C, PRQ, P2_ODD, NO_DBG	K60, 2P7GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P2, NO_DBG
639-2161	PCBA, MLB, K60, 2.8G, 4C, PRQ, P2_ODD, NO_DBG	K60, 2P8GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P2, NO_DBG
639-2118	PCBA, MLB, K60, 2.5G, 4C, PRQ, P1_ODD	K60, 2P5GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P1, YES_DBG
639-2122	PCBA, MLB, K60, 2.7G, 4C, PRQ, P1_ODD	K60, 2P7GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P1, YES_DBG
639-2119	PCBA, MLB, K60, 2.8G, 4C, PRQ, P1_ODD	K60, 2P8GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P1, YES_DBG
639-2132	PCBA, MLB, K60, 2.5G, 4C, PRQ, P1_ODD, NO_DBG	K60, 2P5GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P1, NO_DBG
639-2133	PCBA, MLB, K60, 2.7G, 4C, PRQ, P1_ODD, NO_DBG	K60, 2P7GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P1, NO_DBG
639-2134	PCBA, MLB, K60, 2.8G, 4C, PRQ, P1_ODD, NO_DBG	K60, 2P8GHZ_SNB_CPU_PRQ, BASIC1, BASIC2, CPUVCORE-3PH, ODD_SATA:P1, NO_DBG

BOM GROUPS

BOM GROUP	BOM OPTIONS
BASIC1	COMMON, ALTERNATE, MXM, FCIM, CPU_1V5_SENSE, CPU_VCCSA_SENSE, 1V05_PCH_SENSE, HUB_USX2061, PRODUCTION, VAXG, SSD
BASIC2	AP, BT, IR, T29
DEV_GROUP	VREFMRGN_A, VREFMRGN_B, DIMM_1V5_SENSE
YES_DBG	XDP, XDP_CONN, XDP_CPU_BPM, MOJOMUX: YES, LPCPLUS: YES
NO_DBG	MOJOMUX: NO, LPCPLUS: NO

CPU SOCKET & ILM SUB-BOMS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0071	1	SOCKET, LGA1155, CPU-LF	U1000	CRITICAL	TYCO_SOCKET
604-1474	1	ASSY, PURCHASED, ILM, TYCO	ILM	CRITICAL	TYCO_SOCKET
511S0073	1	SOCKET, LGA1155, CPU-LF	U1000	CRITICAL	MOLEX_SOCKET
604-1161	1	ASSY, PURCHASED, ILM, MOLEX	ILM	CRITICAL	MOLEX_SOCKET

BOM NUMBER	BOM NAME	BOM OPTIONS
085-2452	SUB ASSY, CPU SOCKET, K60, TYCO	TYCO_SOCKET
085-2453	SUB ASSY, CPU SOCKET, K60, MOLEX	MOLEX_SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
085-2452	1	TYCO CPU SOCKET AND ILM	SKT_ILM	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
085-2453	085-2452		SKT_ILM	MOLEX ALTERNATE

BOARD STACK-UP

TOP	SIGNAL
2	GROUND
3	SIGNAL
4	POWER
5	POWER
6	SIGNAL
7	GROUND
BOTTOM	SIGNAL

COMMON

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4088	1	IC, COUGAR POINT, SL14F, BD82Z68, PRQ, B3	U1800	CRITICAL	
353S3055	1	IC, PI3VEDP212, X2 DP MUX, QFN	U9390	CRITICAL	
338S0753	1	IC, FW643, 1394B_PCIE, PHY/LINK	U4100	CRITICAL	
825-7122	1	MLB LABEL, 48.0X4.8	X14	CRITICAL	
343S0534	1	IC, BCM57765, ENET&SD, 8X8	U3900	CRITICAL	
RAW: 335S0807	341T0184	1	FLASH, EFI BOOTROM, K60/K62	U6100	CRITICAL
RAW: 335S0539	341T0328	1	SFLASH ENET 2MBIT, CIV	U3990	CRITICAL
	338S0945	1	T29 ROUTER, IC ASSP	U9700	CRITICAL
	341T0257	1	IC, T29, SERIAL EEPROM	U9790	CRITICAL
RAW: 335S0550	341T0326	1	IC, MCU, 32B, LPC1112A, 16KB/2KB, HVQFN25	U9330	CRITICAL
RAW: 337S3997	341T0330	1	IC, MXM SYS ROM, 24C02	U8570	CRITICAL
RAW: 335S0709	341T0185	1	IC, SMC, K60	U4900	CRITICAL
RAW: 338S0878					K60

CPU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4043	1	SNB, SR005, PRQ, D2, 2.5, 65W, 4+1, 6M, LGA	CPU	CRITICAL	2P5GHZ_SNB_CPU_PRQ
337S4062	1	SNB, SR009, PRQ, D2, 2.7, 65W, 4+1, 6M, LGA	CPU	CRITICAL	2P7GHZ_SNB_CPU_PRQ
337S4061	1	SNB, SR00E, PRQ, D2, 2.8, 65W, 4+1, 8M, LGA	CPU	CRITICAL	2P8GHZ_SNB_CPU_PRQ

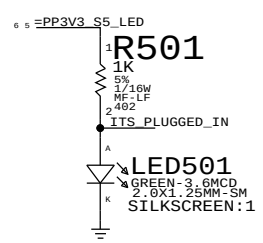
K60 ALTERNATE PARTS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
128S0298	128S0293			330UF
371S0679	371S0652			PIN DIODE
377S0107	377S0066			USB DIODE
376S0972	376S0612			ROHM TRA-BJT

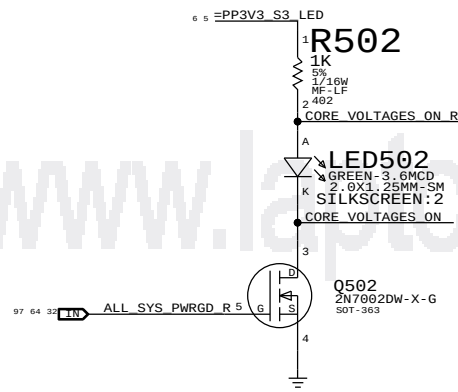
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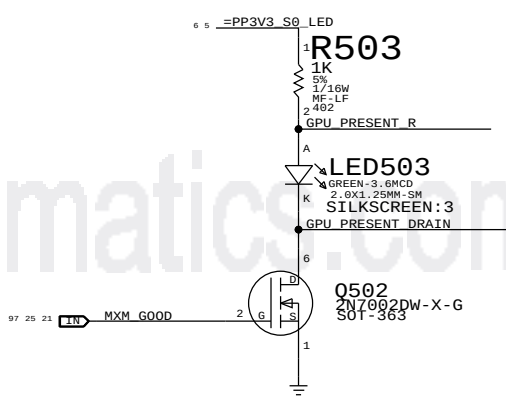
S5 Led



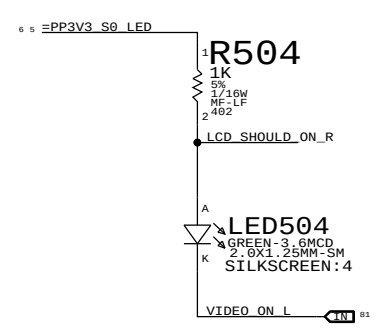
ALL_SYS_PWRGD Led



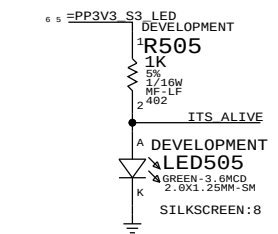
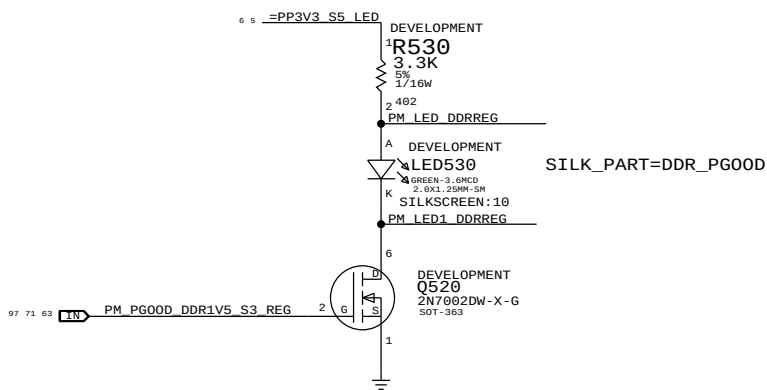
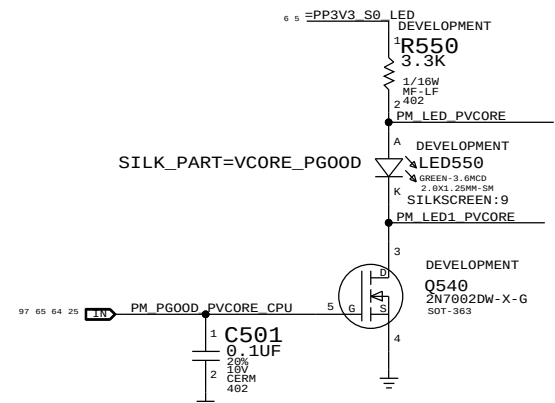
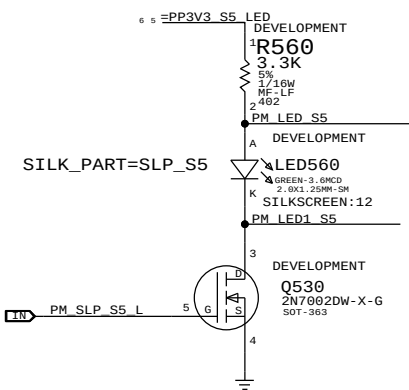
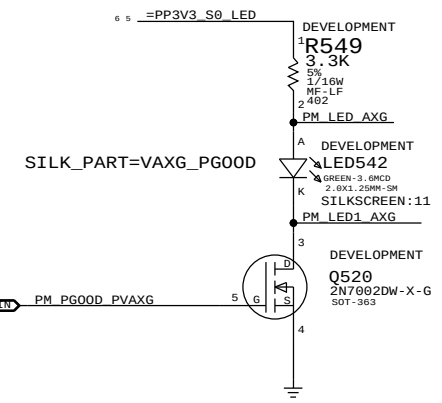
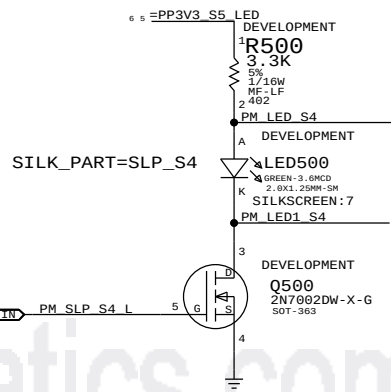
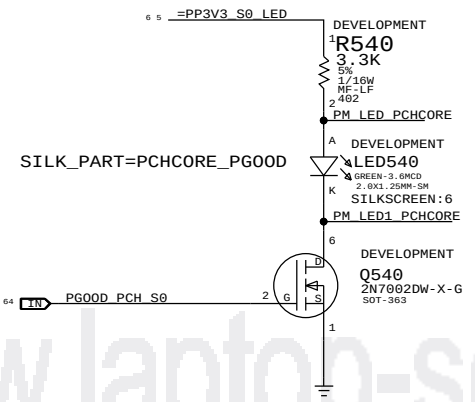
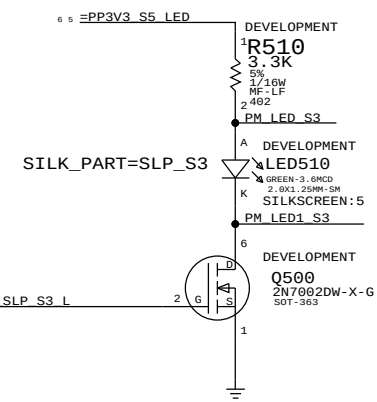
MXM PWR GOOD Led



VIDEO ON Led



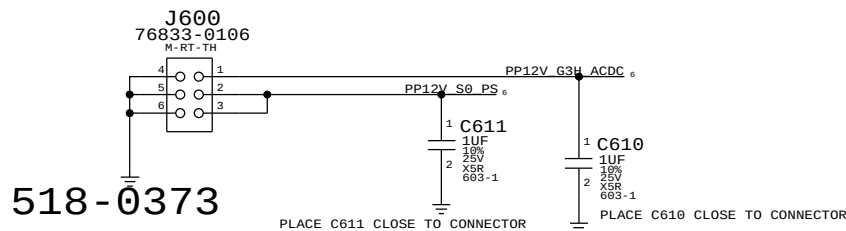
PROTO DEBUG LEDS ARE SHOWN BELOW



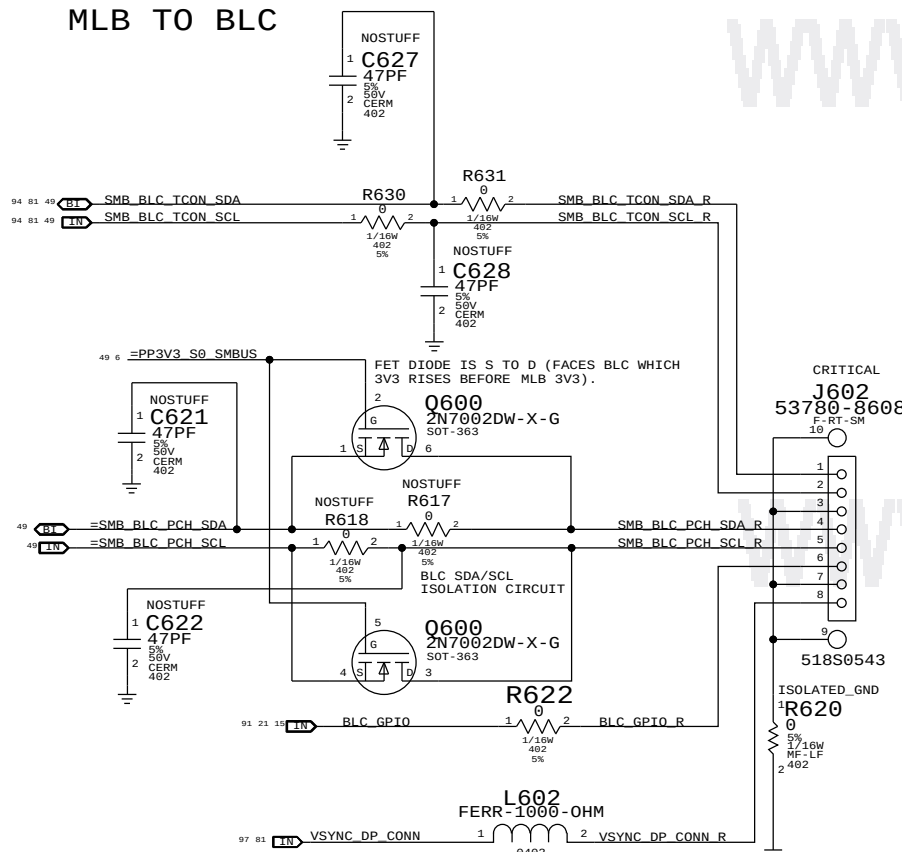
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Apple Inc.		Apple Inc.	
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051-8115		051-8115	
REVISION		REVISION	
11.1.0		11.1.0	
BRANCH		BRANCH	
PAGE		PAGE	
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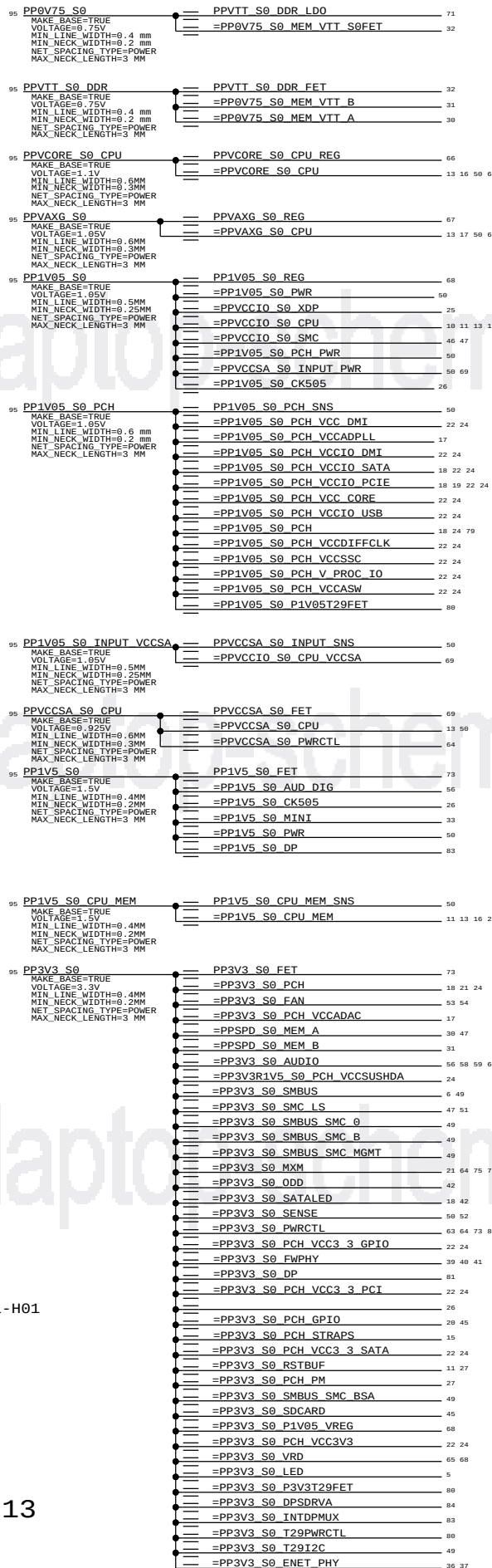
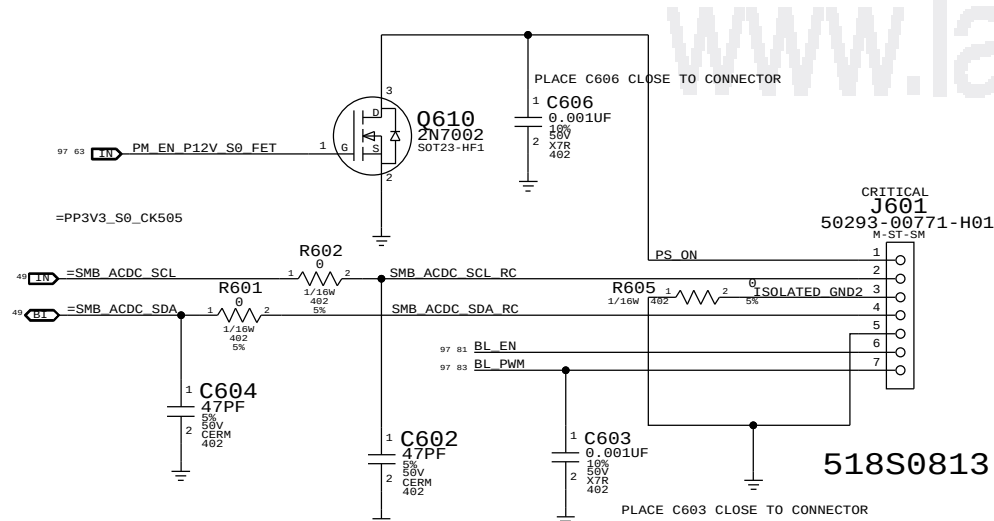
POWER SUPPLY TO MLB



MLB TO BLC

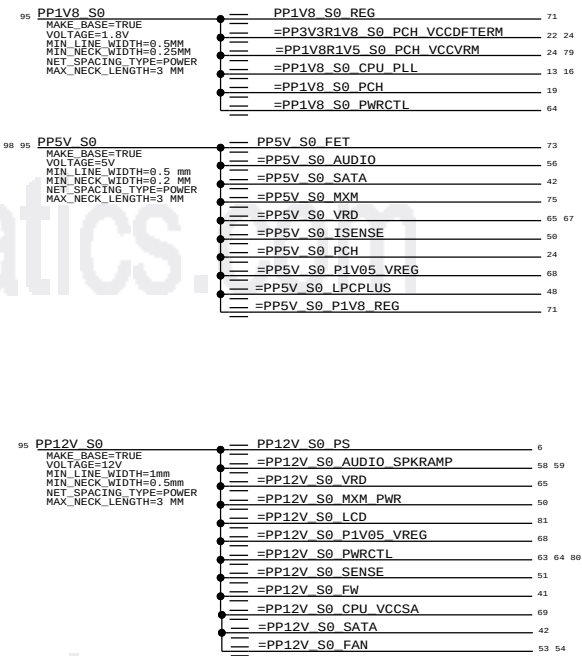


MLB TO PS



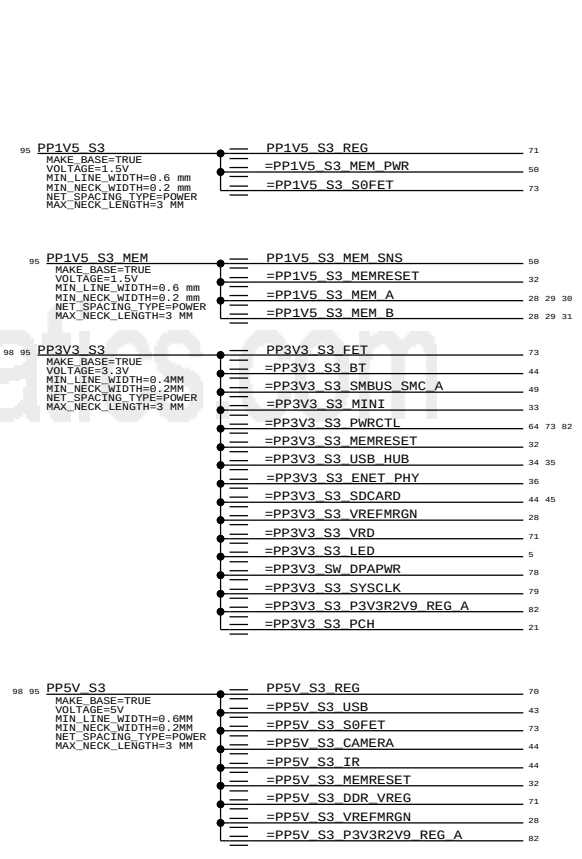
"S0" RAILS

ONLY ON IN RUN



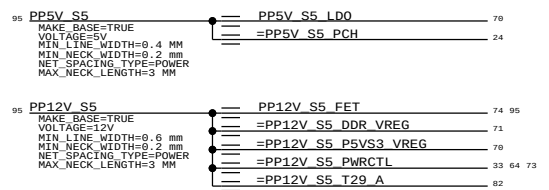
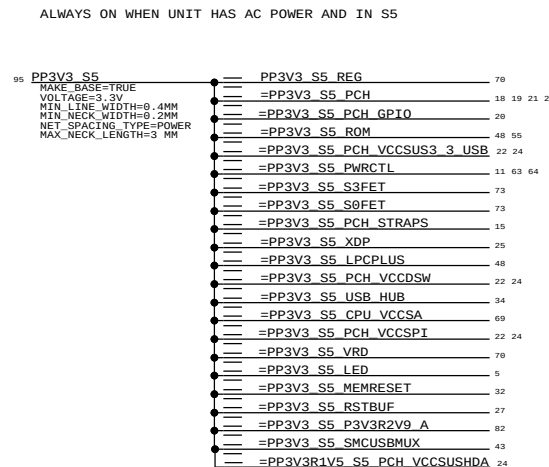
"S3" RAILS

ON IN RUN AND SLEEP



"S5" RAILS

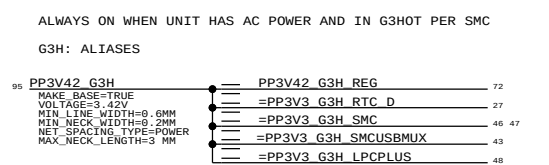
ALWAYS ON WHEN UNIT HAS AC POWER AND IN S5



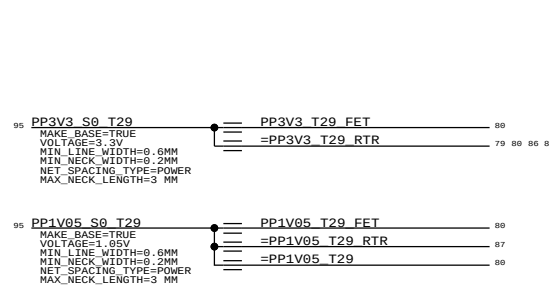
"G3H" RAILS

ALWAYS ON WHEN UNIT HAS AC POWER AND IN G3HOT PER SMC

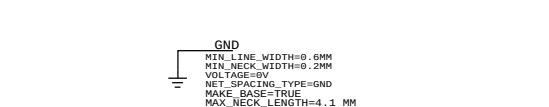
G3H: ALIASES



T29 RAILS

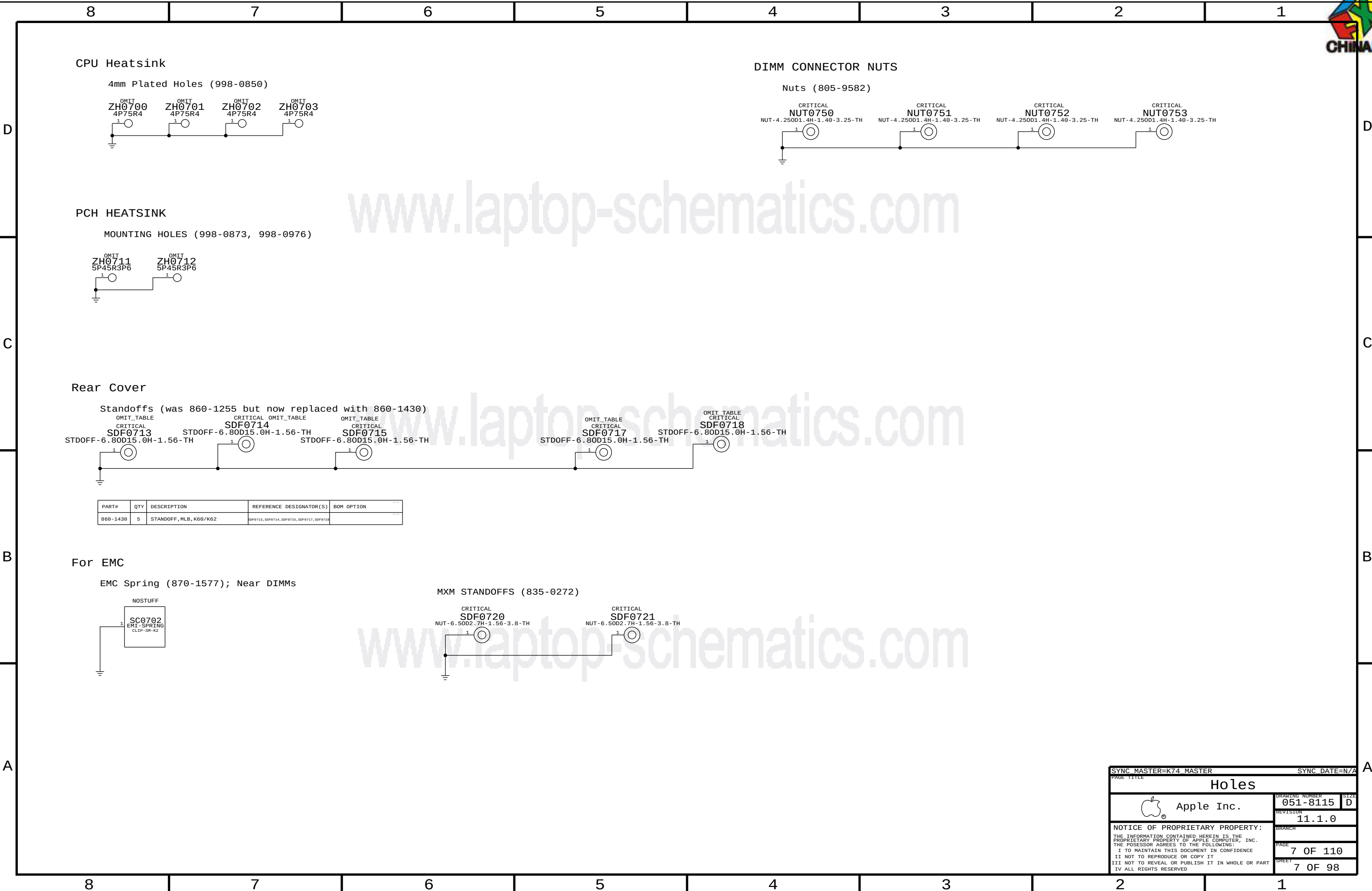


GND RAILS



SYNC MASTER=K60 MARK		SYNC DATE=12/30/2016	
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8	7	6	5	4	3	2	1
UNUSED CPU SIGNALS		NC ON UNUSED PCIE ALIASES		NC ON UNUSED DISPLAY ALIASES		NC ON UNUSED FDI ALIASES	
19 TP_CPU_RSVD<16..1> == NC_CPU_RSVD<16..1> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CPU_RSVD<46..19> == NC_CPU_RSVD<46..19> MAKE_BASE=TRUE NO_TEST=TRUE		19 TP_PCIE_CLK100M_PE5P == NC_PCIE_CLK100M_PE5P MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCIE_CLK100M_PE5N == NC_PCIE_CLK100M_PE5N MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCIE_CLK100M_PE6P == NC_PCIE_CLK100M_PE6P MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCIE_CLK100M_PE6N == NC_PCIE_CLK100M_PE6N MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCIE_CLK100M_PE7P == NC_PCIE_CLK100M_PE7P MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCIE_CLK100M_PE7N == NC_PCIE_CLK100M_PE7N MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PE_RX_N<3..0> == NC_PE_RXN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PE_RX_P<3..0> == NC_PE_RXP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PE_TX_N<3..0> == NC_PE_TXN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PE_TX_P<3..0> == NC_PE_TXP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		19 TP_CRT_IG_DDC_CLK == NC_CRT_IG_DDC_CLK MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CRT_IG_DDC_DATA == NC_CRT_IG_DDC_DATA MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CRT_IG_RED == NC_CRT_IG_RED MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CRT_IG_GREEN == NC_CRT_IG_GREEN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CRT_IG_BLUE == NC_CRT_IG_BLUE MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CRT_IG_HSYNC == NC_CRT_IG_HSYNC MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CRT_IG_VSYNC == NC_CRT_IG_VSYNC MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_MLN<3..0> == NC_DP_IG_B_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_MLP<3..0> == NC_DP_IG_B_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_AUX_N == NC_DP_IG_B_AUXN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_AUX_P == NC_DP_IG_B_AUXP MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_HPD == NC_DP_IG_B_HPD MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_DDC_CLK == NC_DP_IG_B_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_B_DDC_DATA == NC_DP_IG_B_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_MLN<3..0> == NC_DP_IG_C_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_MLP<3..0> == NC_DP_IG_C_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_AUX_N == NC_DP_IG_C_AUXN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_AUX_P == NC_DP_IG_C_AUXP MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_HPD == NC_DP_IG_C_HPD MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_CTRL_CLK == NC_DP_IG_C_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_C_CTRL_DATA == NC_DP_IG_C_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_MLN<3..0> == NC_DP_IG_D_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_MLP<3..0> == NC_DP_IG_D_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_AUXN == NC_DP_IG_D_AUXN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_AUXP == NC_DP_IG_D_AUXP MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_HPD == NC_DP_IG_D_HPD MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_CTRL_CLK == NC_DP_IG_D_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_DP_IG_D_CTRL_DATA == NC_DP_IG_D_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SDVO_TVCLKINN == NC_SDVO_TVCLKINN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SDVO_TVCLKINP == NC_SDVO_TVCLKINP MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SDVO_STALLN == NC_SDVO_STALLN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SDVO_STALLP == NC_SDVO_STALLP MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SDVO_INTN == NC_SDVO_INTN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_SDVO_INTP == NC_SDVO_INTP MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCH_L_BKLTCTL == NC_PCH_L_BKLTCTL MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCH_L_BKLTEN == NC_PCH_L_BKLTEN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCH_L_VDD_EN == NC_PCH_L_VDD_EN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCH_CLKOUT_DPN == NC_PCH_CLKOUT_DPN MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCH_CLKOUT_DPP == NC_PCH_CLKOUT_DPP MAKE_BASE=TRUE NO_TEST=TRUE		19 TP_CPU_FDI_TX_N<7..0> == NC_CPU_FDI_TXN<7..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_CPU_FDI_TX_P<7..0> == NC_CPU_FDI_TXP<7..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCH_FDI_RX_N<7..0> == NC_PCH_FDI_RXN<7..0> MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_PCH_FDI_RX_P<7..0> == NC_PCH_FDI_RXP<7..0> MAKE_BASE=TRUE NO_TEST=TRUE	
NC ON UNUSED PCI ALIASES		NC ON UNUSED USB ALIASES		NC ON UNUSED SATA ALIASES			
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NC ON UNUSED MEM ALIASES							
12 TP_MEM_A_DQ_CB<7..0> == NC_MEM_A_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE 12 TP_MEM_A_DQS_N<8> == NC_MEM_A_DQSN<8> MAKE_BASE=TRUE NO_TEST=TRUE 12 TP_MEM_A_DQS_P<8> == NC_MEM_A_DQSP<8> MAKE_BASE=TRUE NO_TEST=TRUE 12 TP_MEM_B_DQ_CB<7..0> == NC_MEM_B_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE 12 TP_MEM_B_DQS_N<8> == NC_MEM_B_DQSN<8> MAKE_BASE=TRUE NO_TEST=TRUE 12 TP_MEM_B_DQS_P<8> == NC_MEM_B_DQSP<8> MAKE_BASE=TRUE NO_TEST=TRUE							
NC ON UNUSED MISC ALIASES							
19 TP_HDA_SDIN1 == NC_HDA_SDIN1 MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_HDA_SDIN2 == NC_HDA_SDIN2 MAKE_BASE=TRUE NO_TEST=TRUE 19 TP_HDA_SDIN3 == NC_HDA_SDIN3 MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCH_PWM0 == NC_PCH_PWM0 MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCH_PWM1 == NC_PCH_PWM1 MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCH_PWM2 == NC_PCH_PWM2 MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCH_PWM3 == NC_PCH_PWM3 MAKE_BASE=TRUE NO_TEST=TRUE 21 TP_PCH_SST == NC_PCH_SST MAKE_BASE=TRUE NO_TEST=TRUE 18 TP_PCH_CL_CLK1 == NC_PCH_CL_CLK1 MAKE_BASE=TRUE NO_TEST=TRUE 18 TP_PCH_CL_DATA1 == NC_PCH_CL_DATA1 MAKE_BASE=TRUE NO_TEST=TRUE 18 TP_PCH_CL_RST1 == NC_PCH_CL_RST1 MAKE_BASE=TRUE NO_TEST=TRUE							

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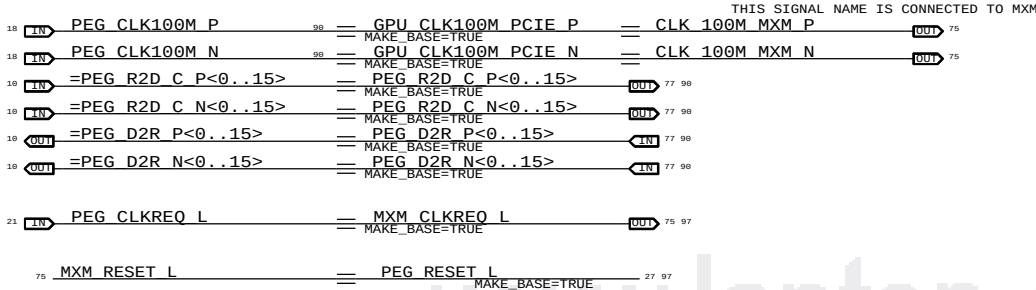
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SYNC MASTER=K62		SYNC DATE=01/06/2011	
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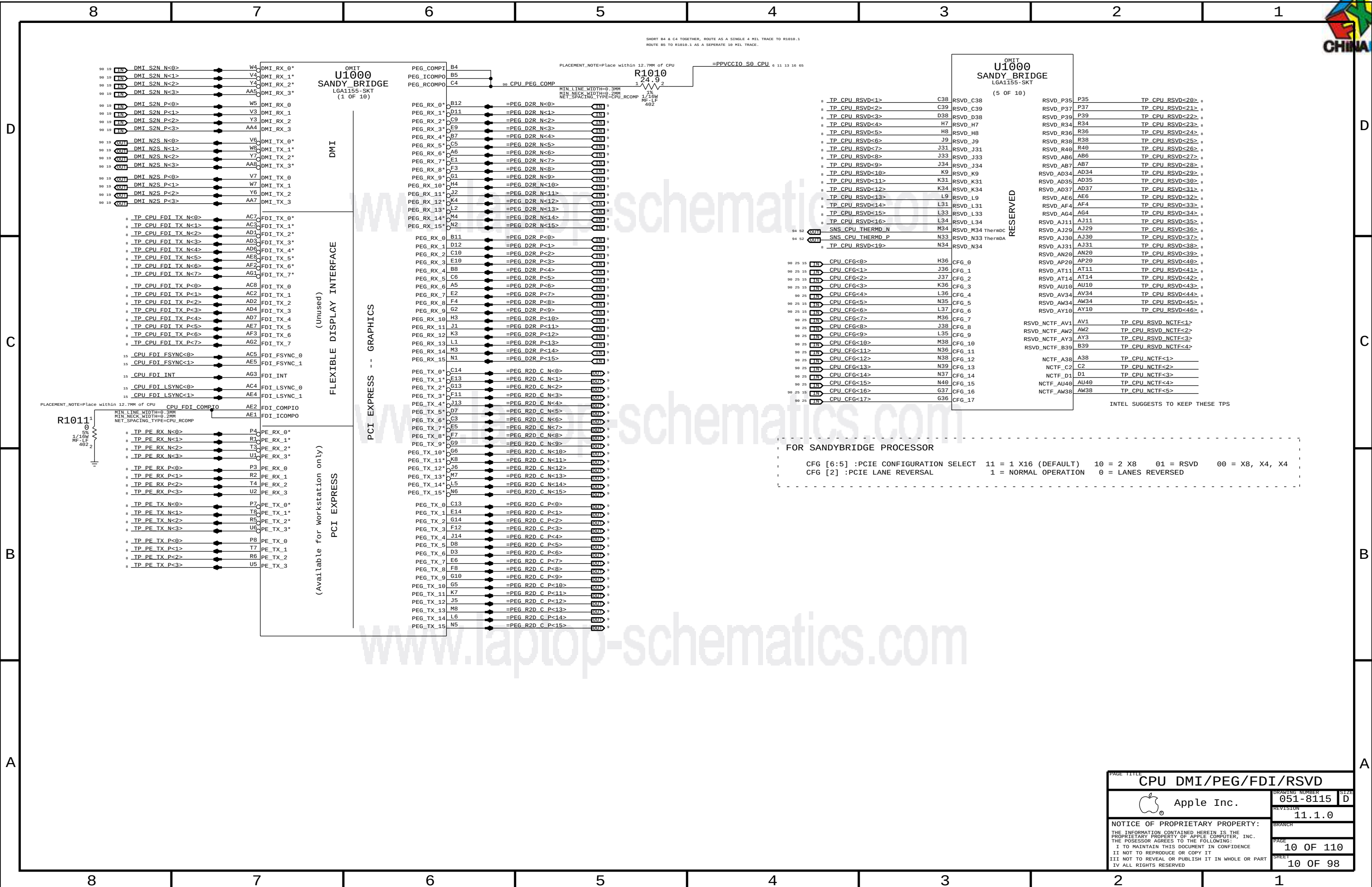


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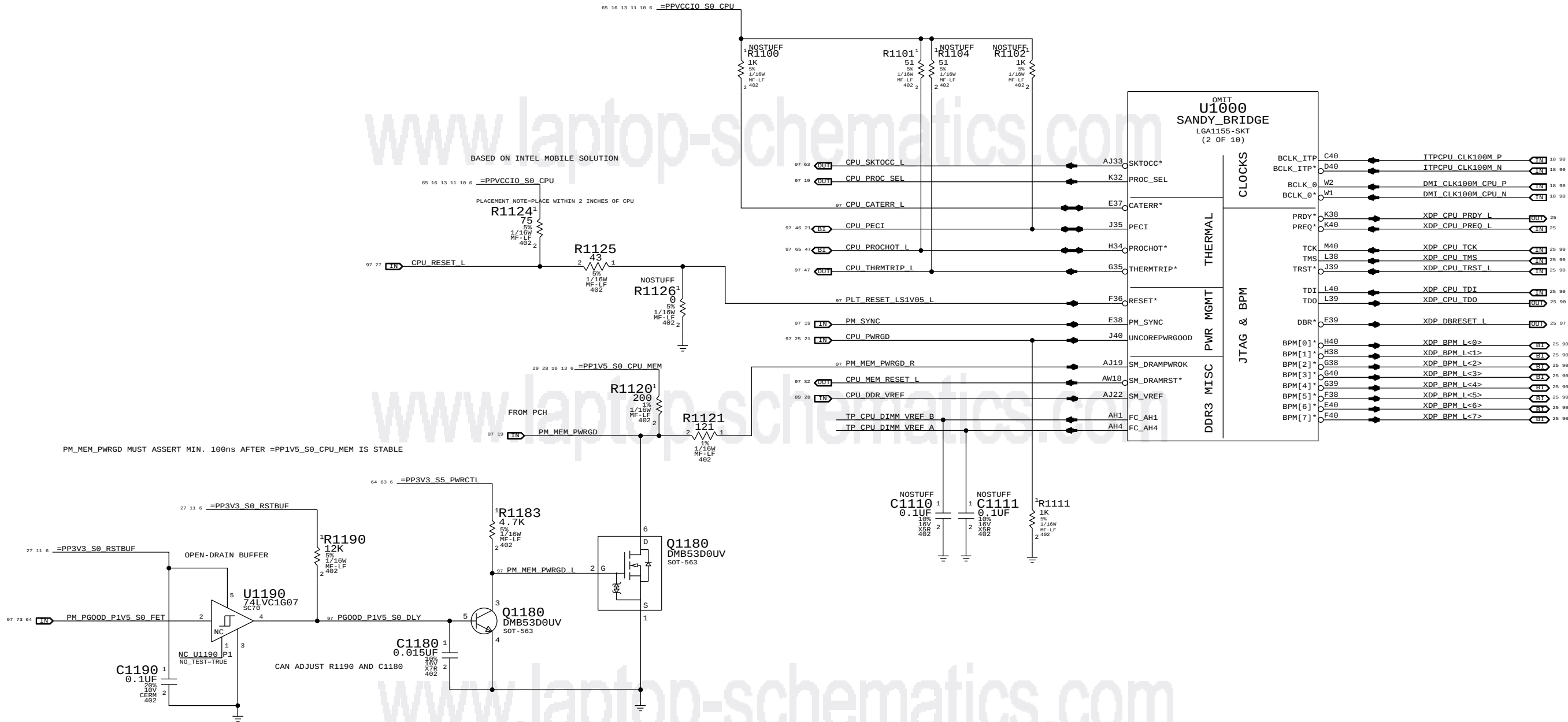
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PAGE TITLE				
CPU CLOCK/MISC/JTAG				
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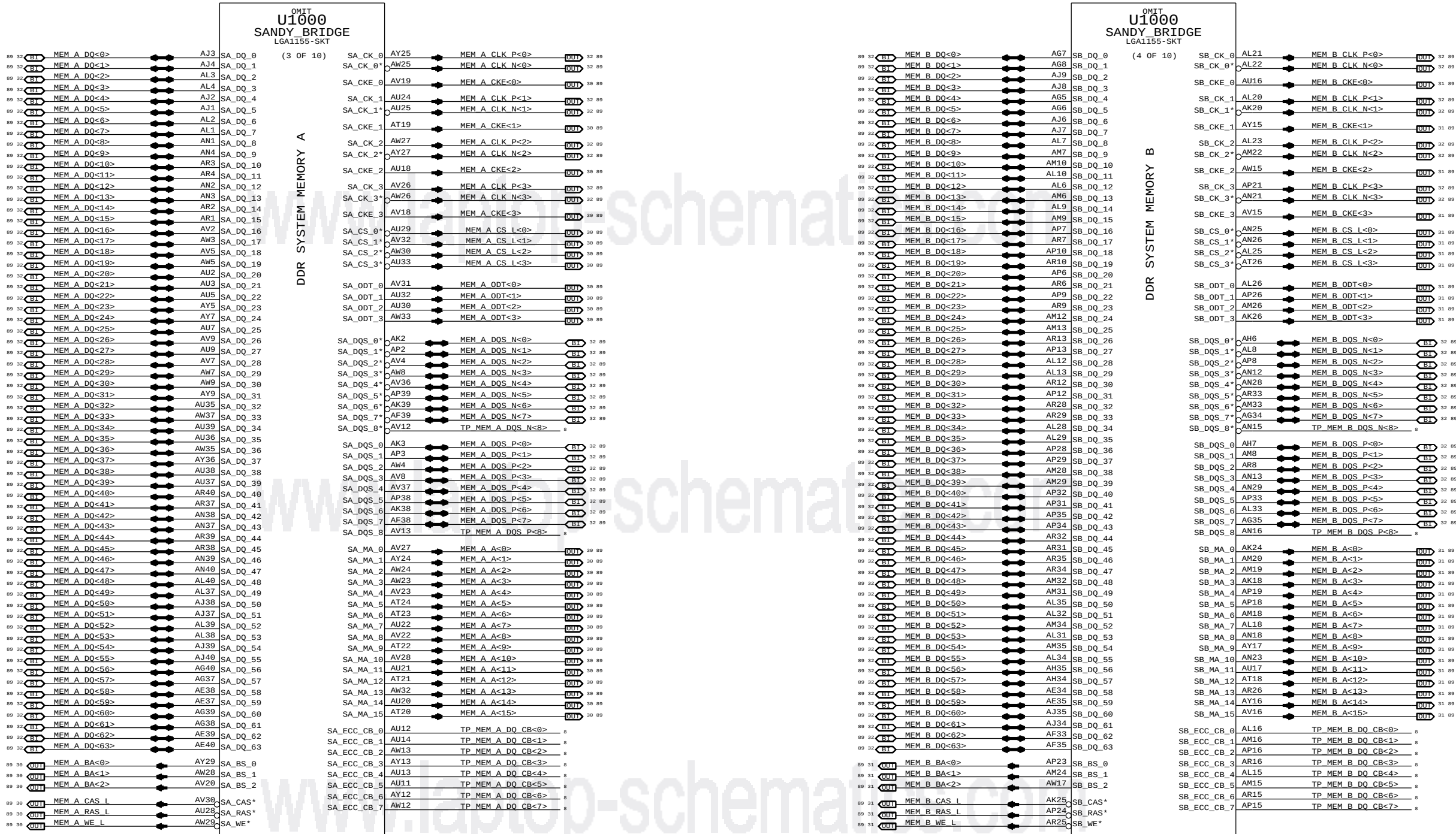
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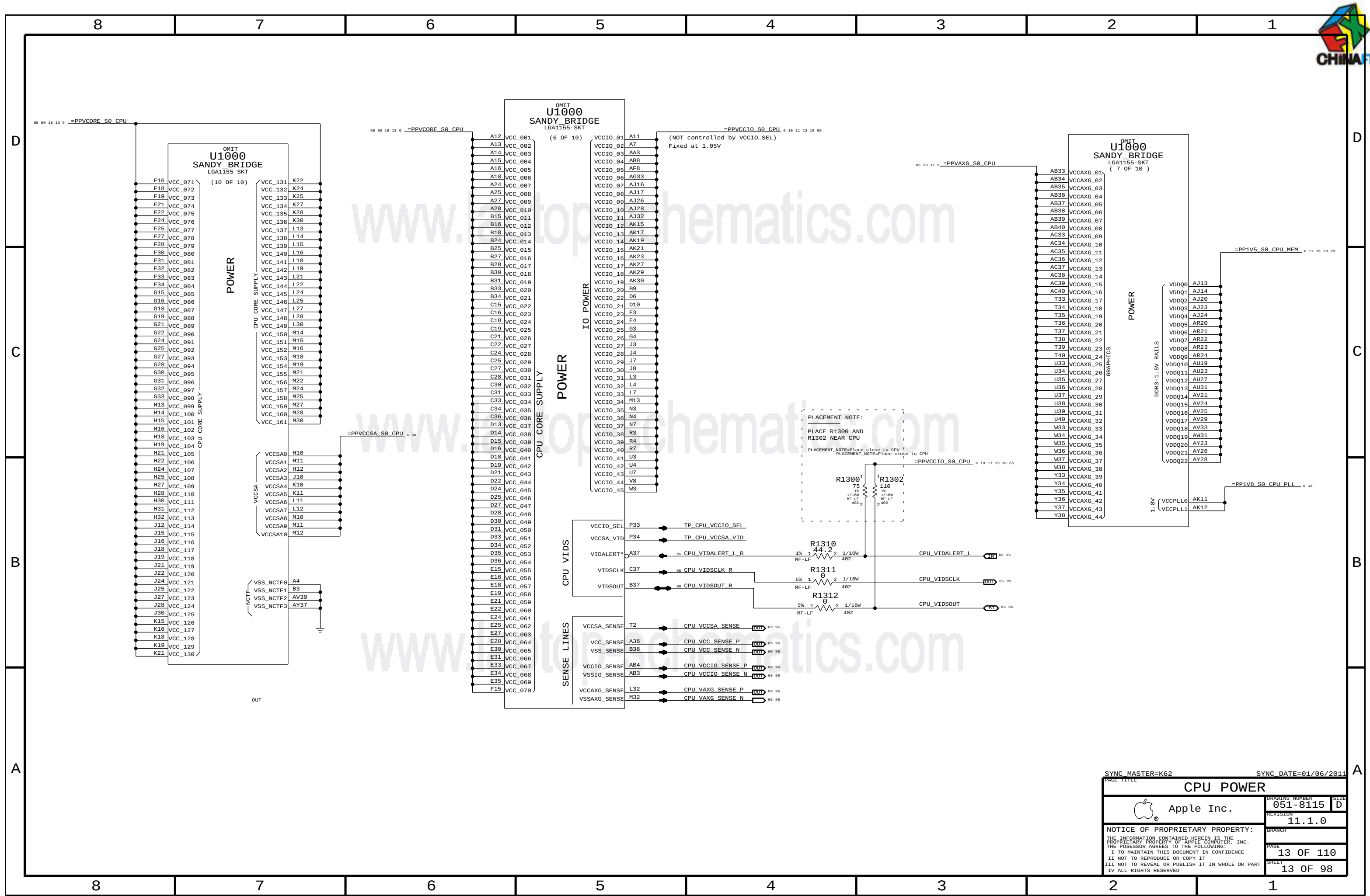
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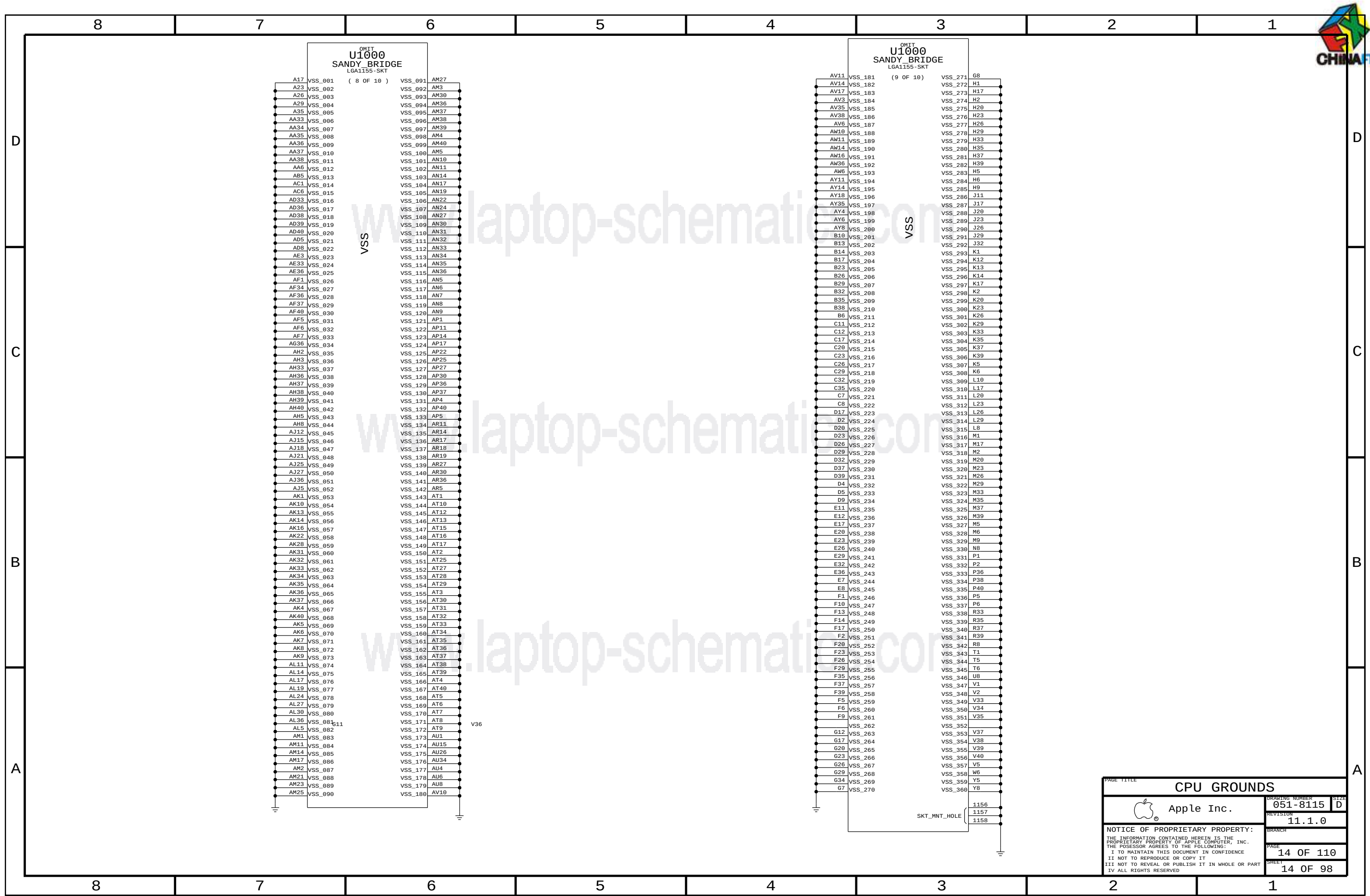
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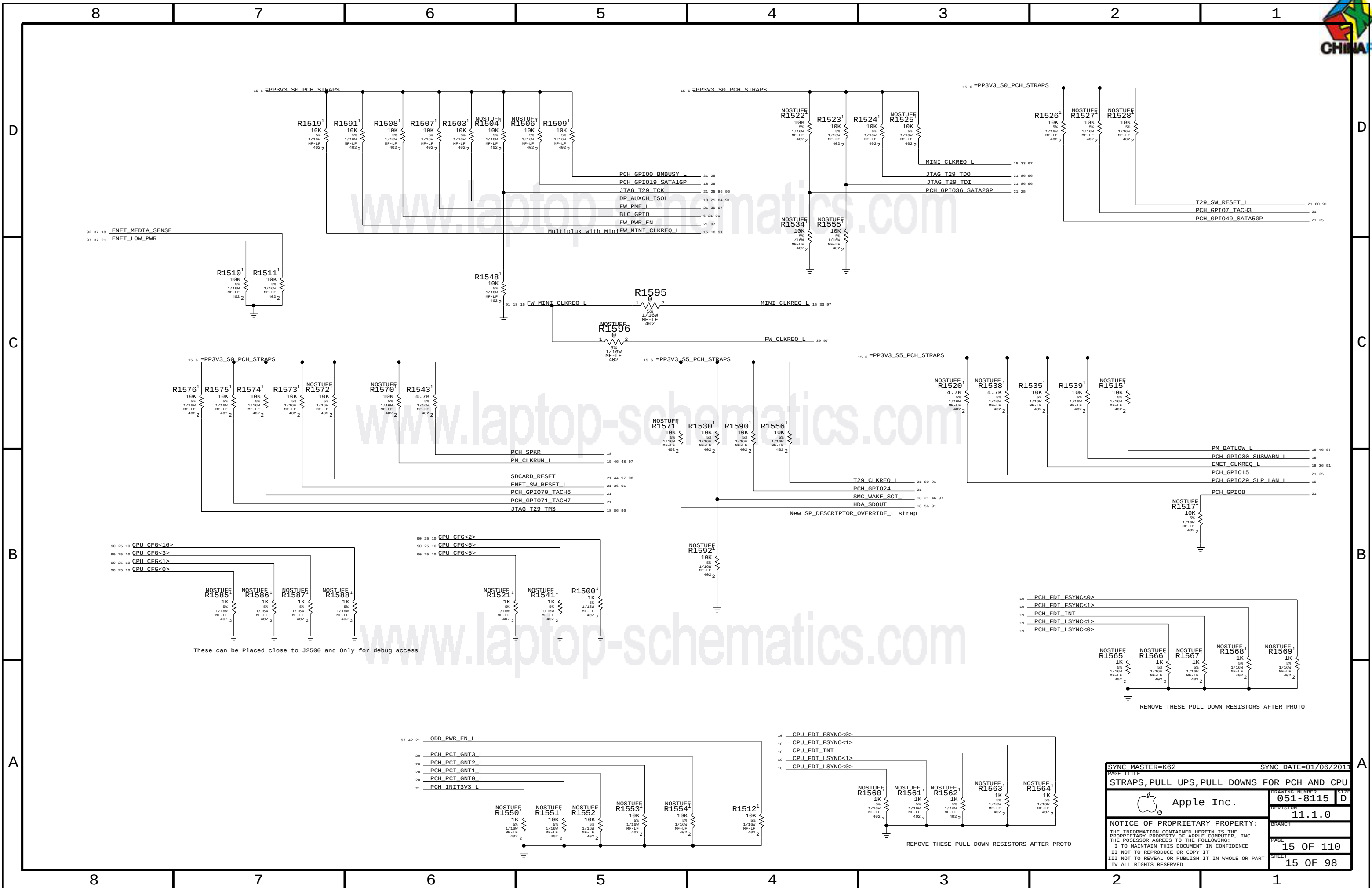




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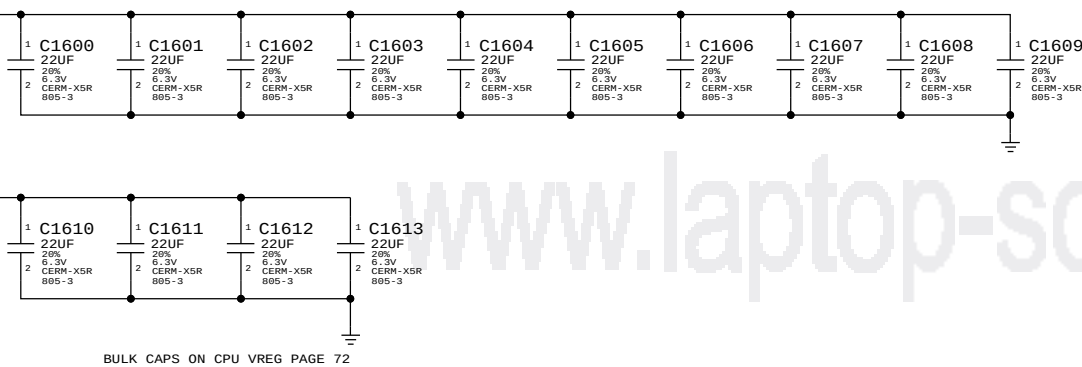


CPU VCORE DECOUPLING

14x 22UF,0805 INTEL RECOMMENDATION 18X 22UF 0805 (14 Inside cavity and 4 North of processor)

PLACEMENT_NOTE (C1600-C1613):

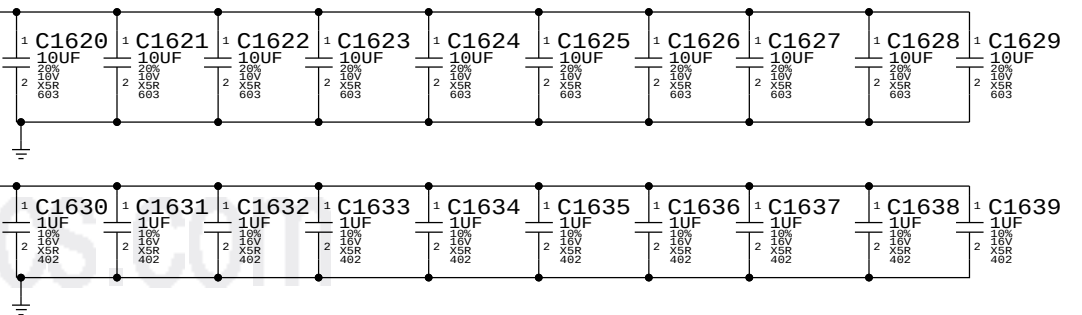
Place inside socket cavity



BULK CAPS ON CPU VREG PAGE 72

10x 10UF and 10x 1UF CAPACITORS

Place inside socket cavity

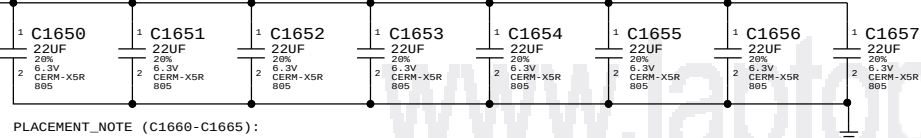


CPU VCCIO DECOUPLING

8X 22UF 0805, 6X 10UF 0805 INTEL RECOMMENDATION 9X22UF 0805,16X 0805 placeholders

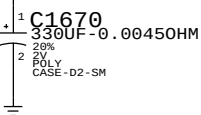
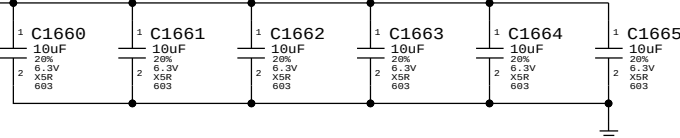
PLACEMENT_NOTE (C1650-C1657):

Place under socket cavity on secondary side.



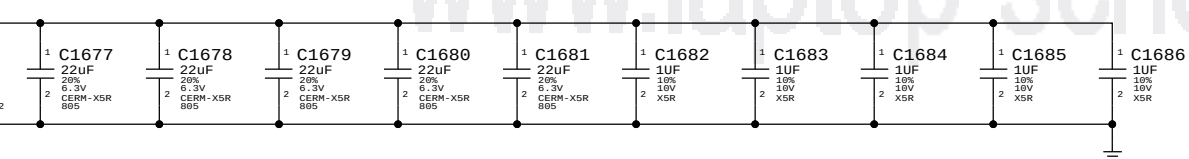
PLACEMENT_NOTE (C1660-C1665):

Place at edge of socket.



Memory (CPU VCCDDR) DECOUPLING

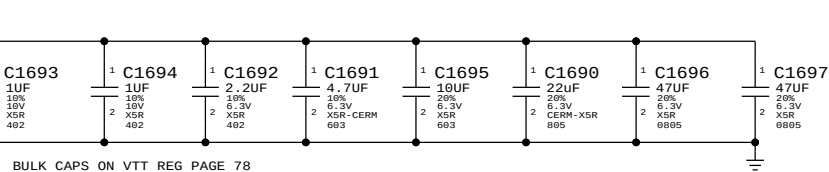
6x 22uf 0805, 5x 1uf 0402. INTEL RECOMMENDATION 9X 22uf 0805



Note: VCCSA decoupling is on regulator page

PLL (CPU VCCSFR) DECOUPLING

2x 47uF, 1x 22uF 0805, 1x 10uF 0603, 1x 4.7uF 0603, 1x 2.2uF 0402, 2x 1uF 0402. INTEL RECOMMENDATION 10x 10uF 0805



BULK CAPS ON VTT REG PAGE 78

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CPU NON-GFX DECOUPLING			
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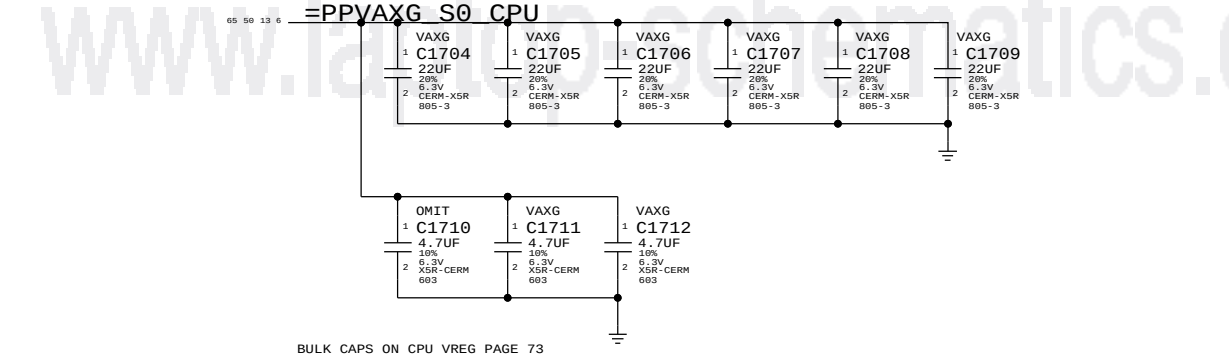


VAXG DECOUPLING

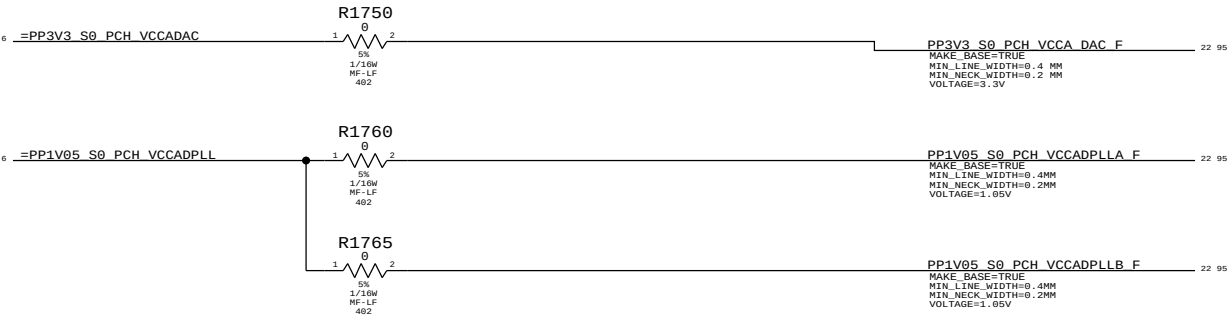
INTEL RECOMMENDATION 6X22UF 0805,3X 4.7UF

PLACEMENT_NOTE (C1704-C1709):

Place inside socket cavity



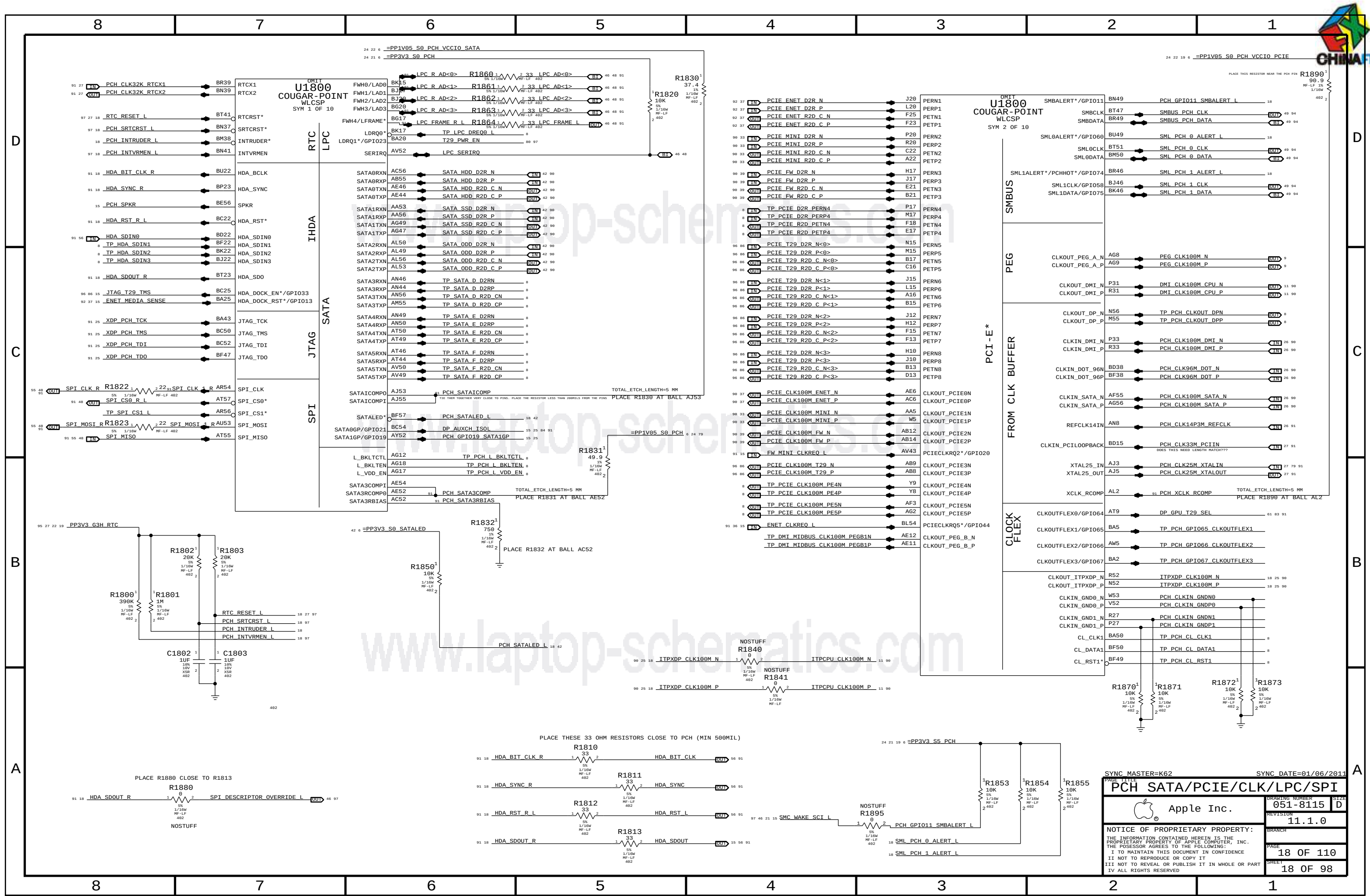
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
138S0586	1	CAP, 4.7UF, 10%, 6.3V, 0603	C1710	VAXG
113S0022	1	RES, 0 OHM, 5%, 0603	C1710	NO_VAXG



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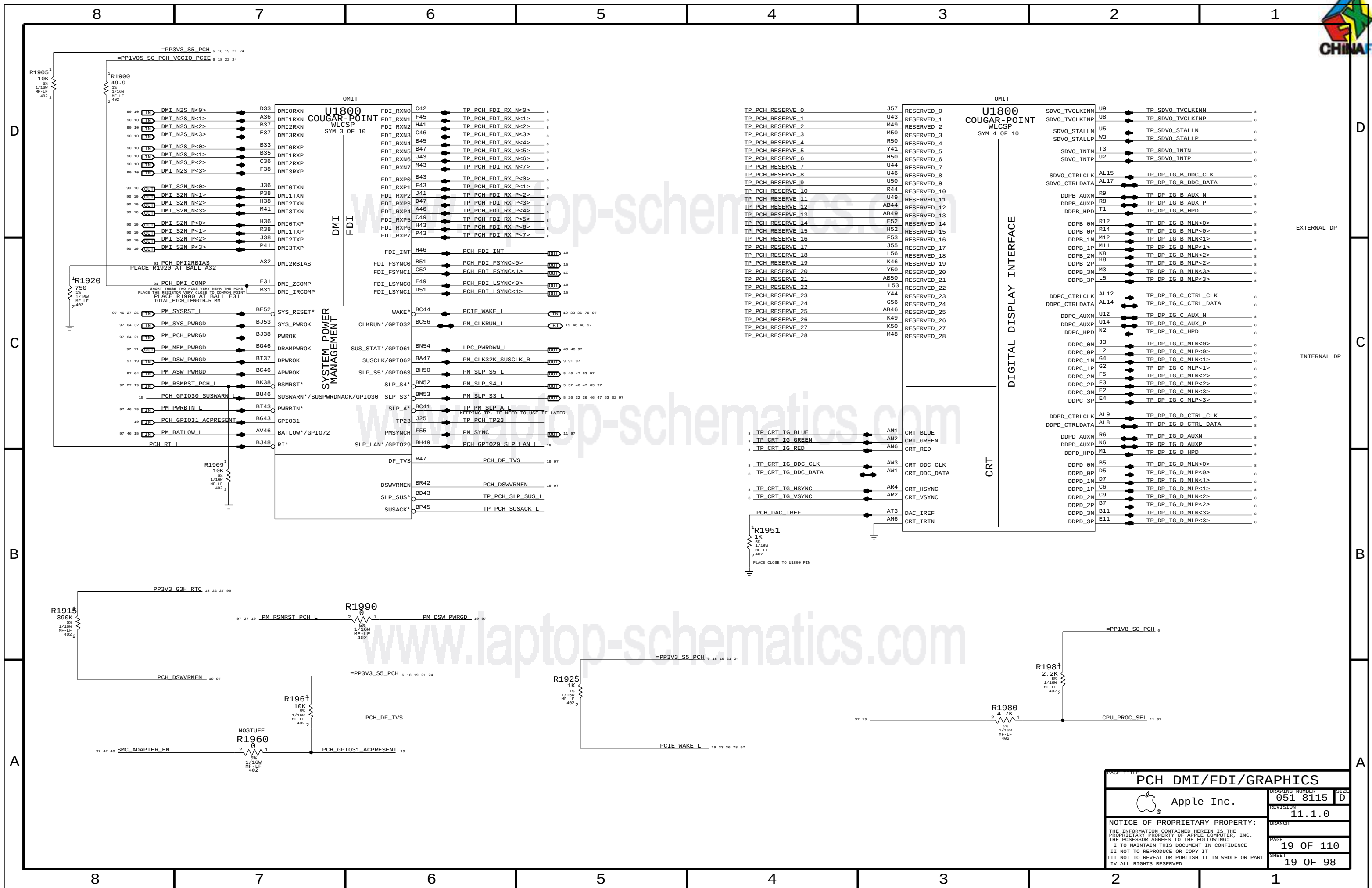
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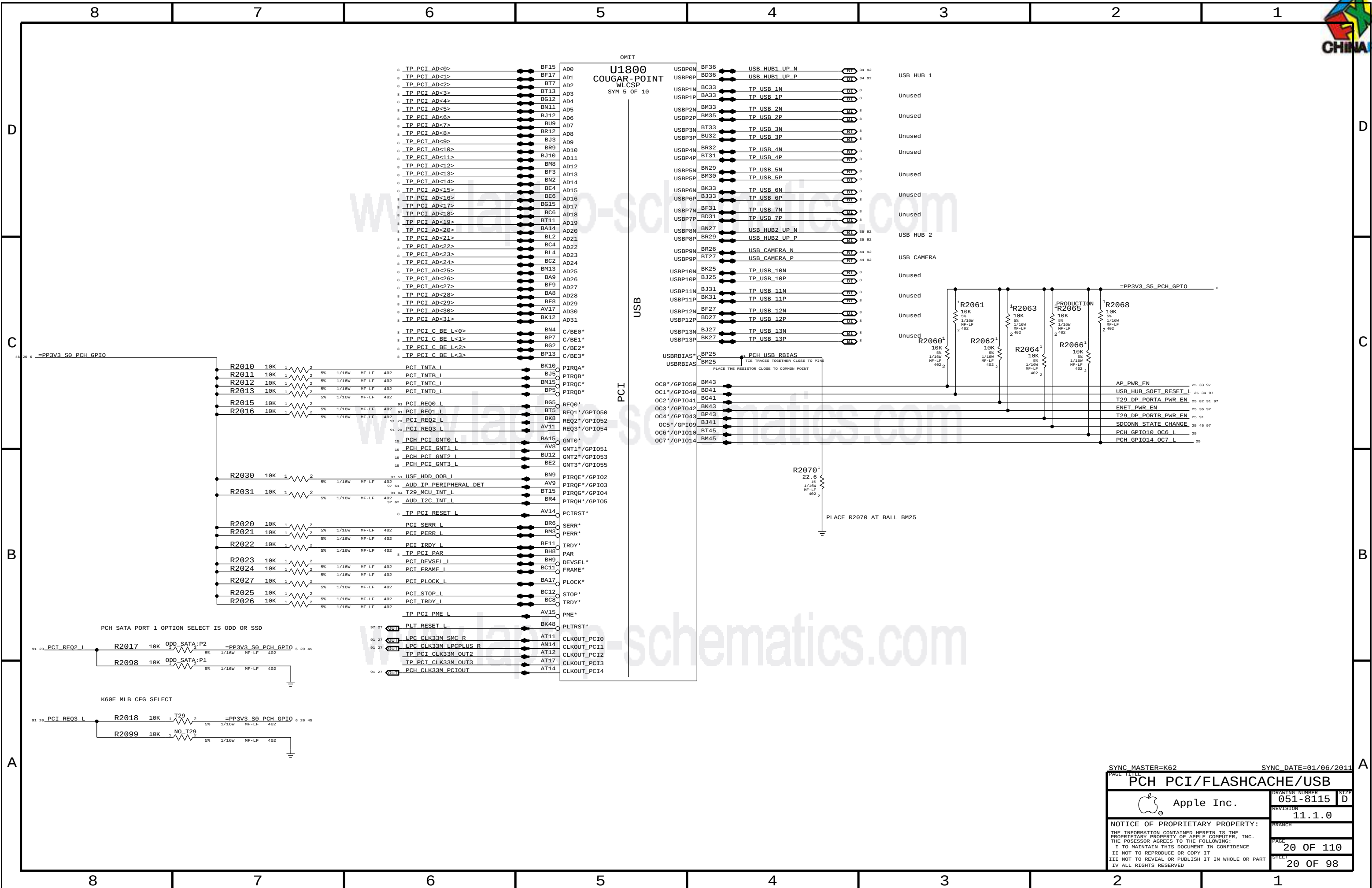


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PAGE TITLE		PCH SATA/PCIE/CLK/LPC/SPI	
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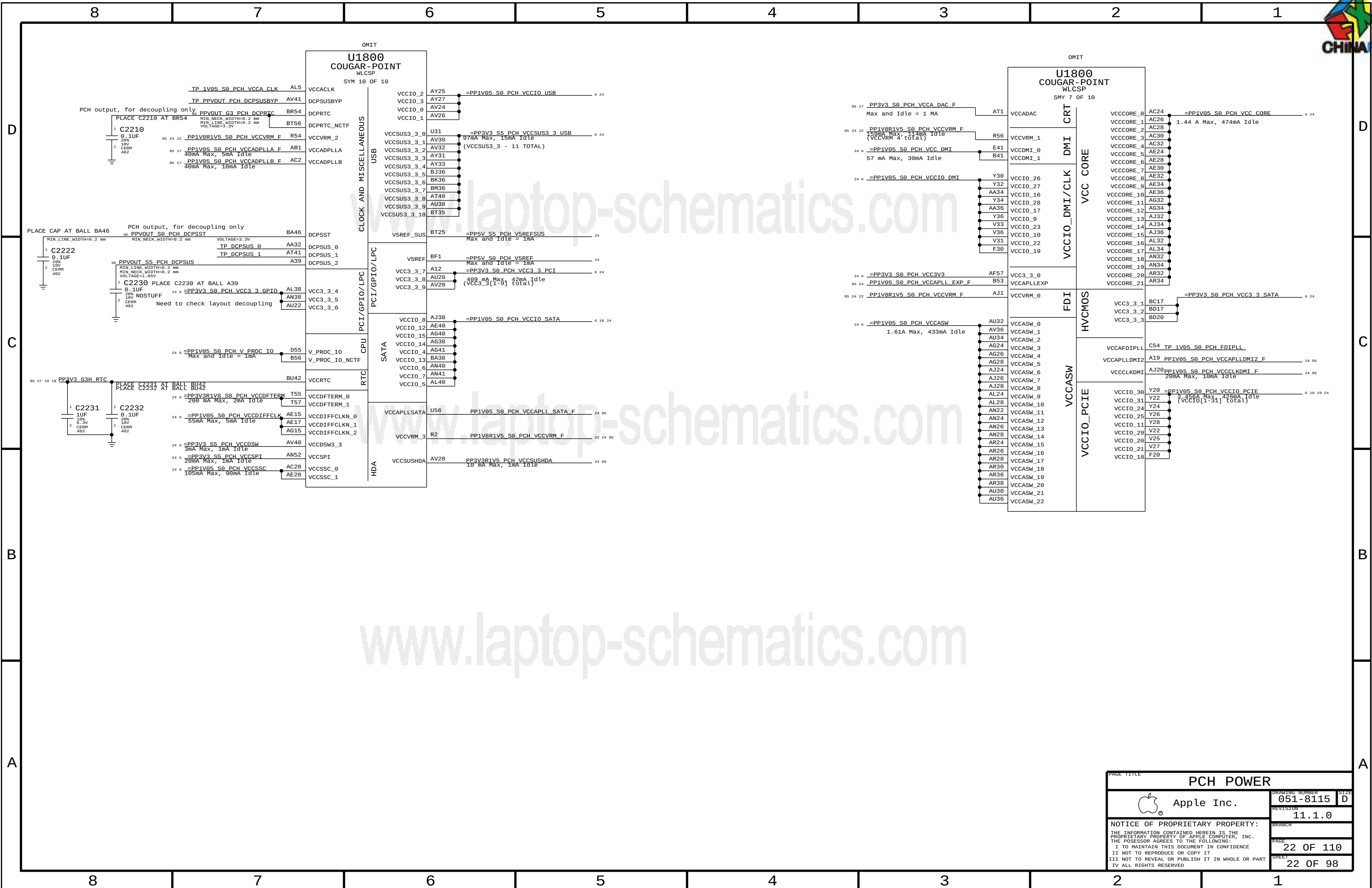
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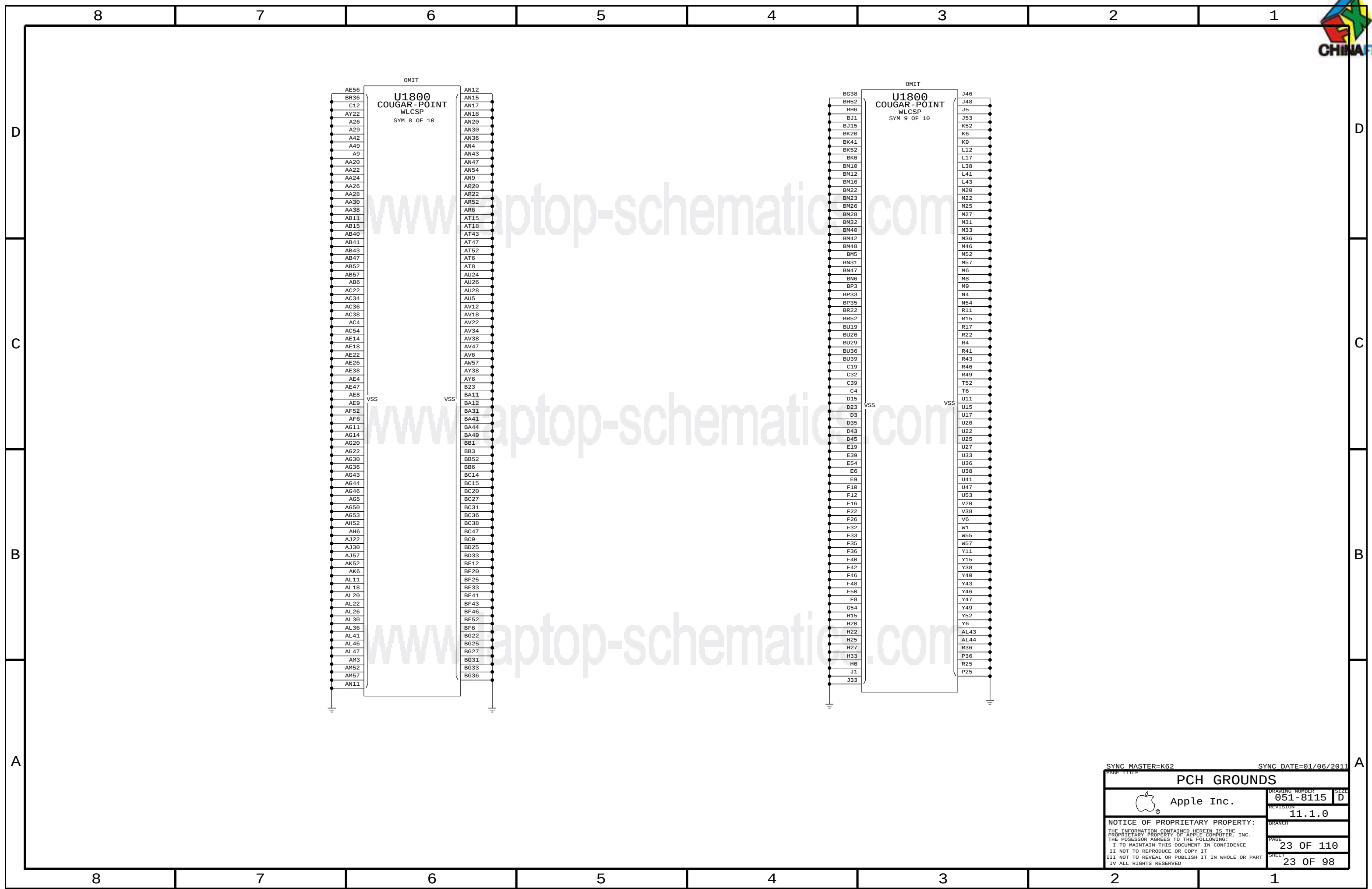
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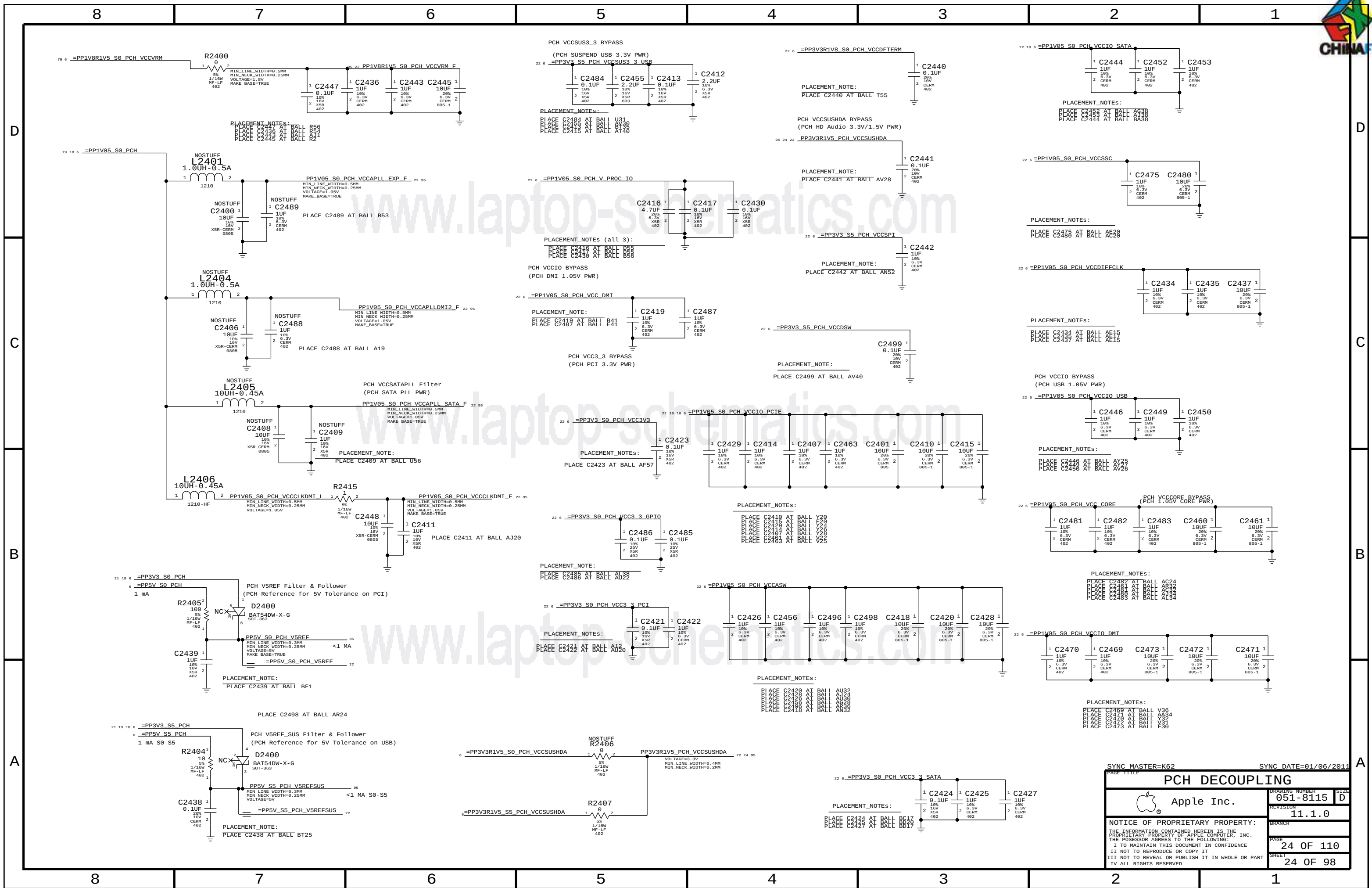
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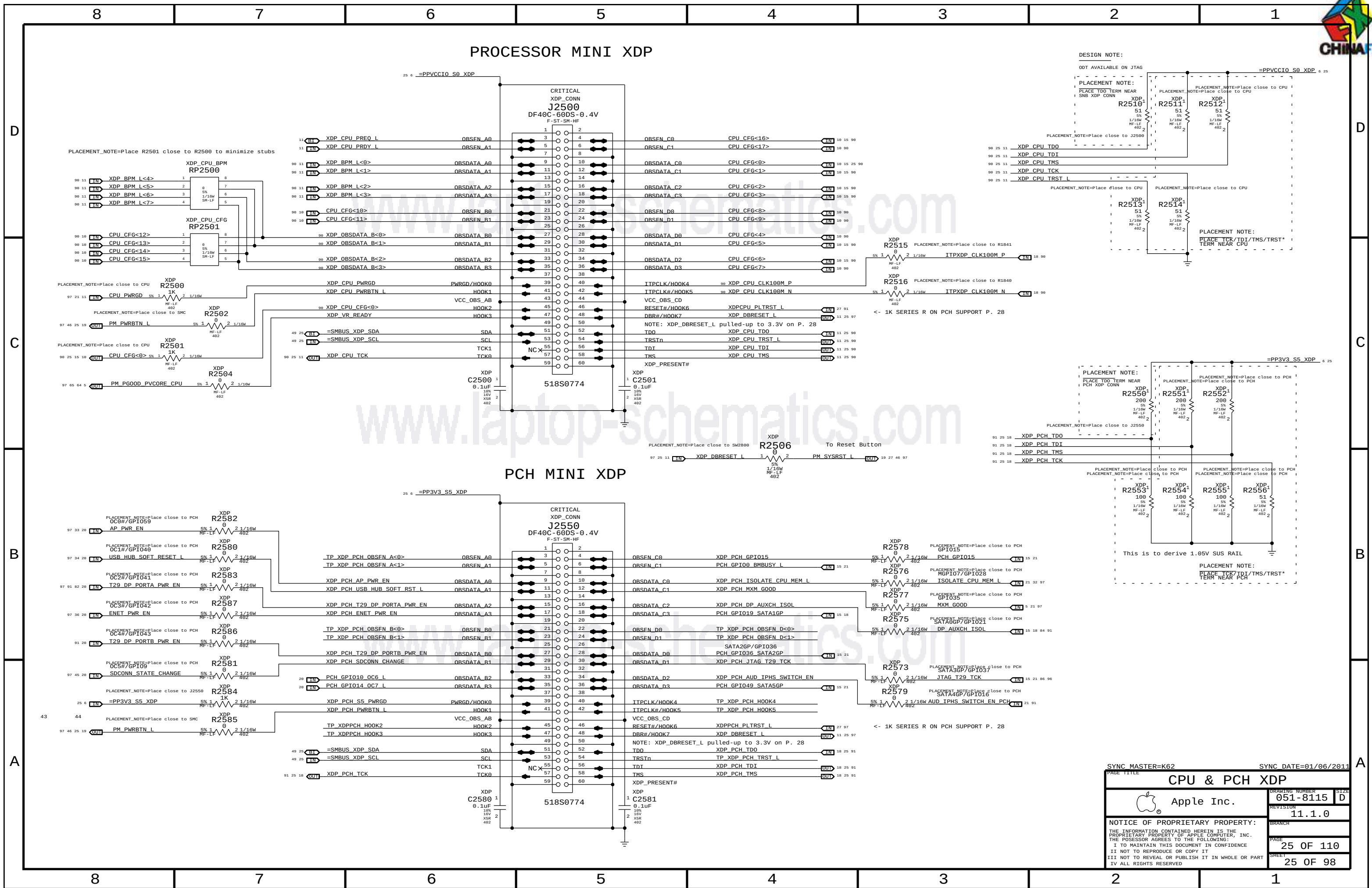
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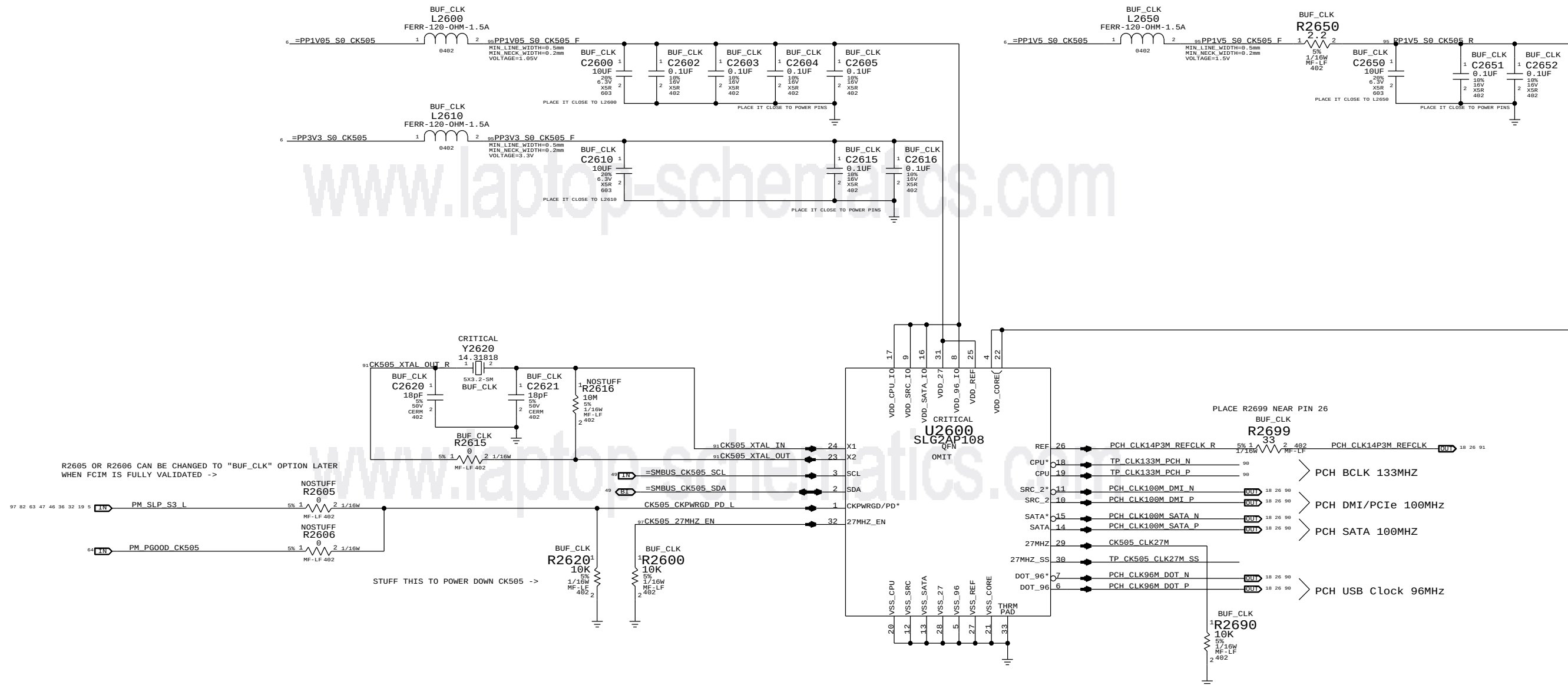




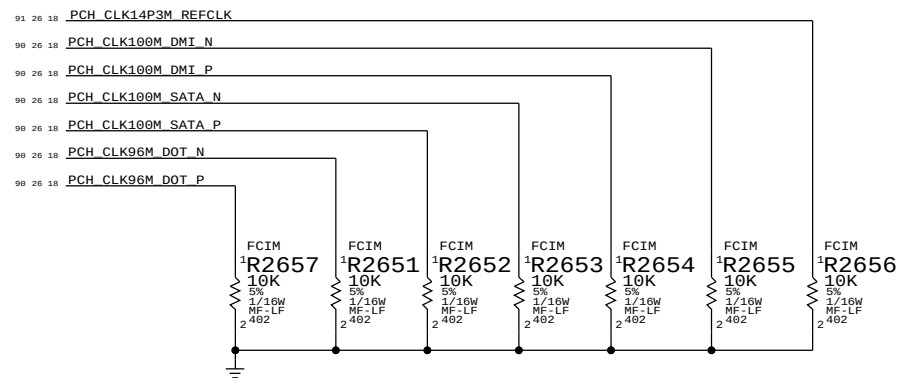
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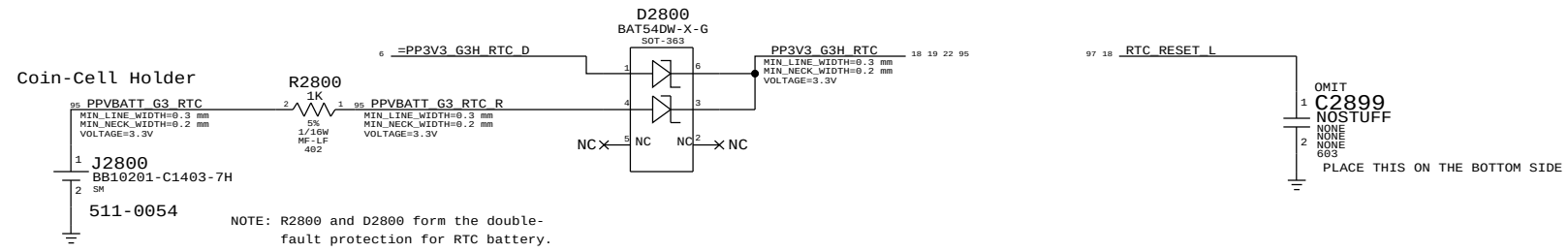
UNUSED clock terminations for FCIM MODE



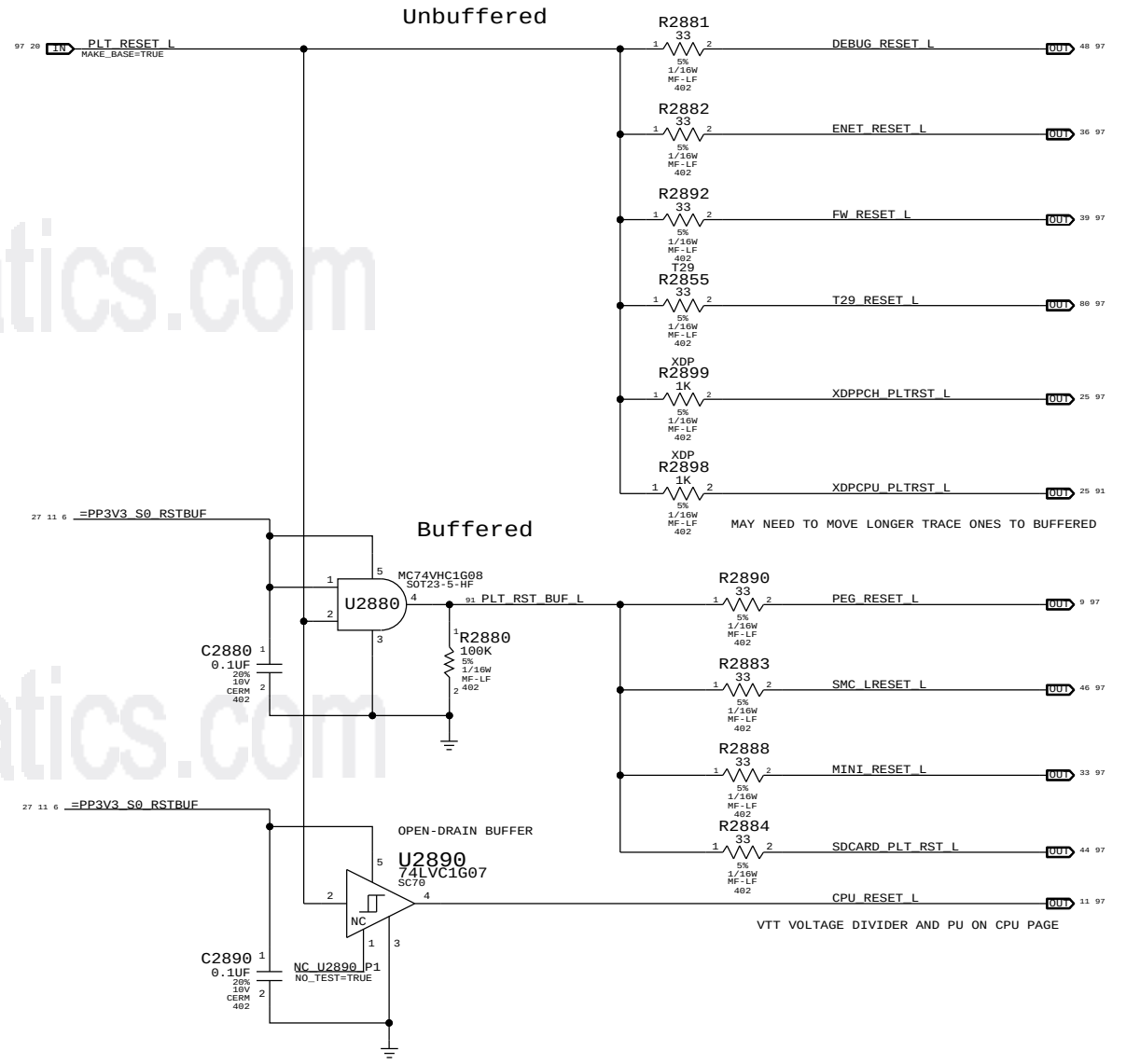
PAGE TITLE		SYNC DATE=01/06/2011	
CLOCK (CK505)		DRAWING NUMBER	051-8115
Apple Inc.		REVISION	11.1.0
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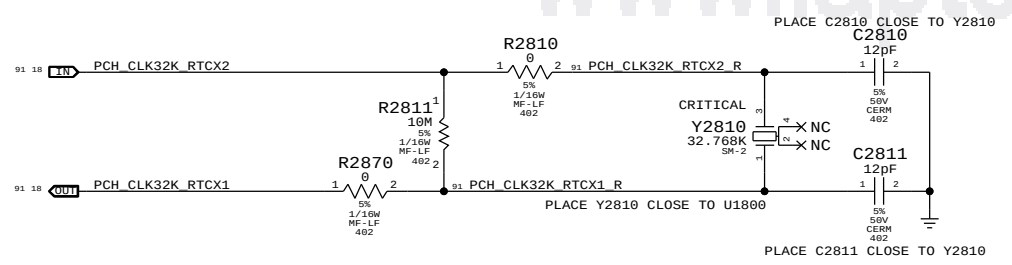
RTC Power Sources



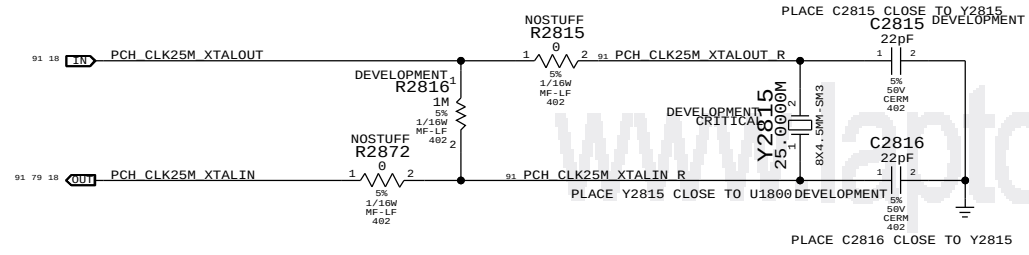
Platform Reset Connections



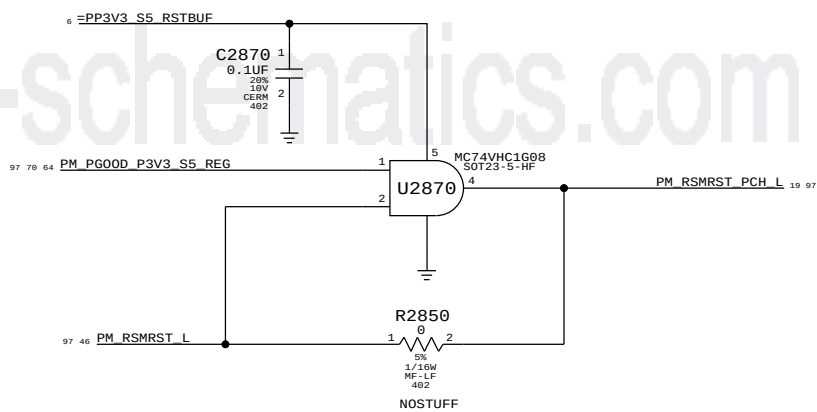
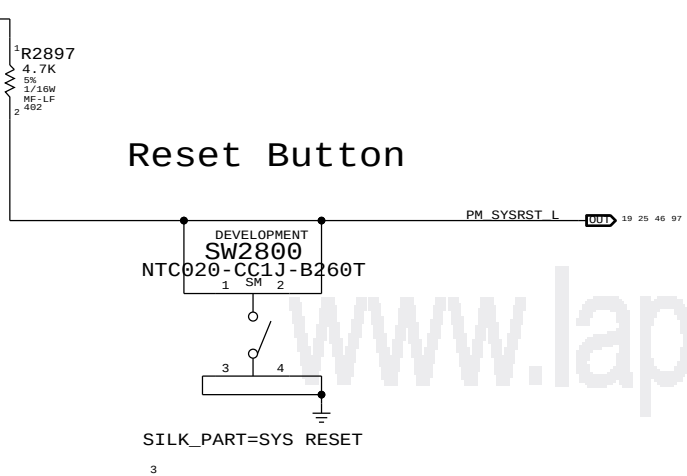
PCH RTC Crystal



PCH 25MHZ CRYSTAL

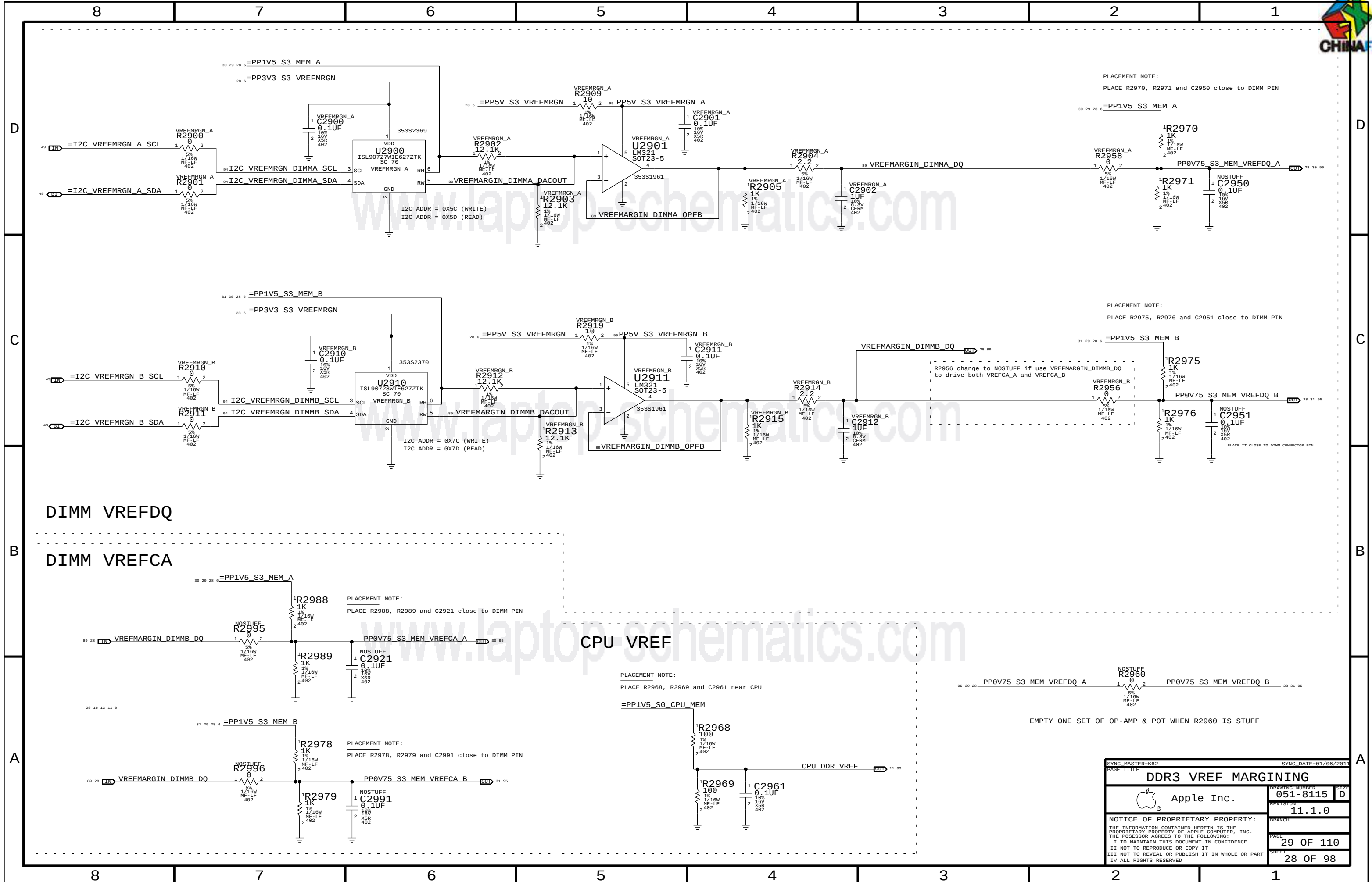


Reset Button



SMC PROVIDES RSMRST_L DE-ASSERTION DELAY UPON ENTRY TO S5
SMC PROVIDES RSMRST_L ASSERTION TIMING REQUIREMENTS UPON EXPECTED EXIT FROM S5
SMC MAY FORCE A RSMRST_L ASSERTION WITHOUT AN S5 POWER TRANSITION IN SOME ERROR CASES
PGOOD PROVIDES RSMRST_L ASSERTION TIMING REQUIREMENTS UPON AN UN-EXPECTED EXIT FROM S5 (POWER LOSS)

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		CHIPSET SUPPORT	
Apple Inc.		DRAWING NUMBER	051-8115
		REVISION	11.1.0
		BRANCH	
		PAGE	28 OF 110
		SHEET	27 OF 98
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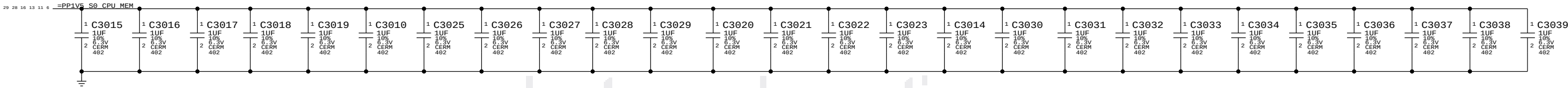


DIMM A (CLOSER TO CPU)

CAPS TO STITCH 1V5_CPU_MEM TO GND NEAR DIMM

DIMM B (FURTHER FROM CPU)

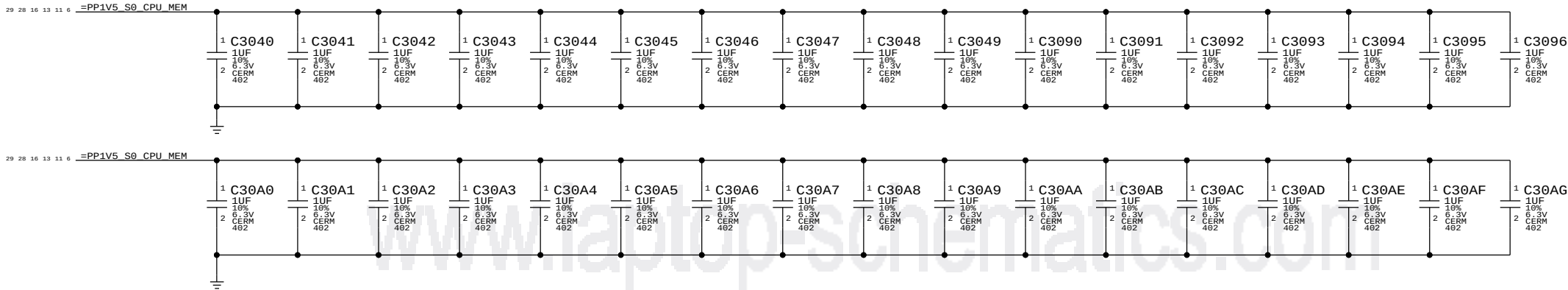
D



D

C

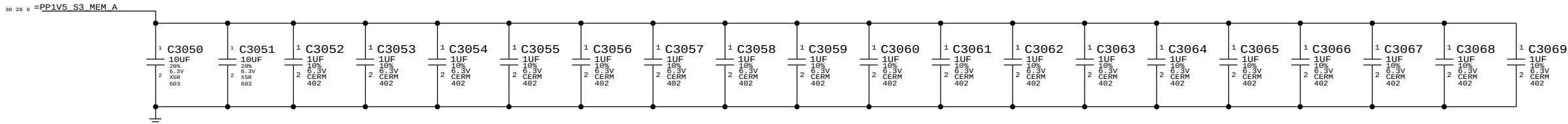
EXTRA DECOUPLING CAPS FOR 1V5_CPU_MEM RAIL



C

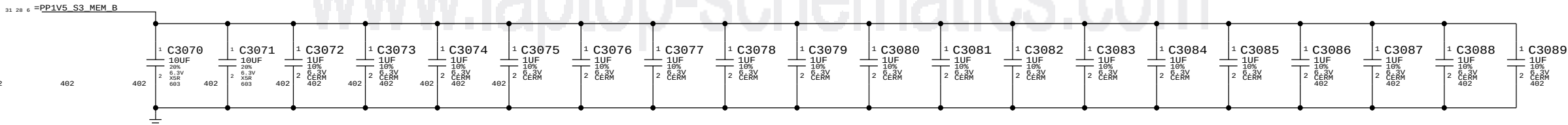
B

DECOUPLING CAPS FOR 1V5_S3_MEM AT CHANNEL A DIMM CONNECTOR



B

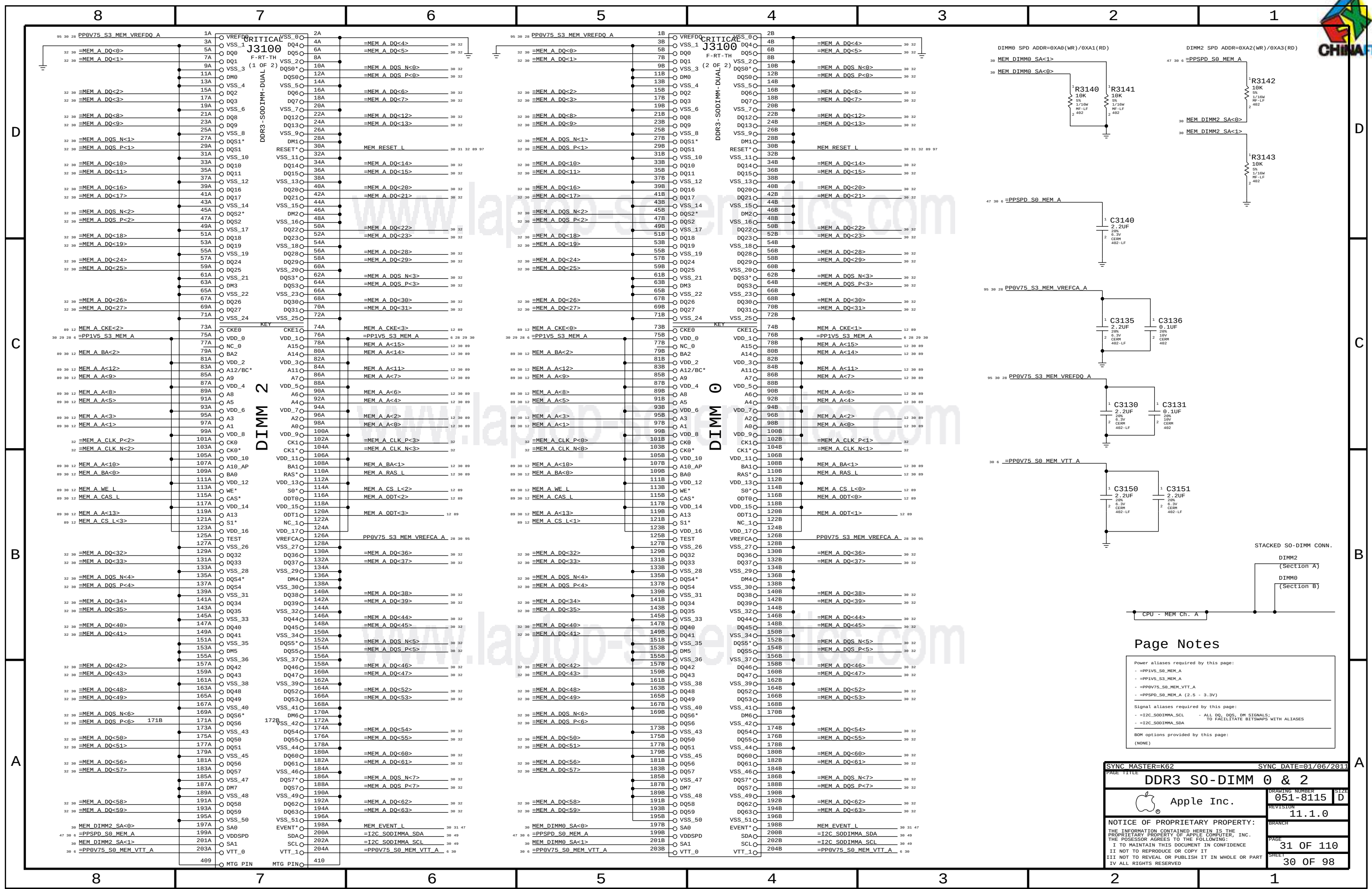
DECOUPLING CAPS FOR 1V5_S3_MEM AT CHANNEL B DIMM CONNECTOR



A

A

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		MEMORY CAPS	
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		REVISION	11.1.0
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		PAGE	30 OF 110
		SHEET	29 OF 98



Page Notes

Power aliases required by this page:

- PP1V5_S0_MEM_A
- PP1V5_S3_MEM_A
- PP0V75_S0_MEM_VTT_A
- PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

- I2C_S0DIMMA_SCL - ALL DQ, DQS, DM SIGNALS; TO FACILITATE BITSTREAMS WITH ALIASES
- I2C_S0DIMMA_SDA

BOM options provided by this page:

(NONE)

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
DDR3 SO-DIMM 0 & 2		DRAWING NUMBER	051-8115
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Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PP1V5_S3_MEM_B
- =PP8V75_S0_MEM_VTT_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_S0IMMB_SCL - ALL DQ, DQS, DM SIGNALS;
TO FACILITATE BITSWAPS WITH ALIASES
- =I2C_S0IMMB_SDA

BDM options provided by this page:

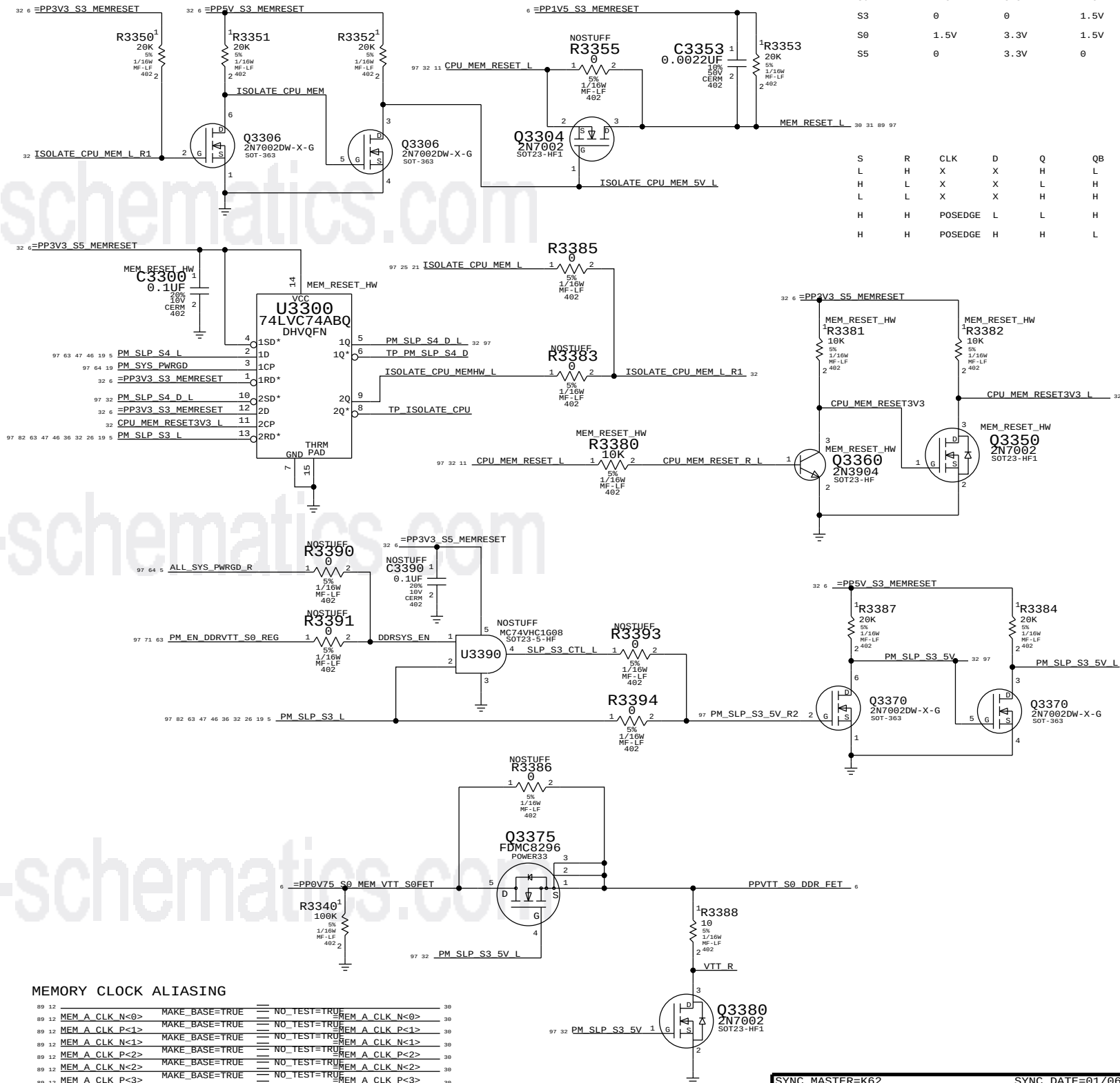
(NONE)

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		PAGE NO.	
DDR3 SO-DIMM		1 & 3	
 Apple Inc.		DRAWING NUMBER 051-8115	
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		BRANCH PAGE 32 OF 110	
		SHEET 31 OF 98	



DDR3 RESET SUPPORT

SNB? CANNOT CONTROL THIS SIGNAL DIRECTLY SINCE IT MUST BE HIGH IN SLEEP AND CPU MEM RAILS ARE NOT POWERED IN SLEEP.



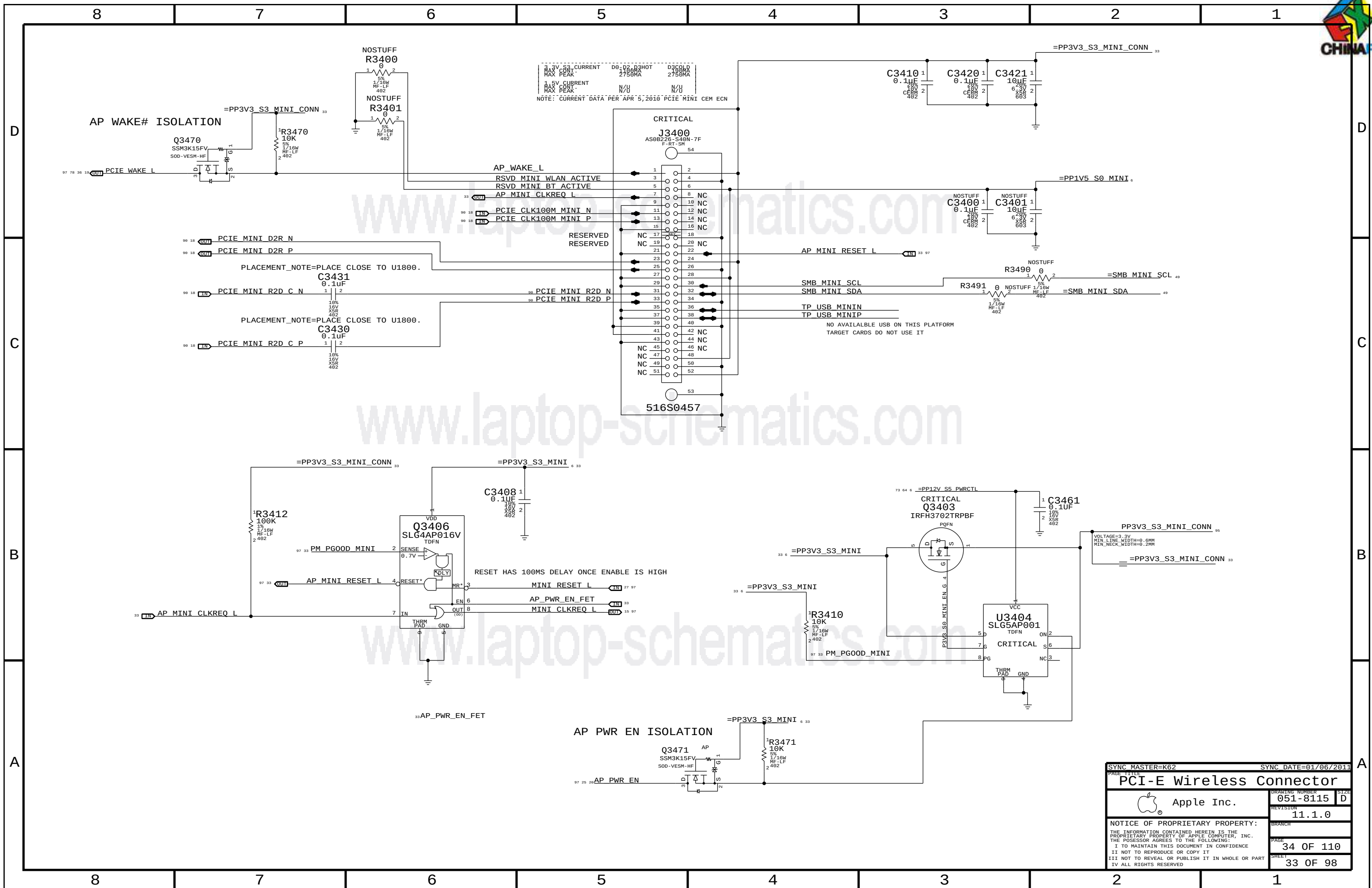
MEMORY CLOCK ALIASING

MEM A CLK N<0>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM A CLK N<0>	MEM B CLK N<0>
MEM A CLK P<1>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM A CLK P<1>	MEM B CLK P<1>
MEM A CLK N<1>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM A CLK N<1>	MEM B CLK N<1>
MEM A CLK P<2>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM A CLK P<2>	MEM B CLK P<2>
MEM A CLK N<2>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM A CLK N<2>	MEM B CLK N<2>
MEM A CLK P<3>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM A CLK P<3>	MEM B CLK P<3>
MEM A CLK N<3>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM A CLK N<3>	MEM B CLK N<3>
MEM B CLK P<0>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM B CLK P<0>	MEM B CLK P<0>
MEM B CLK N<0>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM B CLK N<0>	MEM B CLK N<0>
MEM B CLK P<1>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM B CLK P<1>	MEM B CLK P<1>
MEM B CLK N<1>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM B CLK N<1>	MEM B CLK N<1>
MEM B CLK P<2>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM B CLK P<2>	MEM B CLK P<2>
MEM B CLK N<2>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM B CLK N<2>	MEM B CLK N<2>
MEM B CLK P<3>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM B CLK P<3>	MEM B CLK P<3>
MEM B CLK N<3>	MAKE_BASE=TRUE	NO_TEST=TRUE	MEM B CLK N<3>	MEM B CLK N<3>

CPU_RESET_L	ISOLATE_L	MEM_RESET
S5	0	3.3V
S0	0	3.3V
S0	1.5V	3.3V
S3	0	1.5V
S0	1.5V	3.3V
S5	0	3.3V

S	R	CLK	D	Q	QB
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H	H
H	H	POSEDGE	L	L	H
H	H	POSEDGE	H	H	L

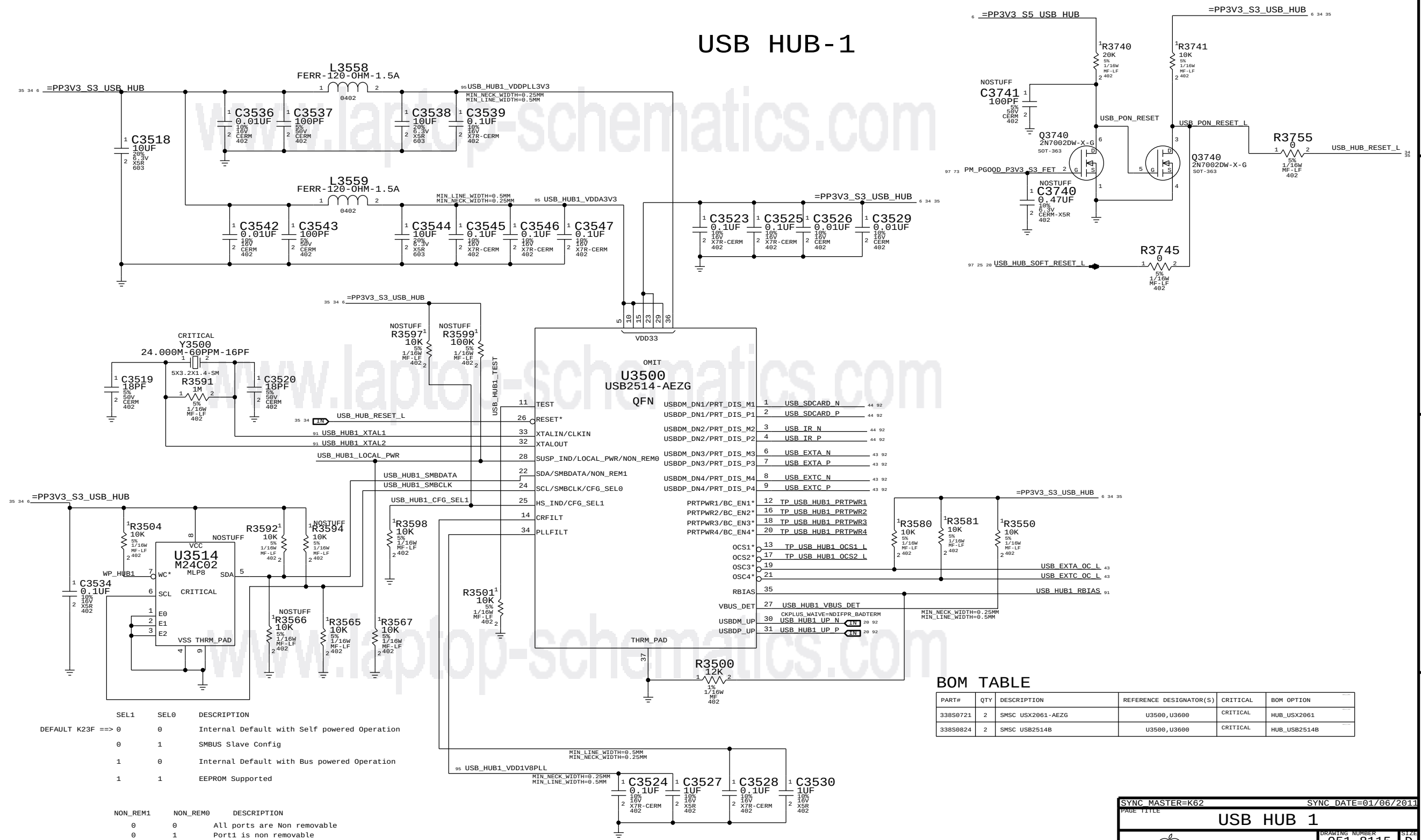
SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		PAGE	
DDR3 SUPPORT AND BITSWAPS		DRAWING NUMBER	
Apple Inc.		051-8115	
REVISION		SIZE	
11.1.0		D	
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PAGE TITLE			
PCI-E Wireless Connector			
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		SIZE	D
		REVISION	11.1.0
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USB HUB - 1



BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S9721	2	SMSC USX2061-AEZG	U3500, U3600	CRITICAL	HUB_USX2061
338S9824	2	SMSC USB2514B	U3500, U3600	CRITICAL	HUB_USB2514B

SEL1	SEL0	DESCRIPTION
DEFAULT K23F ==> 0	0	Internal Default with Self powered Operation
0	1	SMBUS Slave Config
1	0	Internal Default with Bus powered Operation
1	1	EEPROM Supported

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are Non removable
0	1	Port1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port1,2 and 3 are non Removable

SYNC MASTER=K62

SYNC DATE=01/06/2011

USB HUB 1

Apple Inc.

051-8115

11.1.0

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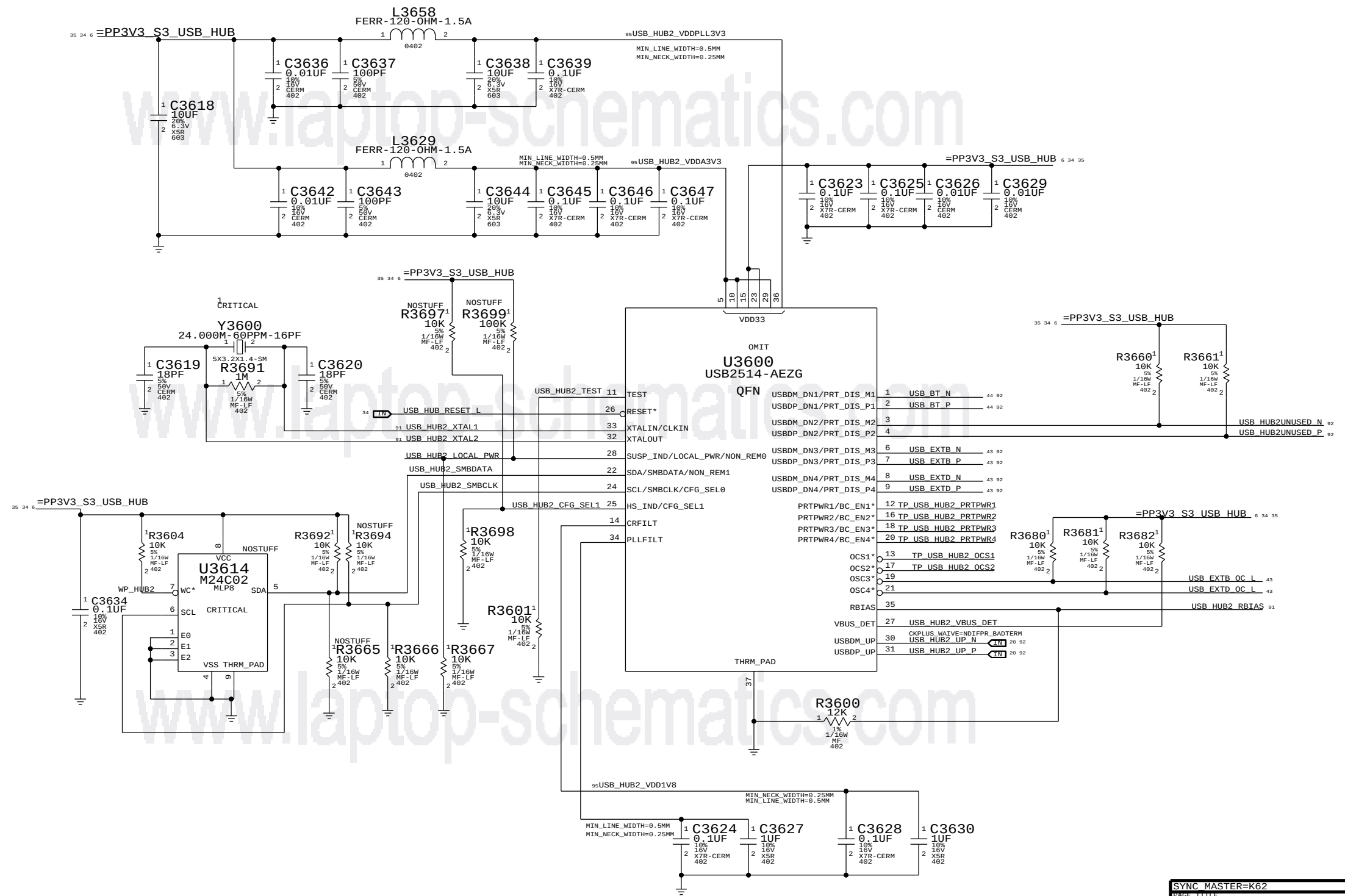
34 OF 98

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USB HUB-2

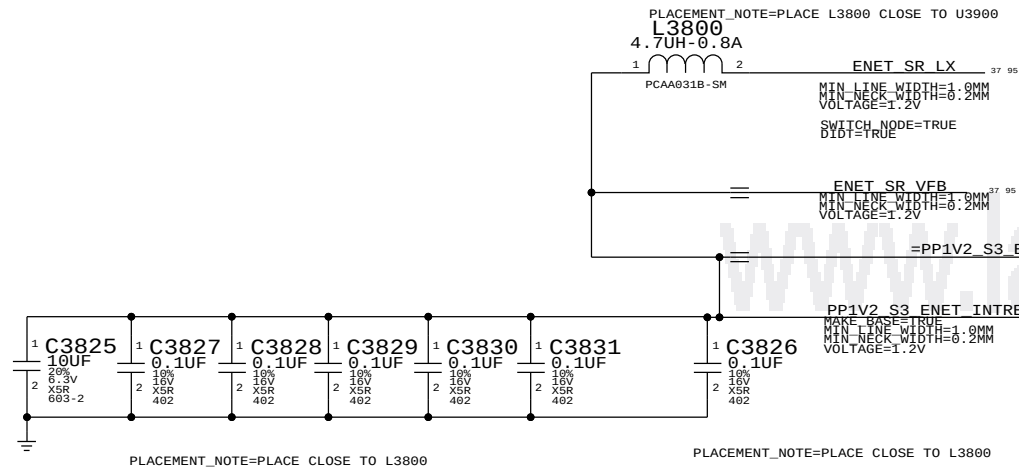


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PAGE TITLE		USB HUB 2	
DRAWING NUMBER		051-8115	SIZE D
REVISION		11.1.0	BRANCH
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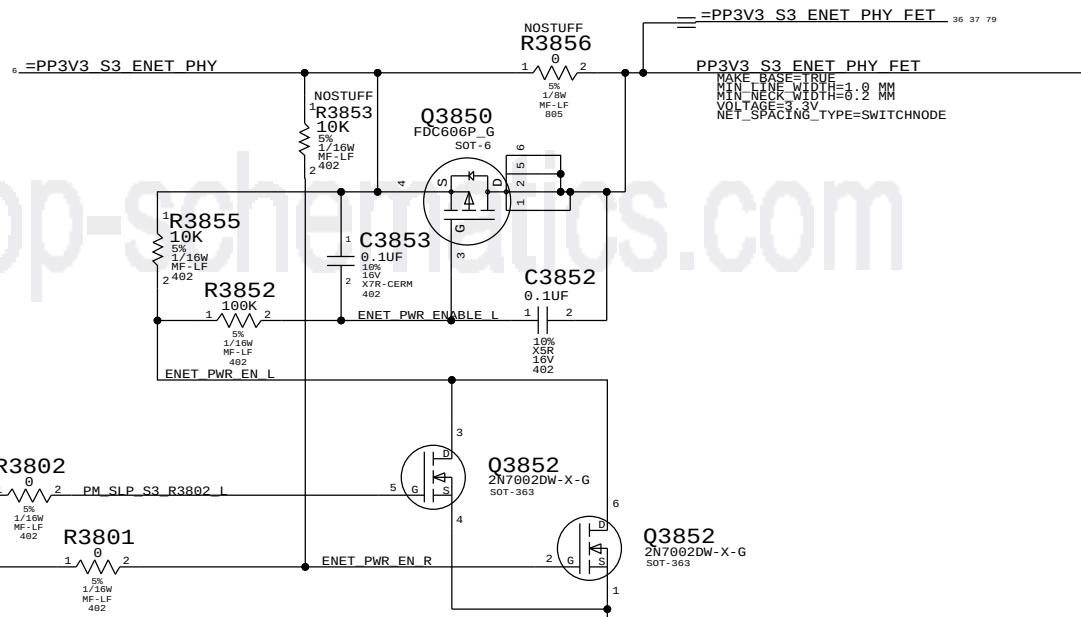
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CAESAR IV 1.2V INT.VR CMPTS

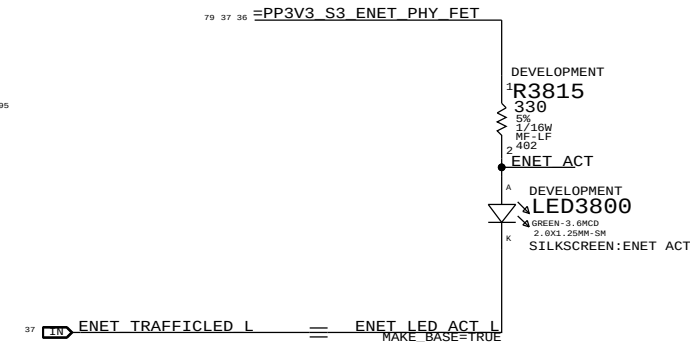


CAESAR IV POWER ENABLE CIRCUIT

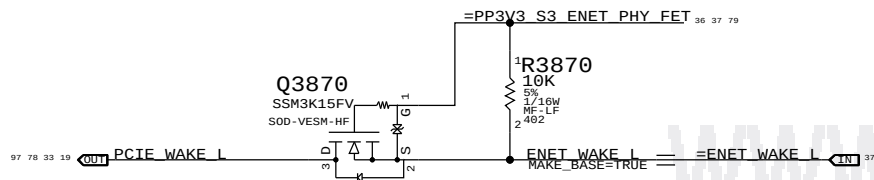
3V3_ENET_PHY_FET = S0 || (S3 POWER && ENET_PWR_EN)



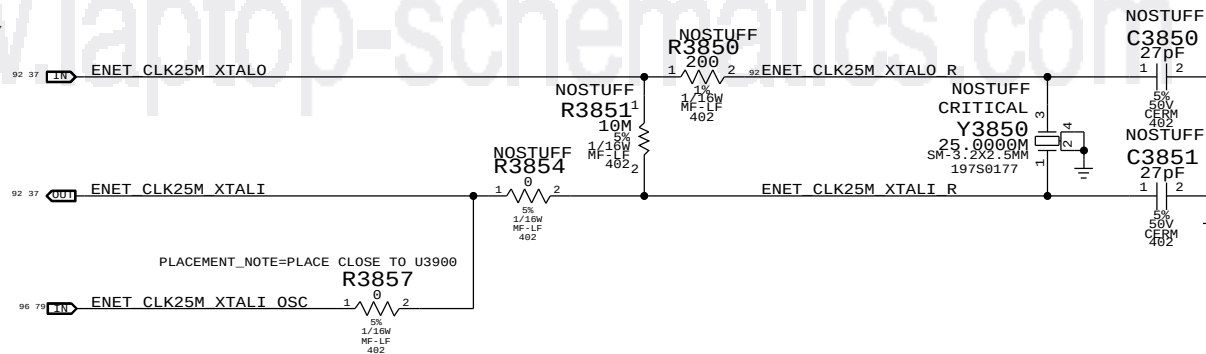
CAESAR IV ACTIVITY LED



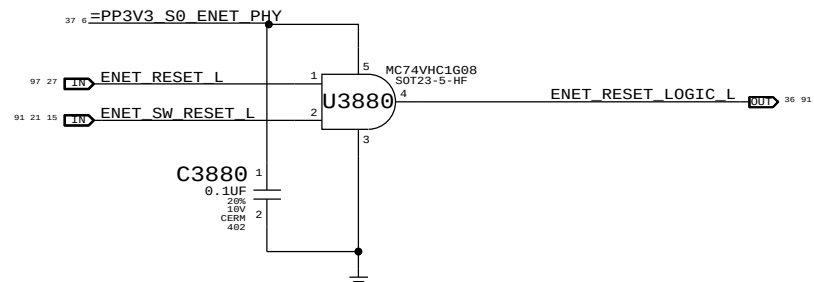
CAESAR IV WAKE# ISOLATION



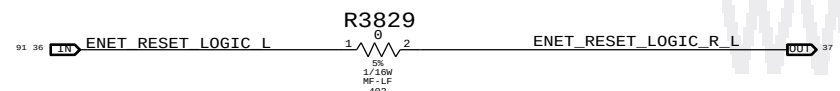
CAESAR IV 25MHZ XTAL



CAESAR IV SW RESET GATING

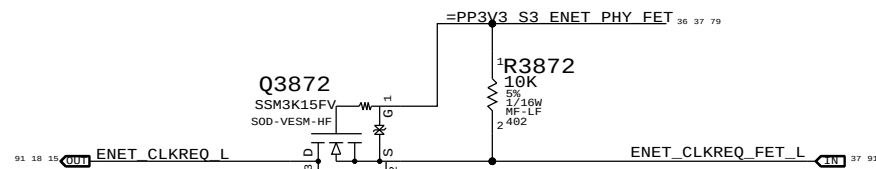


CAESAR IV RESET CONNECTION

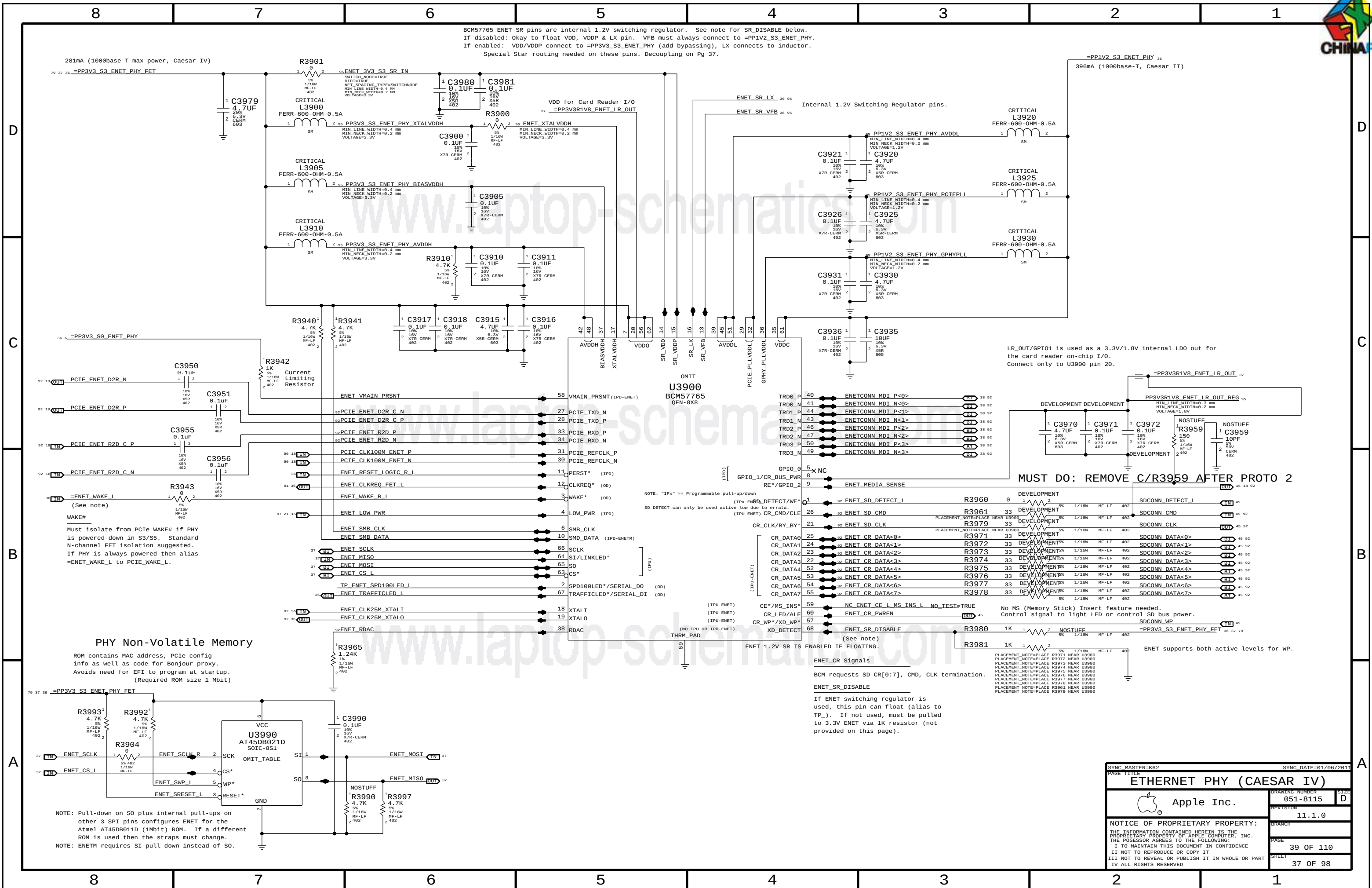


CAESAR IV STRAPS (NONE)

CAESAR IV CLKREQ ISOLATION

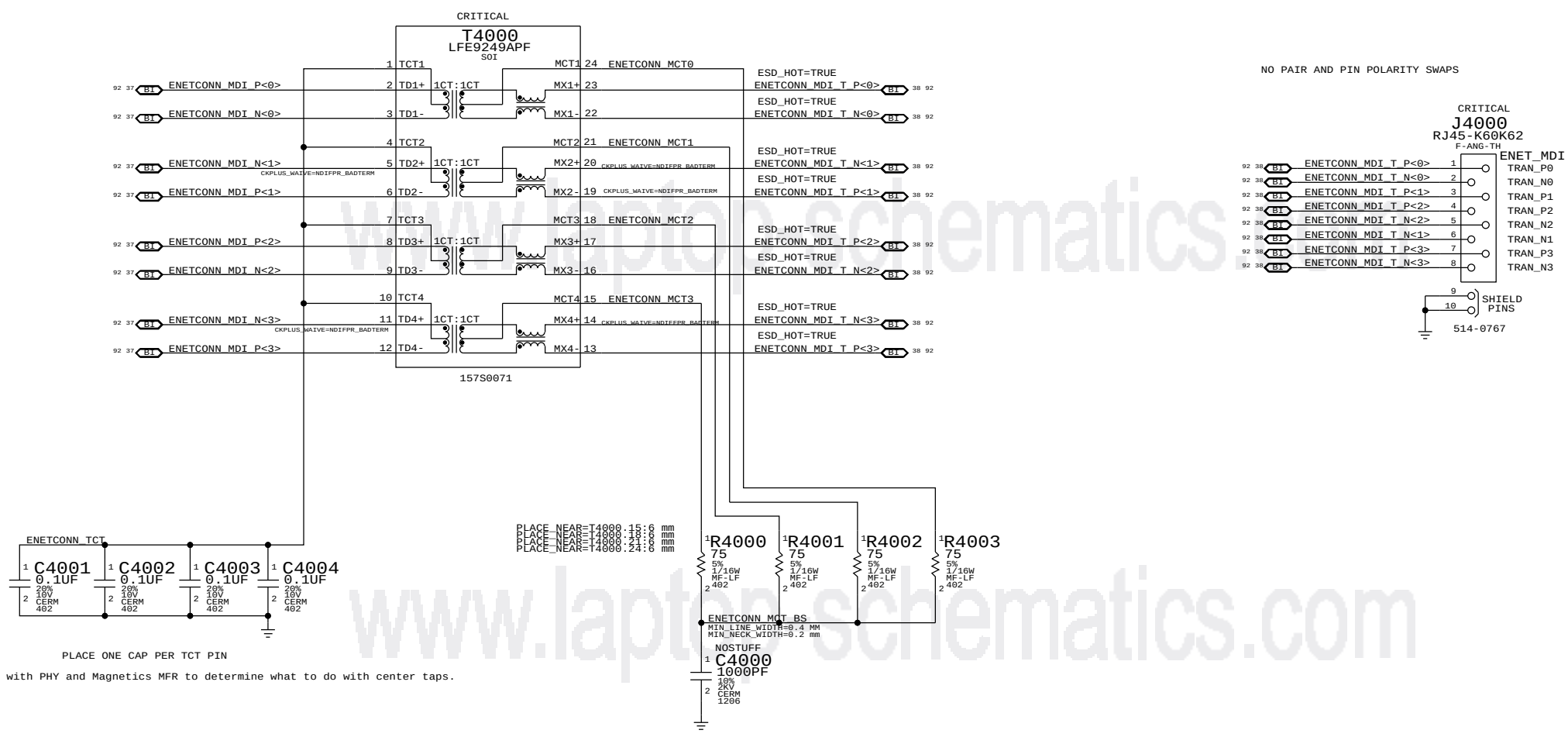


SYNC MASTER=K62		SYNC DATE=01/06/2011	
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CAESAR IV SUPPORT			
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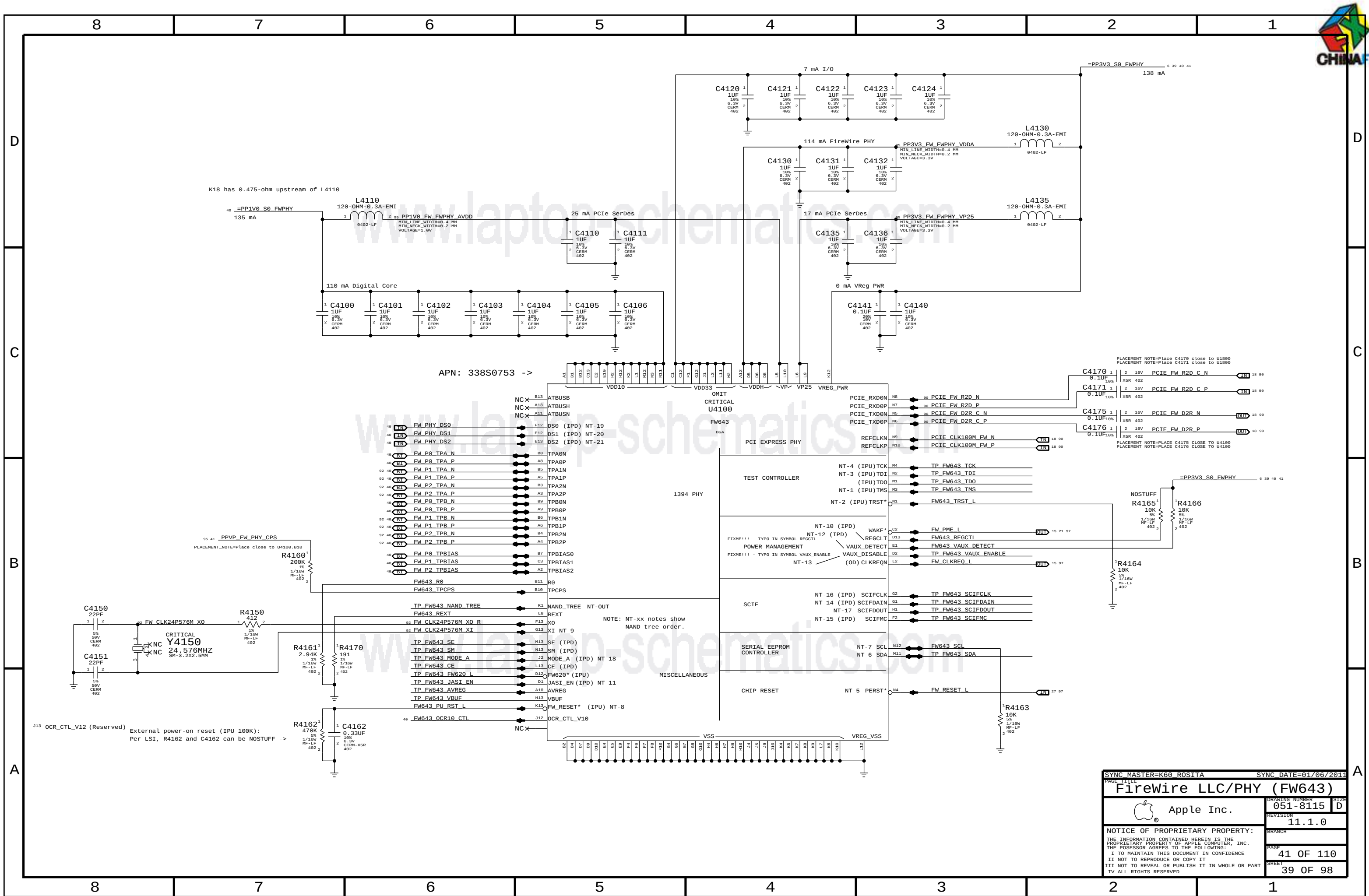
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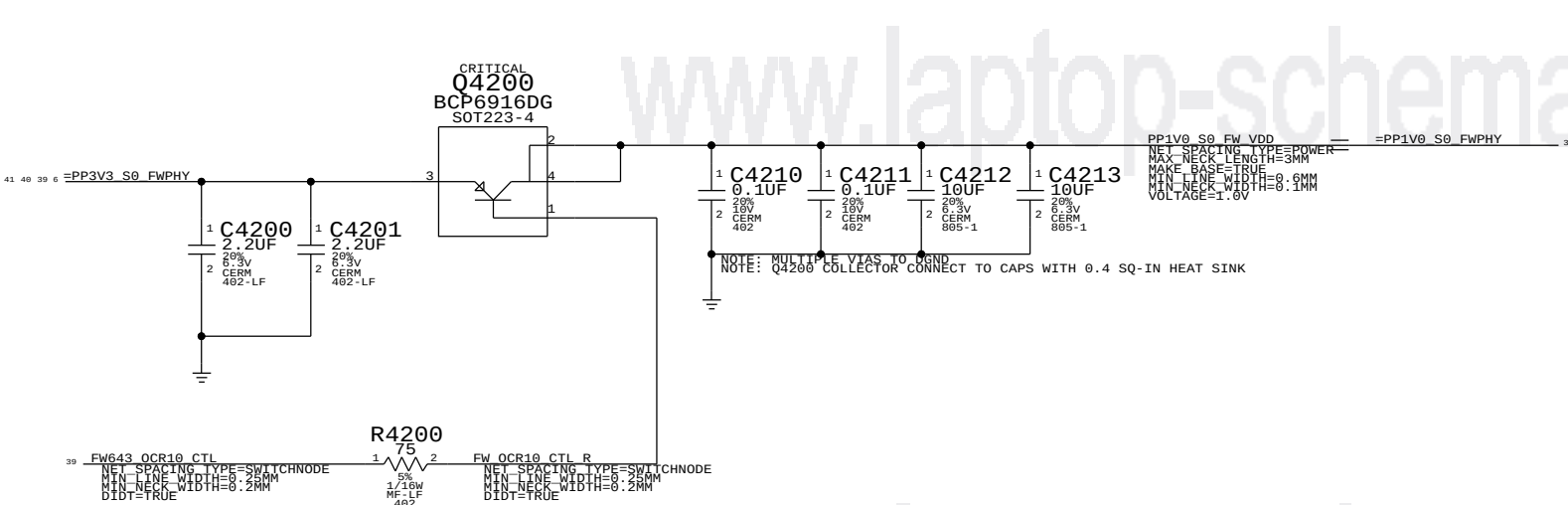
NOTE: Check with PHY and Magnetics MFR to determine what to do with center taps.

NOTE: BOB SMITH TERMINATION FOR EMC.

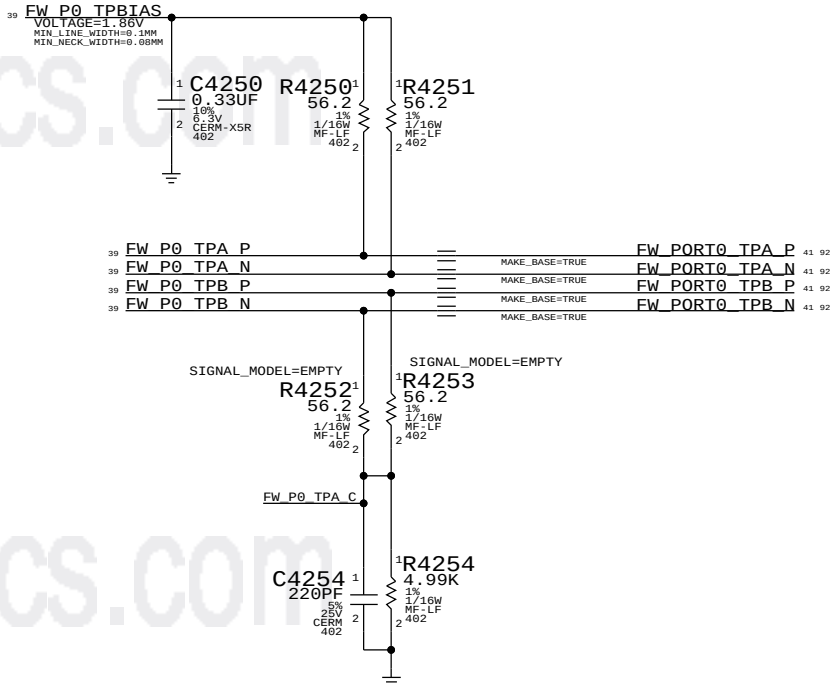
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PAGE TITLE			
Ethernet Connector			
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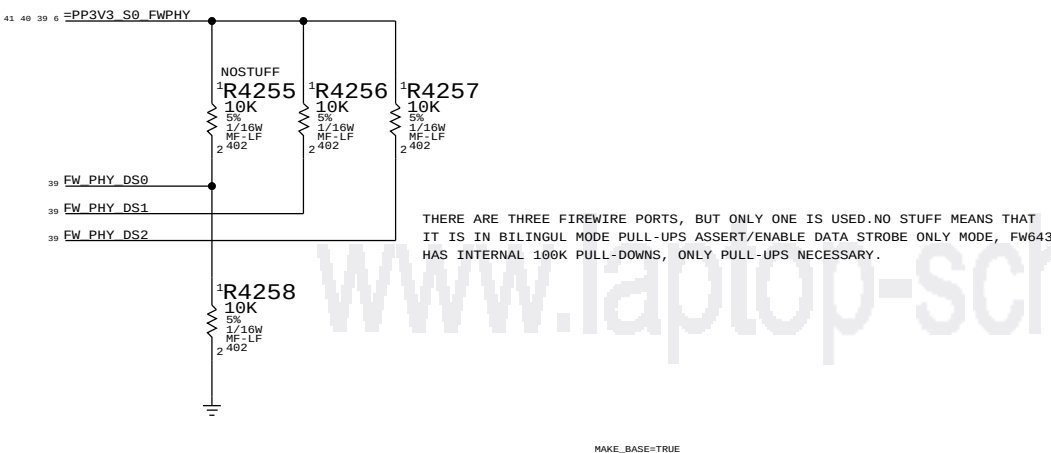
FW643 1.0V GENERATION



Termination Place close to FireWire PHY

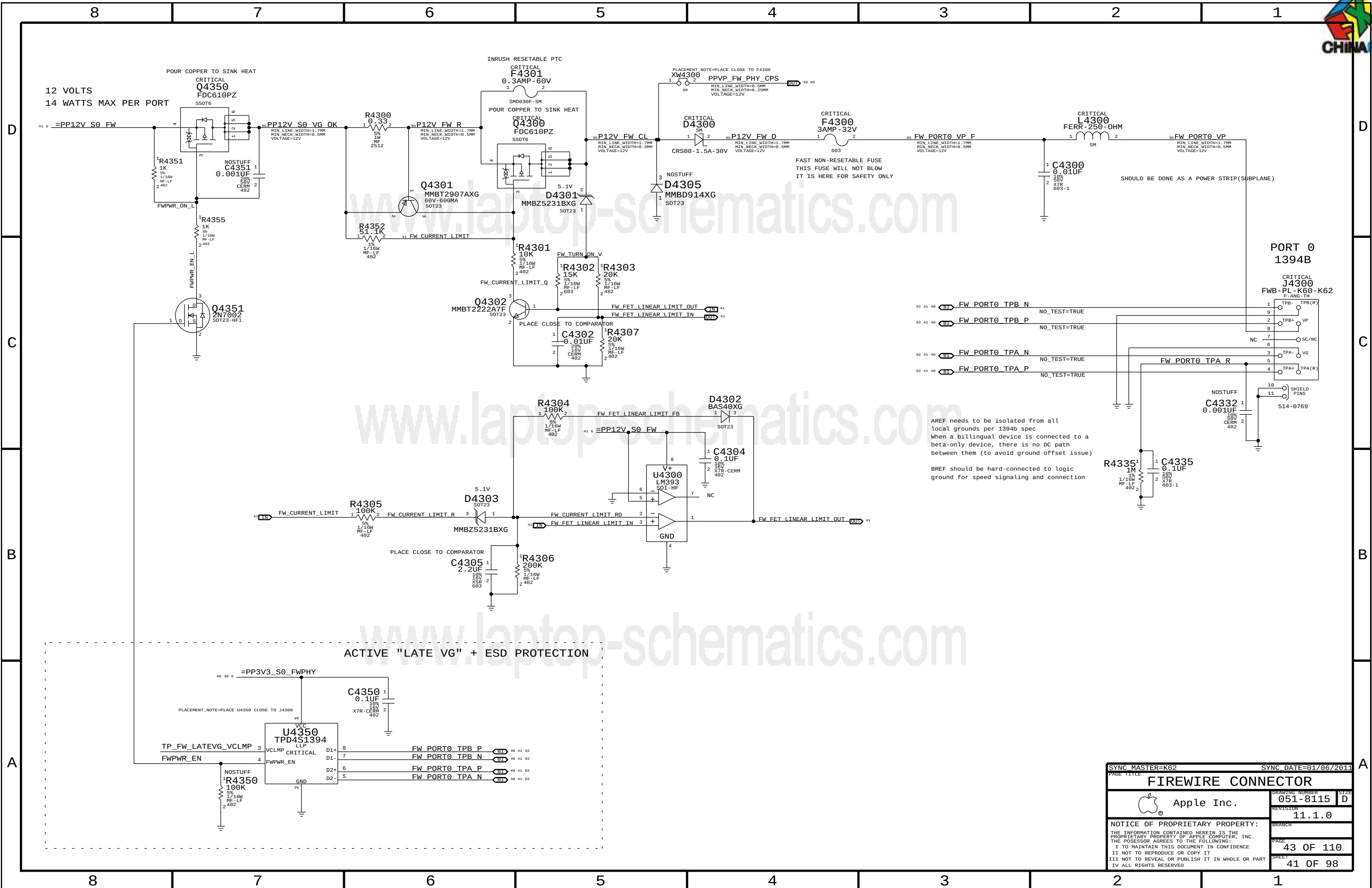


1394 PHY DATA/STROBE OPTIONS



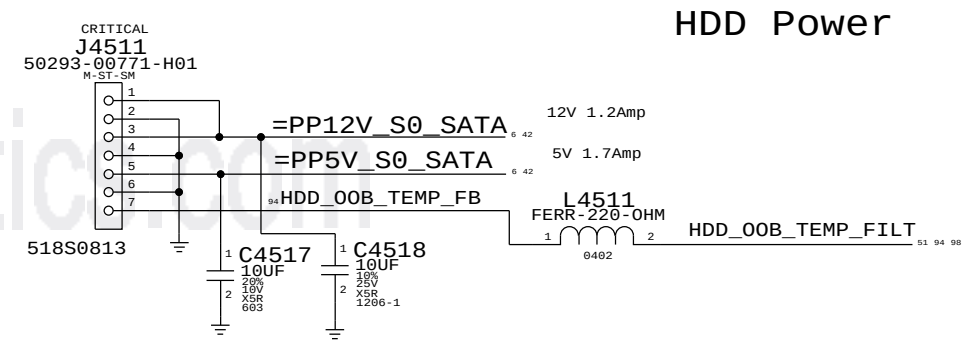
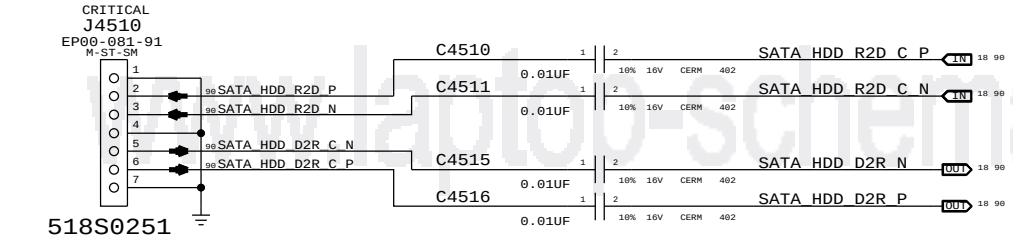
2ND & 3RD TPA/TPB PAIR UNUSED

FW_P1_TPBias	=	NC_FW_PORT1_TPBias
FW_P1_TPA_P	=	NC_FW_PORT1_TPA_P
FW_P1_TPA_N	=	NC_FW_PORT1_TPA_N
FW_P1_TPB_P	=	NC_FW_PORT1_TPB_P
FW_P1_TPB_N	=	NC_FW_PORT1_TPB_N
FW_P2_TPBias	=	NC_FW_PORT2_TPBias
FW_P2_TPA_P	=	NC_FW_PORT2_TPA_P
FW_P2_TPA_N	=	NC_FW_PORT2_TPA_N
FW_P2_TPB_P	=	NC_FW_PORT2_TPB_P
FW_P2_TPB_N	=	NC_FW_PORT2_TPB_N

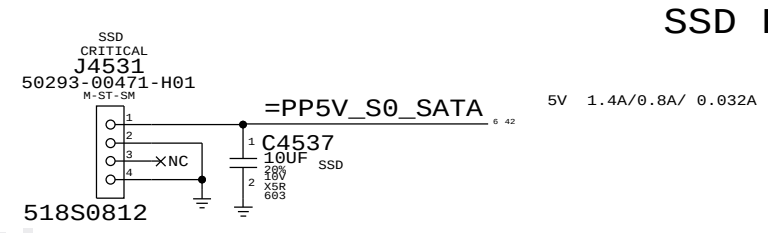
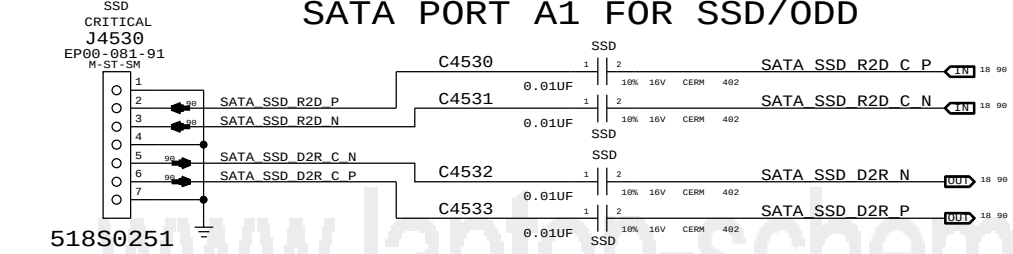




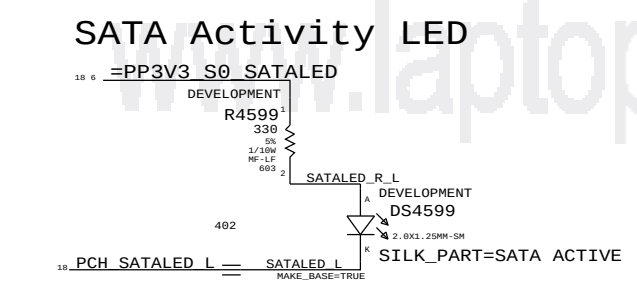
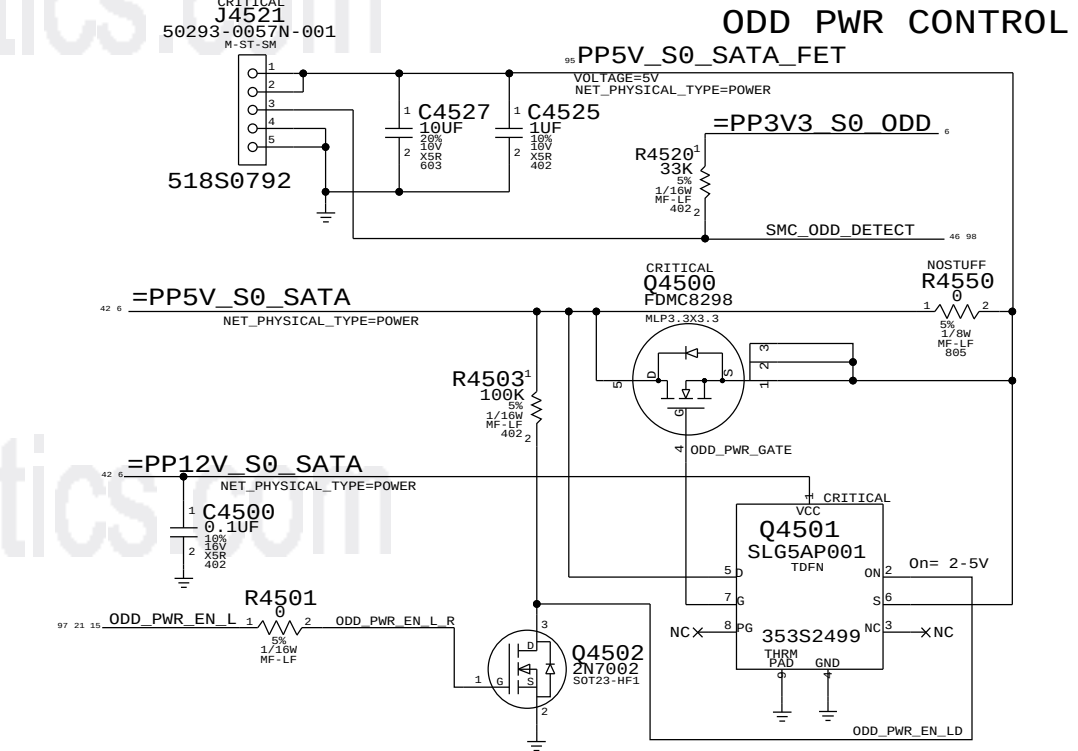
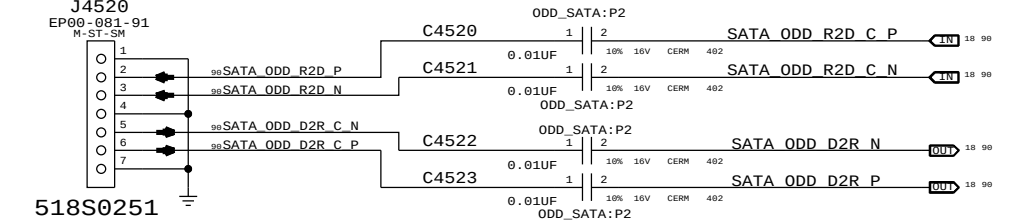
SILKSCREEN:SATA0 SATA PORT A0 FOR HDD



SILKSCREEN:SATA1 SATA PORT A1 FOR SSD/ODD



SILKSCREEN:SATA2 SATA PORT A2 FOR ODD



BOMOPTION OPTIONS FOR SATA PORT A1 AND A2

A1	A2	ODD_SATA:P1	ODD_SATA:P2
SSD	ODD		X
ODD		X	

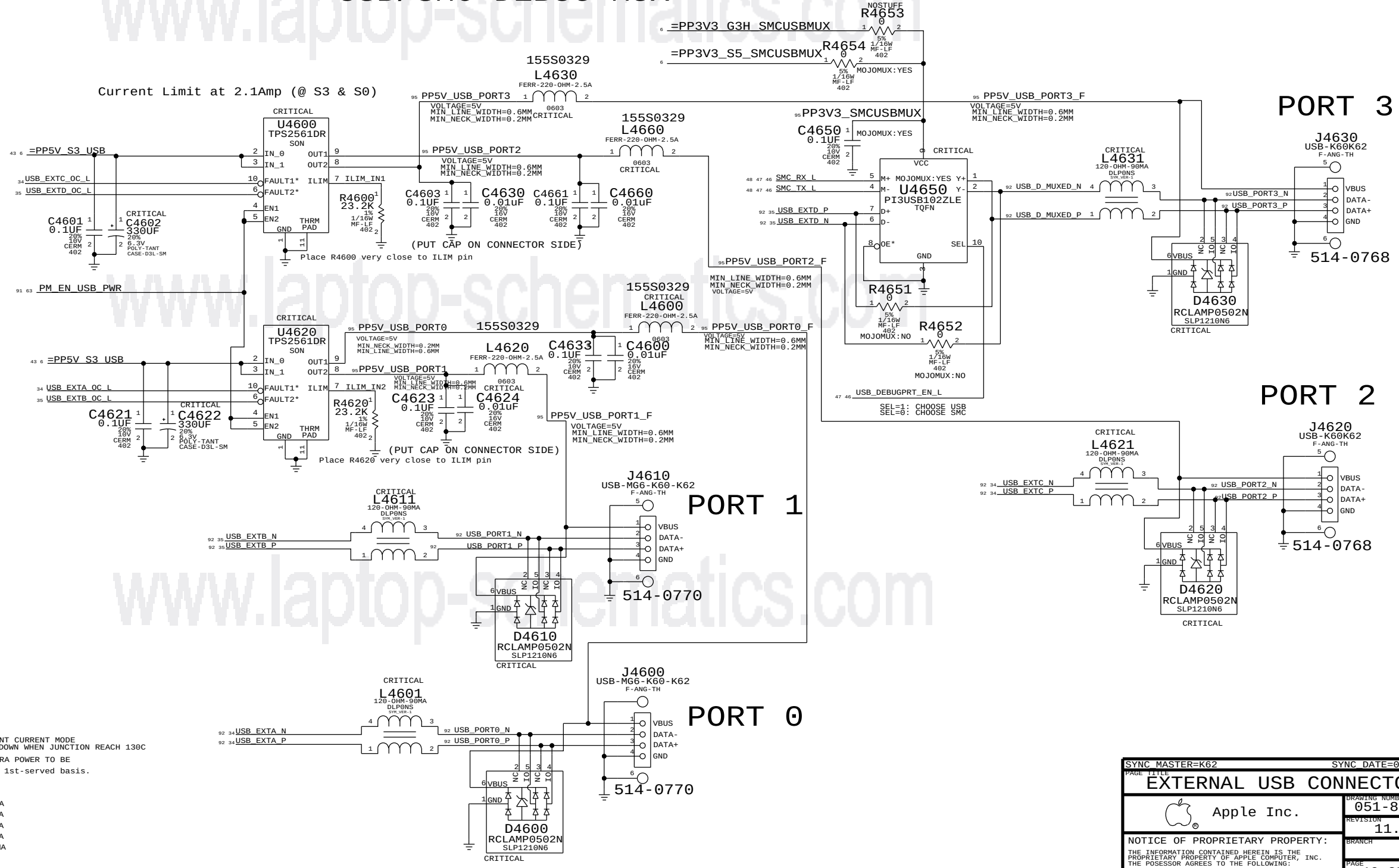
USE OF PORT A2 FOR SSD IS NOT INTENDED VIA BOMOPTION THOUGH MLB SUPPORTS IT.
K60E should stuff S_P1_ODD:YES because it has no SSD option.

PAGE TITLE		SYNC MASTER=K60 JERRY		SYNC DATE=01/06/2011	
SATA Connectors		DRAWING NUMBER		SIZE	
Apple Inc.		051-8115		D	
REVISION		11.1.0		BRANCH	
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USB/SMC DEBUG MUX

ADDED AT EVT & SWITCH TO S5 RAIL



USB PORT POWER:

EACH PORT IS HARDWARE Capable of :

STATE	MAX	MIN	(WITHIN THE TOLERANCE)
S0, S3	2.7A	2.1A	-- PER PORT

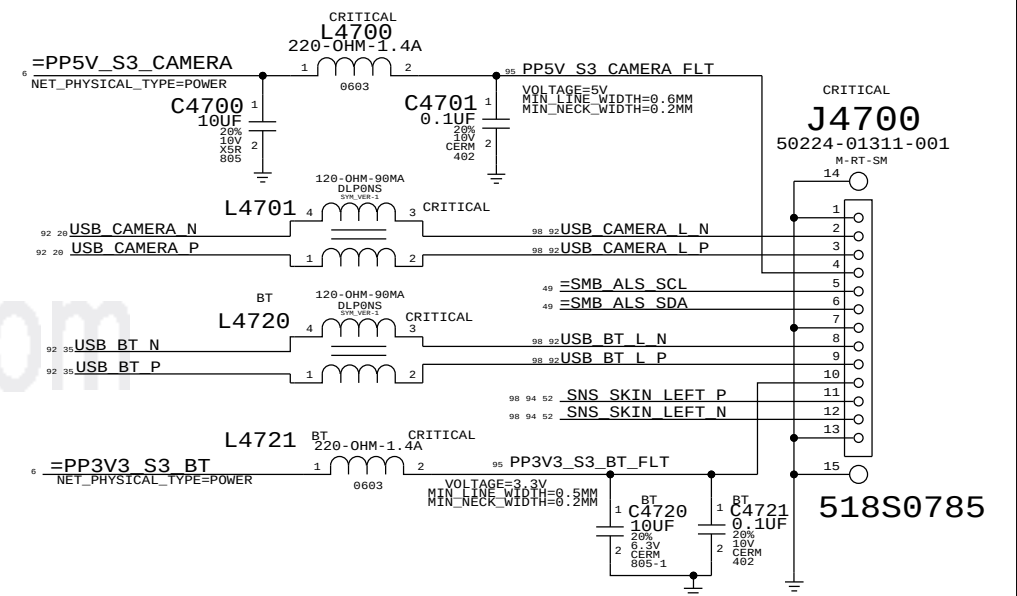
WHEN CURRENT HITS LIMIT, TPS2561 BECOME CONSTANT CURRENT MODE AND STAY AT THE LIMIT LEVEL UNTIL THERMAL SHUTDOWN WHEN JUNCTION REACH 130C

SOFTWARE WILL ALLOW 500MA/PORT, PLUS 2700MA EXTRA POWER TO BE distributed to approved devices on a 1st-come, 1st-served basis.

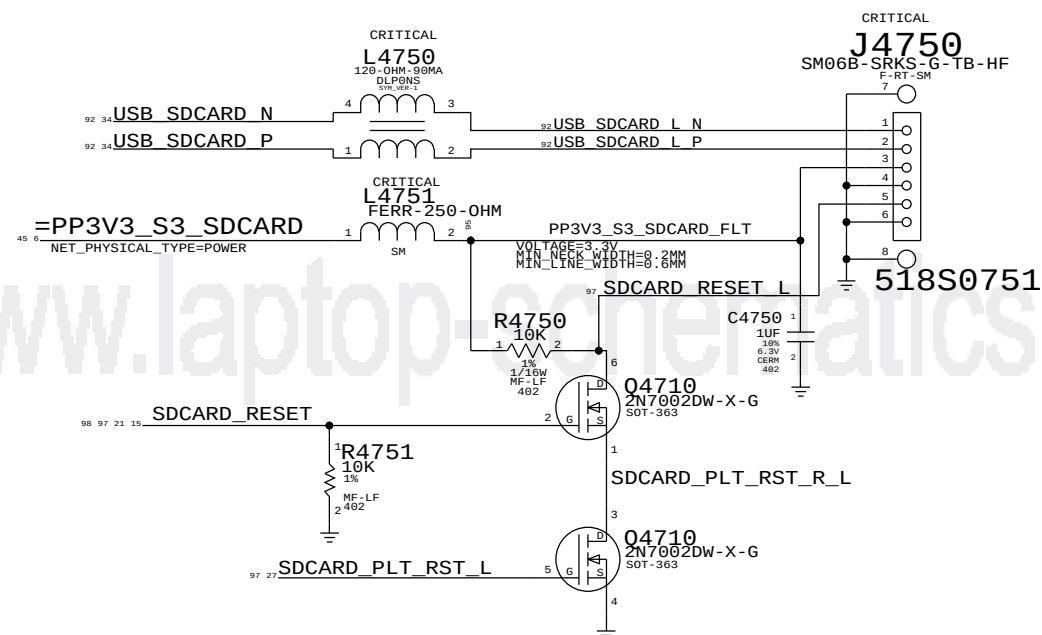
EXAMPLE: Port 1 - iPad fast charging = 2100mA
Port 2 - Wired Keyboard = 1100mA
Port 3 - iPhone fast charging = 1000mA
PORT 4 - USB 2.0 500MA = 500MA
TOTAL: 4700MA

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		EXTERNAL USB CONNECTORS	
Apple Inc.		DRAWING NUMBER	051-8115
		REVISION	11.1.0
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CAMERA/ALS & BLUETOOTH (K37A) CONNECTOR



SD Card Reader Board (Lazarus)

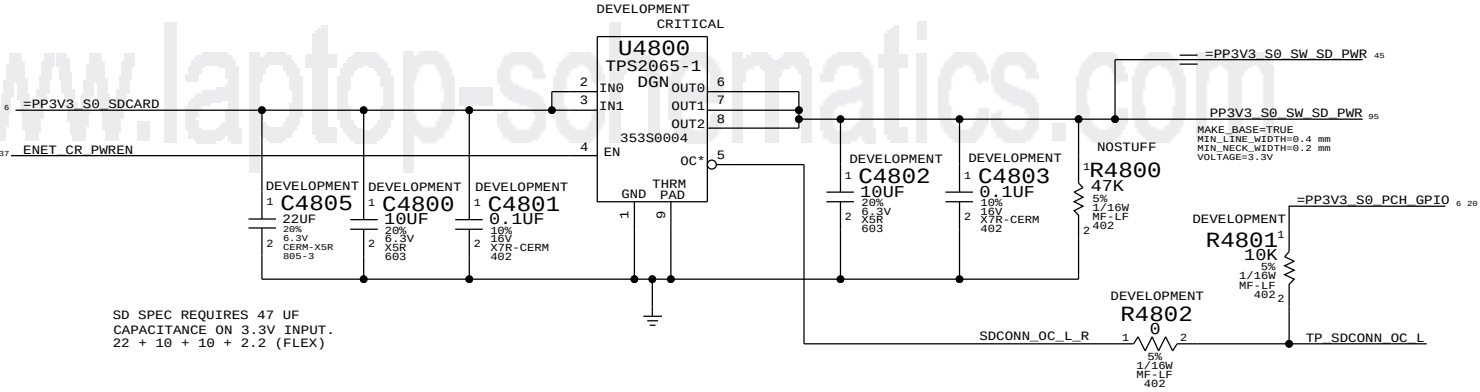


Skin Temp sense at upper Left Screen corner

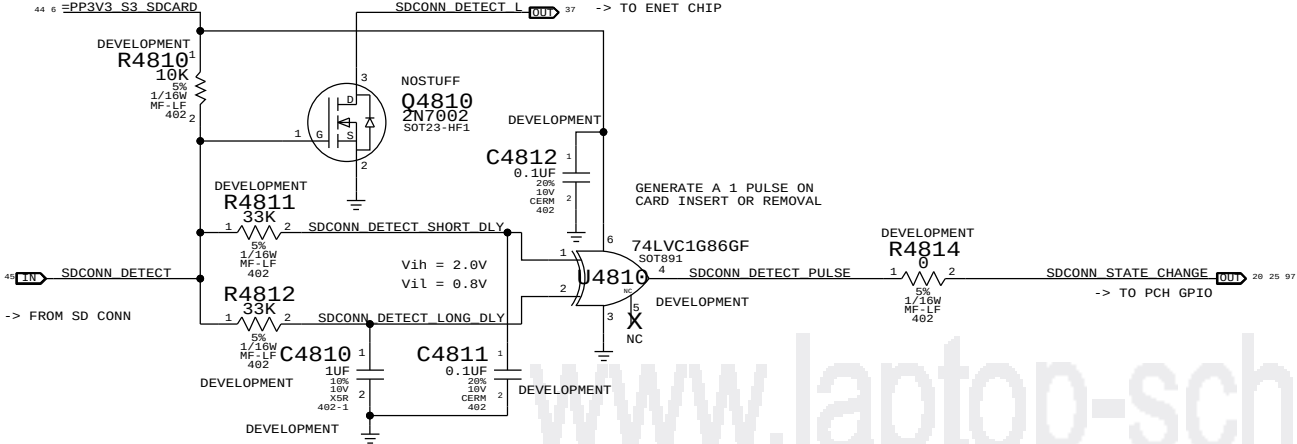


SD CARD 3.3V OVERCURRENT PROTECTION CHIP WITH ACTIVE LOAD DISCHARGE

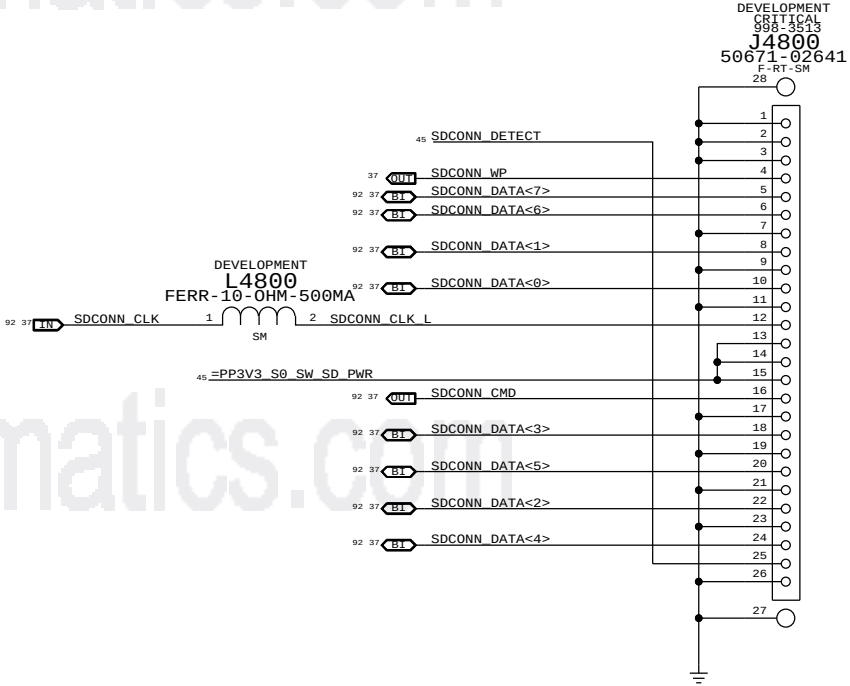
TPS2065-1 (1.0A LIMIT) HAS ACTIVE LOAD DISCHARGE SO R4800 IS NOSTUFF.



SDCONN DETECT DEBOUNCE, INVERSION, AND DETECT-CHANGED PCH GPIO CIRCUIT



SD CARD CONNECTOR

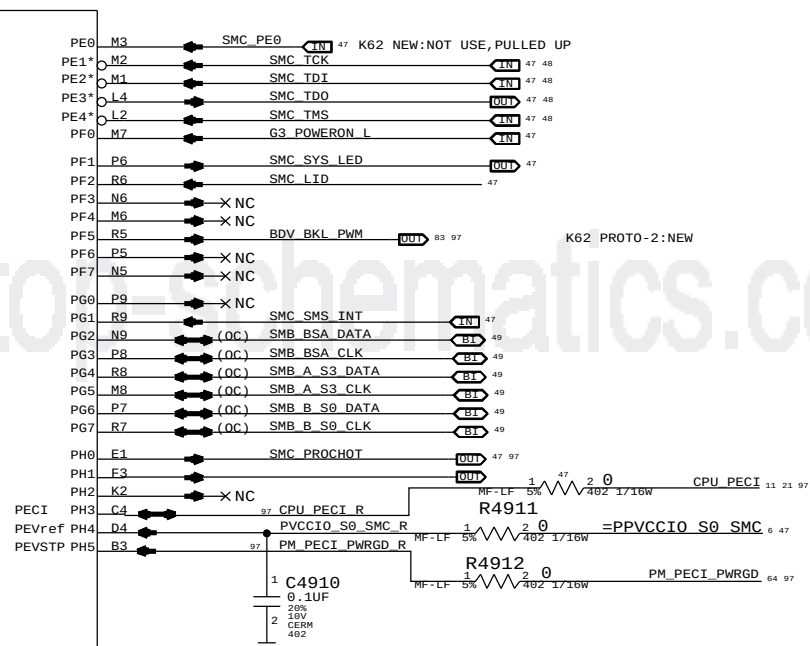
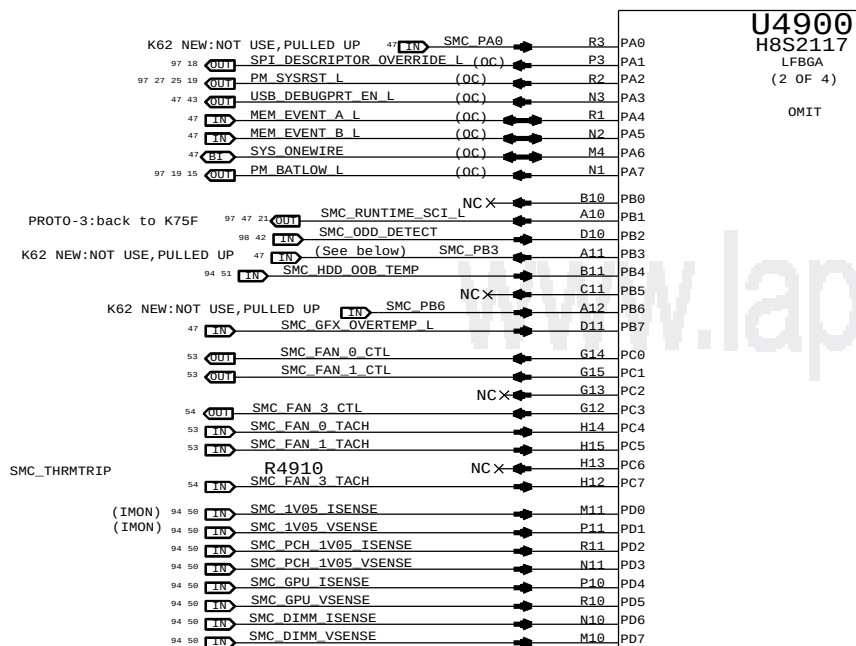
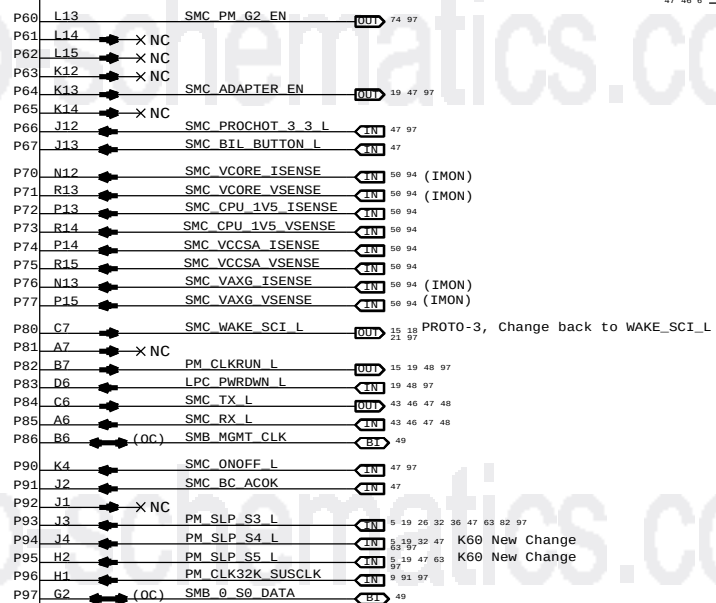
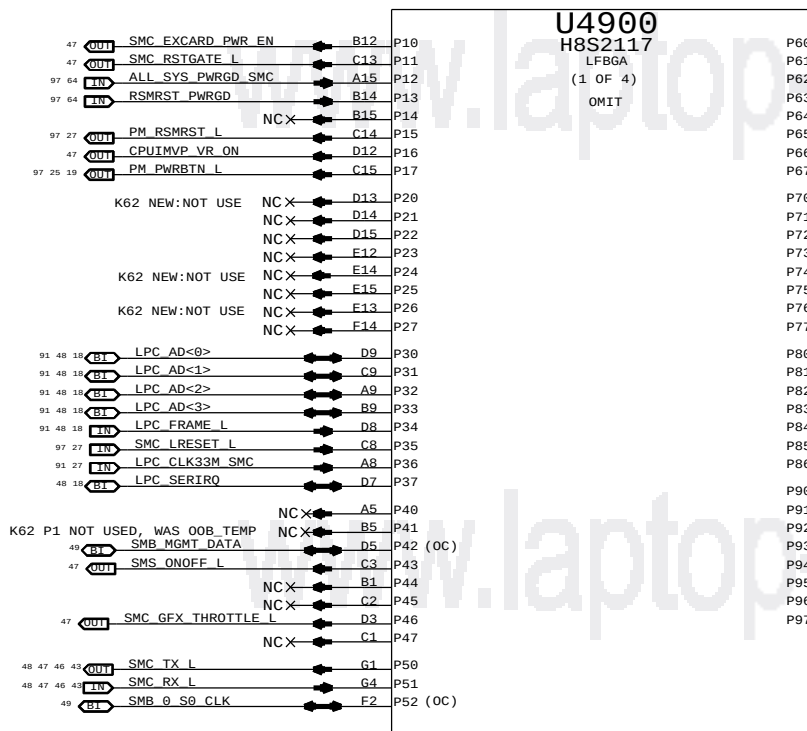


(CARD INSERTED = OPEN)
CAESAR-IV CARD DETECT IS PROGRAMMABLE, BUT A SILICON BUG
MAKES THE ACTIVE-HIGH CASE UNUSABLE.

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
SD READER CONNECTOR			
 Apple Inc.		DRAWING NUMBER	051-8115
		SIZE	D
		REVISION	11.1.0
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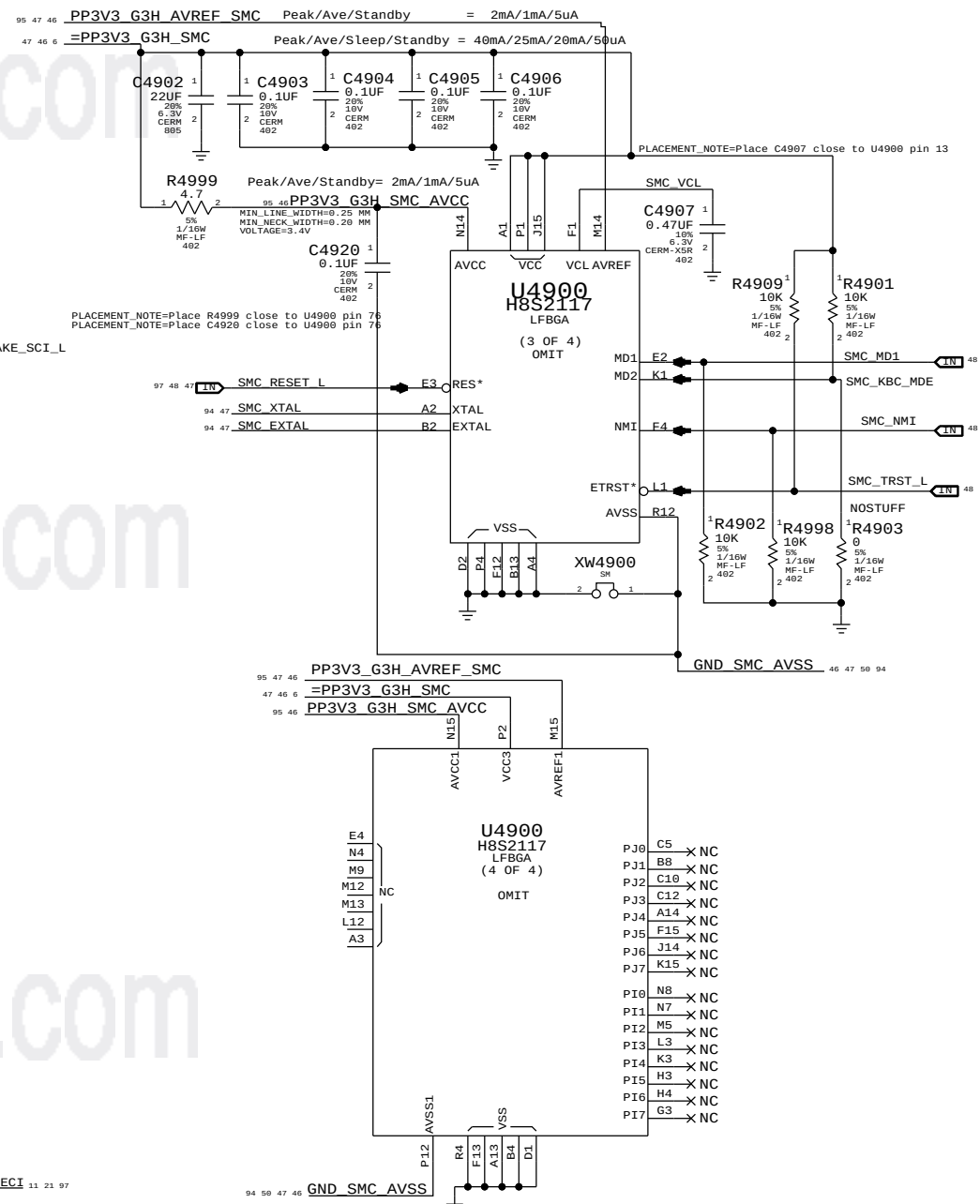
NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

338S0878



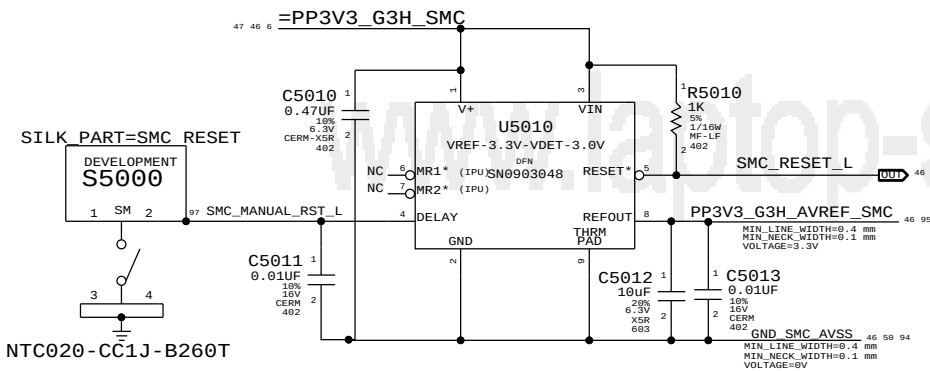
SMC PB3: SMC_IG_THROTTLE_L for MG systems.
Otherwise, TP/NC okay (was ISENSE_CAL_EN)

SMC PG1: SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.





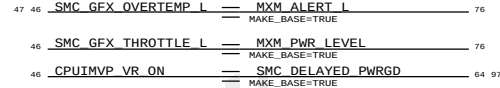
SMC Reset "Button", Supervisor & AVREF Supply



MR1* and MR2* must both be low to cause manual reset.
Used on mobiles to support SMC reset via keyboard.

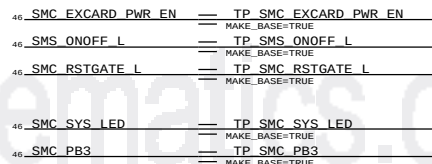
NOTE: Internal pull-ups are to VIN, not V+.

MISC. SIGNAL ALIASES

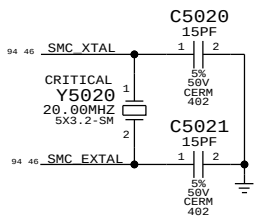


UNUSED PORT 7 ANALOG SENSORS

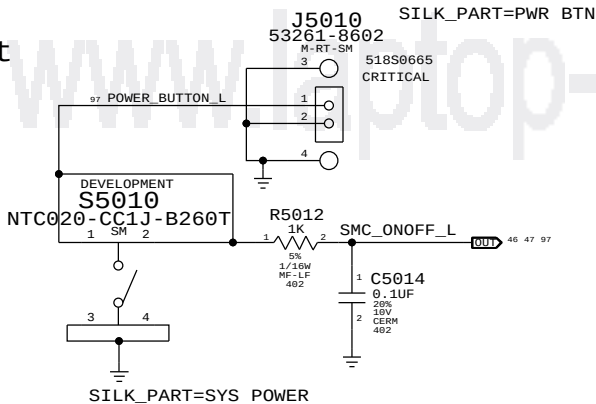
UNUSED TP/NC ALIASES



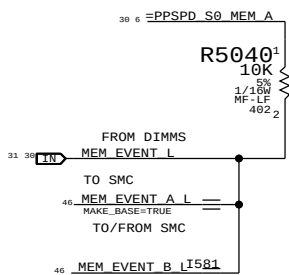
SMC Crystal Circuit



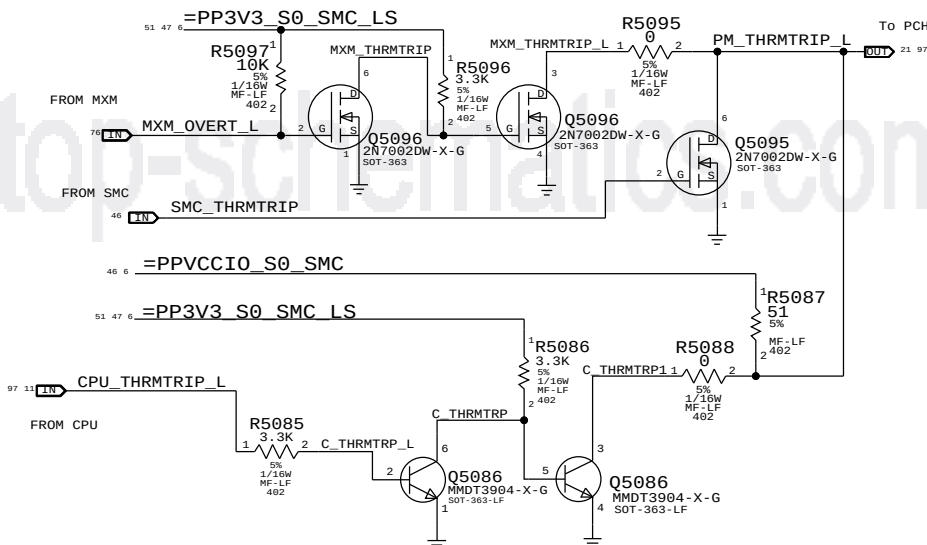
POWER BUTTON



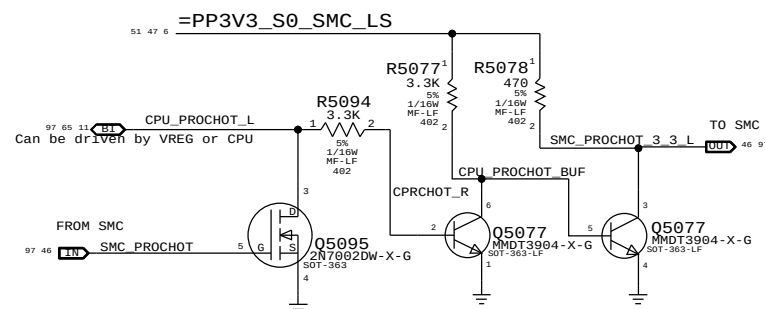
MEM_EVENT



SMC & MXM THERMTRIP LEVEL SHIFTING



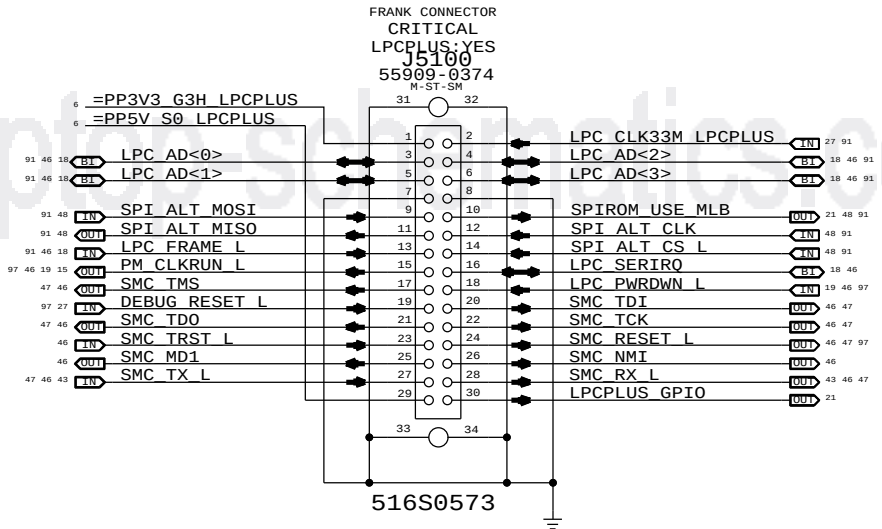
SMC PROCHOT 3.3V LEVEL SHIFTING



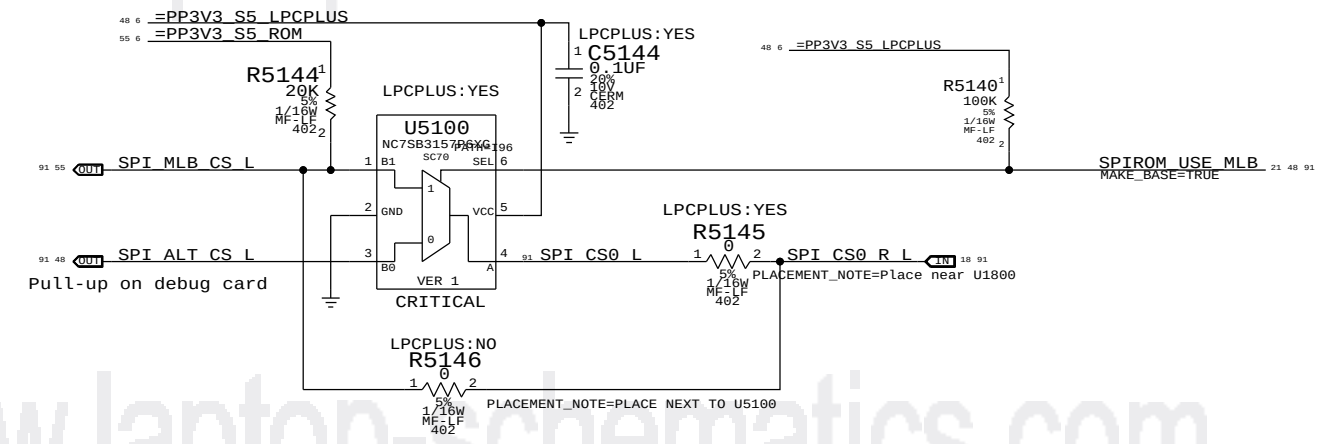
SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		SMC Support	
Apple Inc.		DRAWING NUMBER	051-8115
		REVISION	11.1.0
		BRANCH	
		PAGE	50 OF 110
		SHEET	47 OF 98
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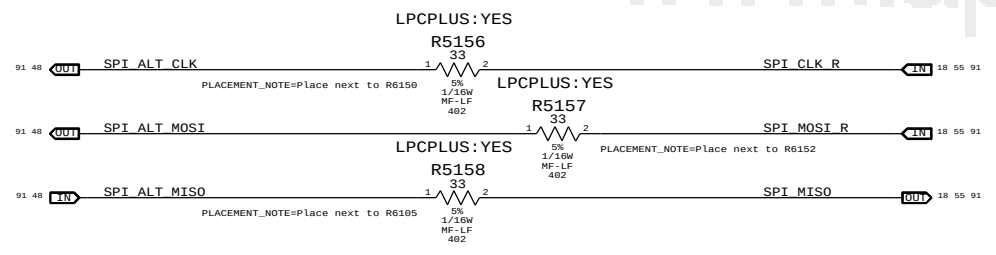
LPC+SPI Connector



Alternate SPI ROM Support



SPI Bus Series Resistance Option



SYNC MASTER=K62 AARON		SYNC DATE=11/30/2009	
PAGE TITLE LPC+SPI Debug Connector			
Apple Inc.		DRAWING NUMBER 051-8115	SIZE D
REVISION 11.1.0		BRANCH	
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PCH "SMBUS" CONNECTIONS

PCH "SML 0" CONNECTIONS

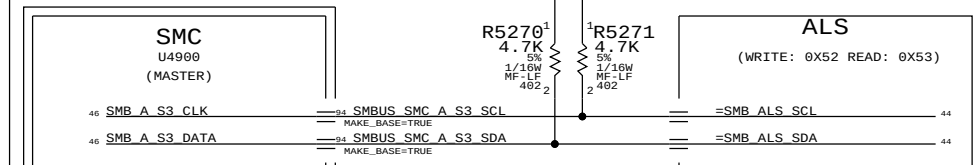
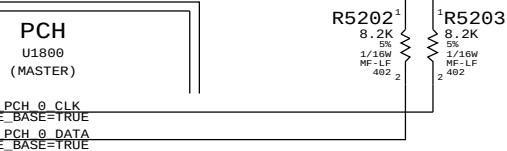
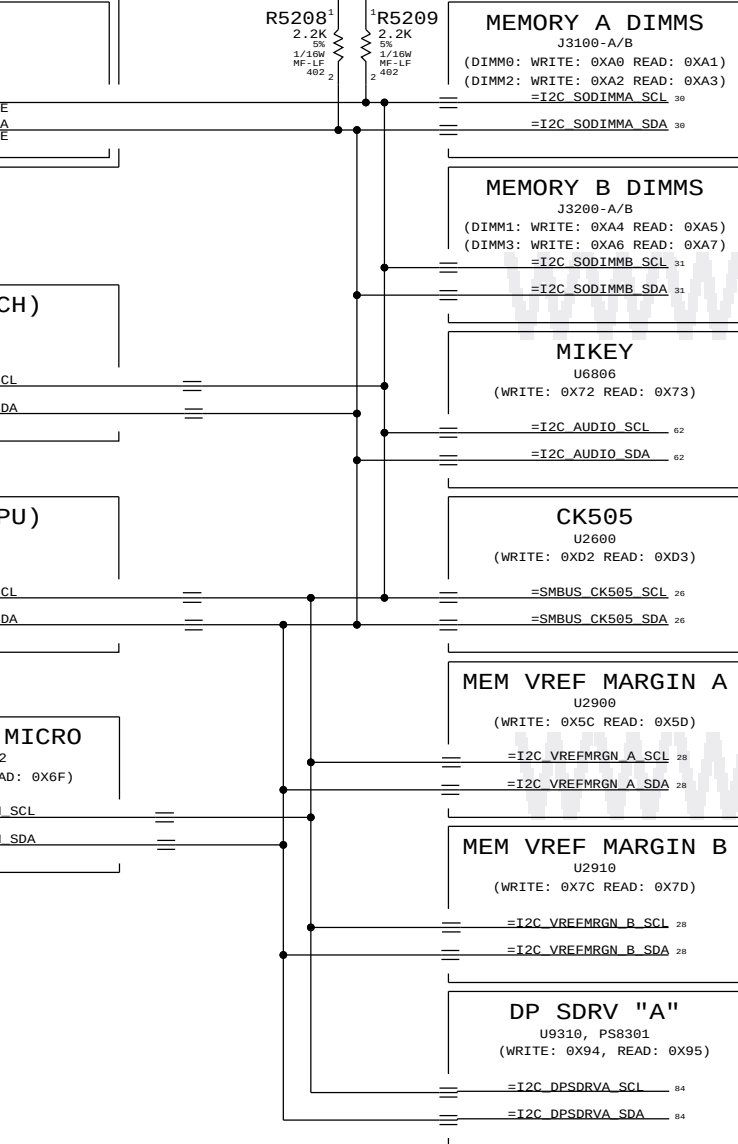
SMC "A" SMBUS CONNECTIONS

NOTE: SMC RMT BUS REMAINS POWERED AND MAY BE ACTIVE IN S3 STATE
BUS A CAN USE EITHER INTERNAL SMC CHANNEL 0 OR 1, K74 CHOOSES 1

49 6 =PP3V3_S0_SMBUS

49 6 =PP3V3_S0_SMBUS

6 =PP3V3_S3_SMBUS_SMC_A

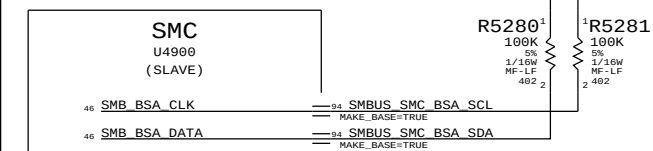


THIS PAGE DIFFERENT BETWEEN K60 AND K62.

SMC SLAVE SMBUS "2" CONNECTIONS

USES INTERNAL SMC CONTROLLER CHANNEL 2 ONLY (NO CONNECTIONS, JUST PULLUP)

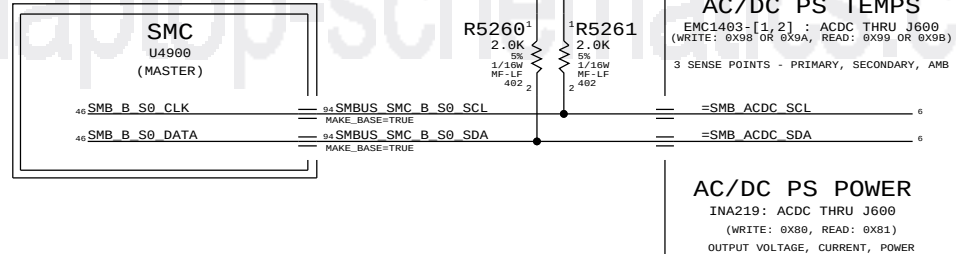
6 =PP3V3_S0_SMBUS_SMC_BSA



SMC "B" SMBUS CONNECTIONS

BUS B CAN USE EITHER INTERNAL SMC CHANNEL 0 OR 1, K60/62 CHOOSES 0

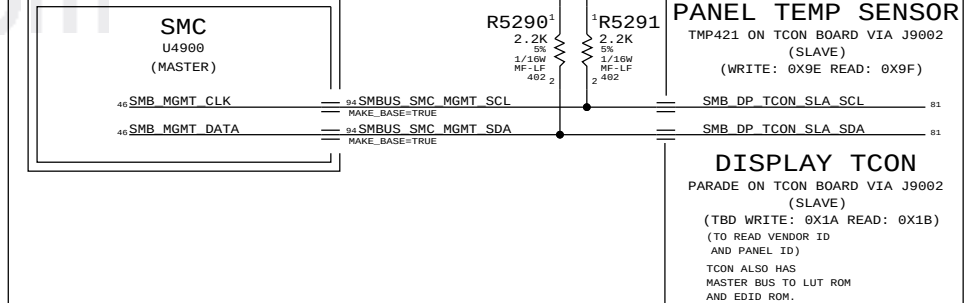
6 =PP3V3_S0_SMBUS_SMC_B



SMC "MANAGEMENT" SMBUS (BUS 1)

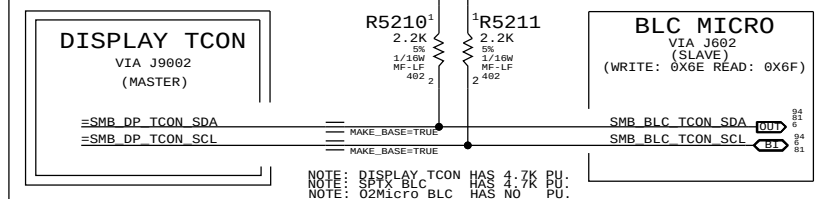
USES INTERNAL SMC CONTROLLER CHANNEL 1 ONLY

6 =PP3V3_S0_SMBUS_SMC_MGMT



DISPLAY TCON TO SPTX OR 02M BLC

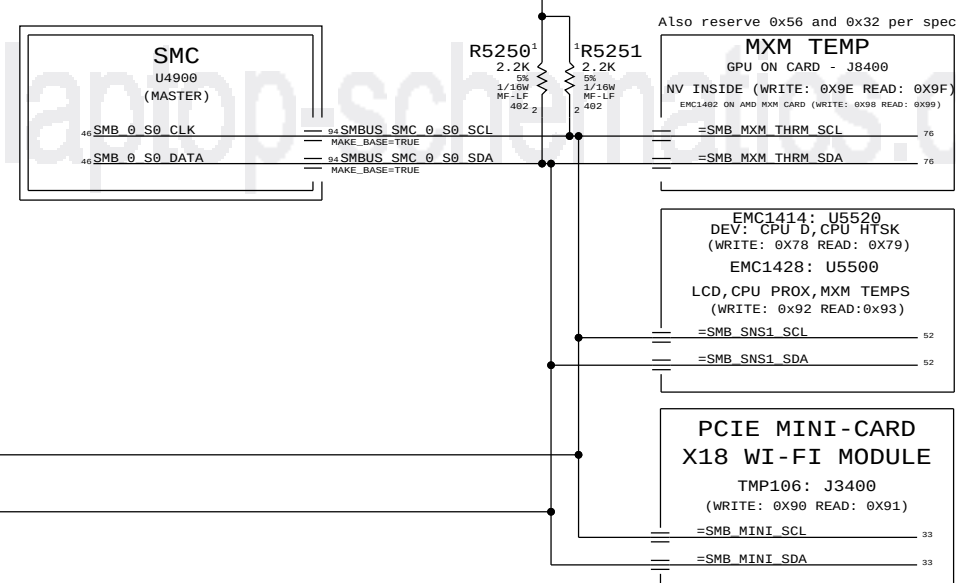
49 6 =PP3V3_S0_SMBUS



SMC "0" SMBUS CONNECTIONS

USES INTERNAL SMC CONTROLLER CHANNEL 0 ONLY

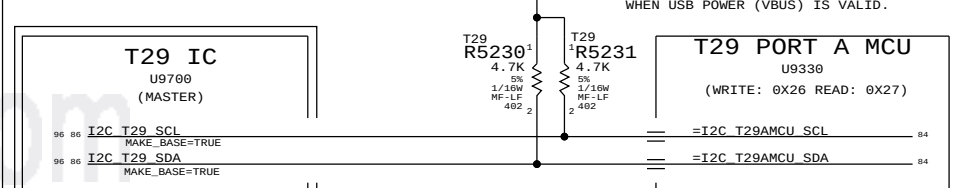
6 =PP3V3_S0_SMBUS_SMC_0



T29 I2C CONNECTIONS

6 =PP3V3_S0_T29I2C

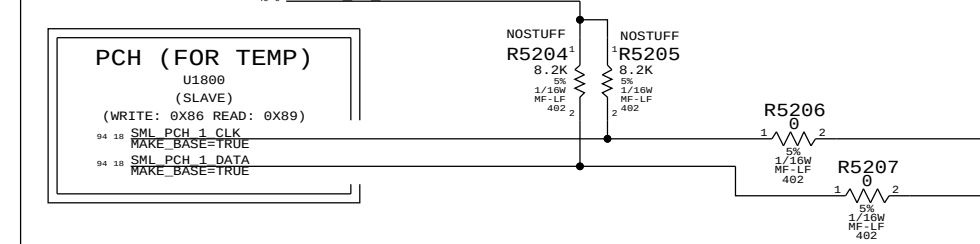
I2C BUS PULL-UP RAIL MUST REFLECT WHEN USB POWER (VBUS) IS VALID.



PCH "SML 1" CONNECTIONS

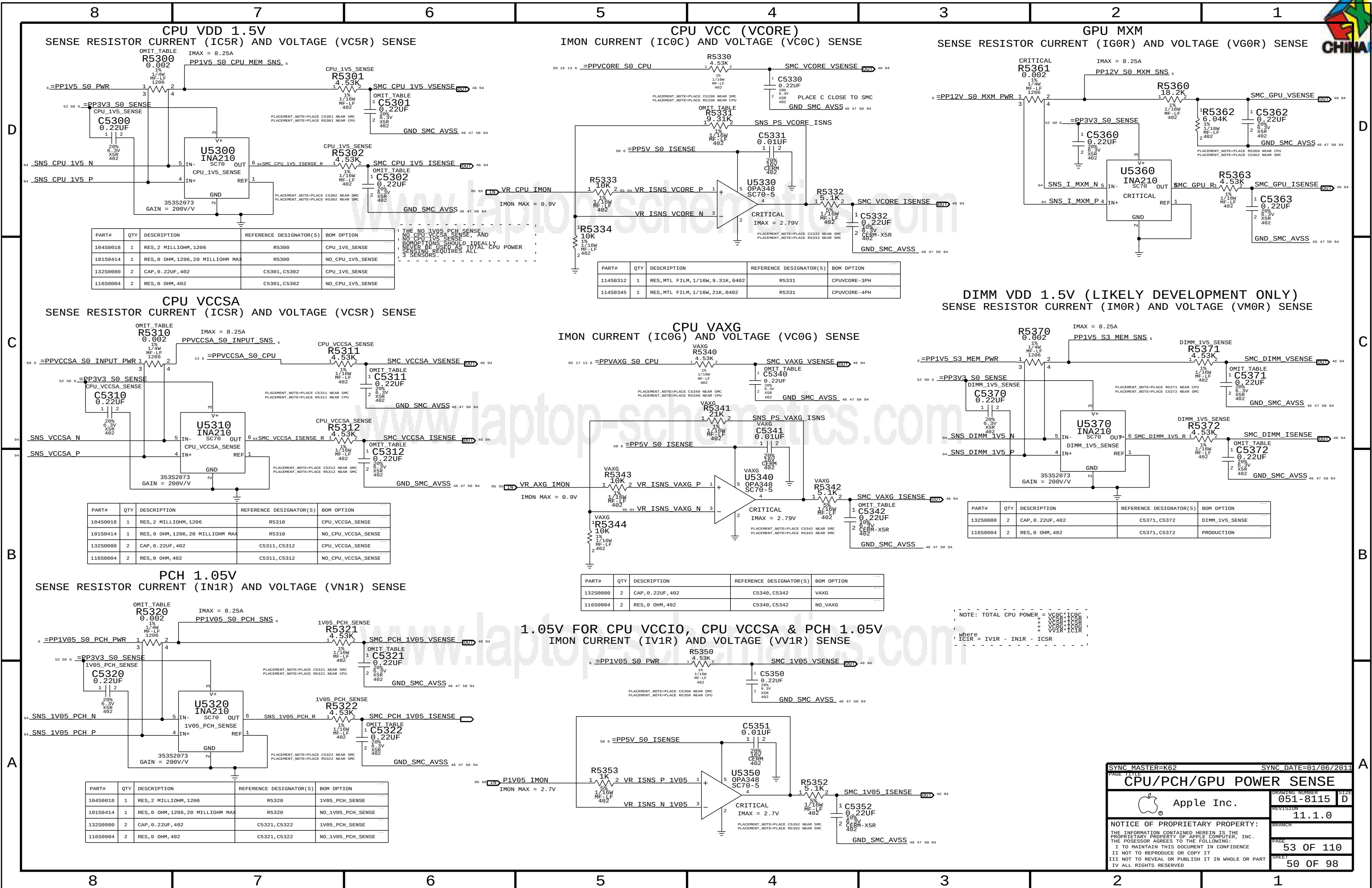
PROD: AMB, L-SKIN, R-SKIN, ODD,

49 6 =PP3V3_S0_SMBUS



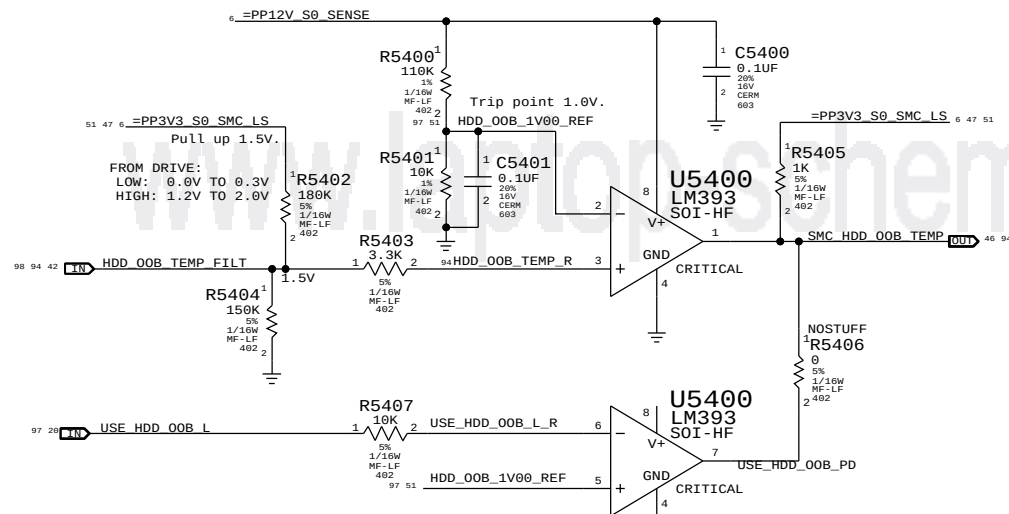
THE PCH address is user programmable by SPI ROM

SYNC MASTER=K60 MARK		SYNC DATE=01/06/2011	
PAGE TITLE		SMBUS CONNECTIONS	
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		REVISION	11.1.0
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HDD OOB TEMPERATURE SENSING



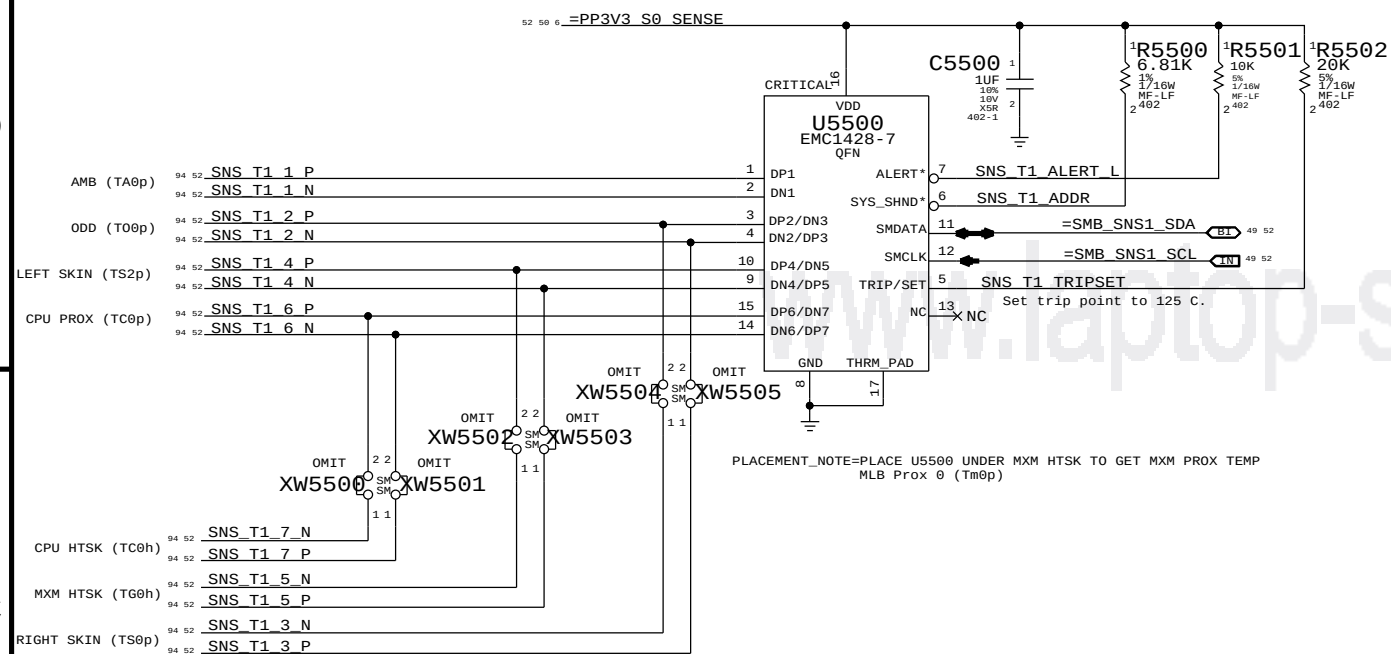
DRIVE ACTIVE = VALID SIGNAL PROTOCOL BETWEEN 0-2.0V.
DRIVE ASLEEP = HDD DRIVES HDD_OOB_TEMP LOW
DRIVE ABSENT = OOB IS PULLED HIGH UNLESS PCH DETERMINES SSD PRESENT AND DRIVES USE_HDD_OOB_L LOW WHICH THEN PULLS HDD_OOB_TEMP LOW.

NOTE: WILL BE CONNECTED TO SATA PWR CONNECTOR PIN 11
THIS PIN IS ORIGINALLY INTENDED FOR HDD LED OUTPUT,
AND ALSO FOR HDD STAGGERED PIN UP (FLOATING) OR IMMEDIATE SPIN-UP (GROUND).
BOTH FUNCTIONS NOT USED.

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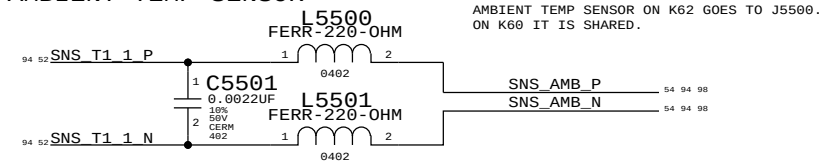
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PAGE TITLE		HDD OOB SENSE	
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		SHEET	51 OF 98

SNS T1: PRODUCTION TEMP SENSOR IC

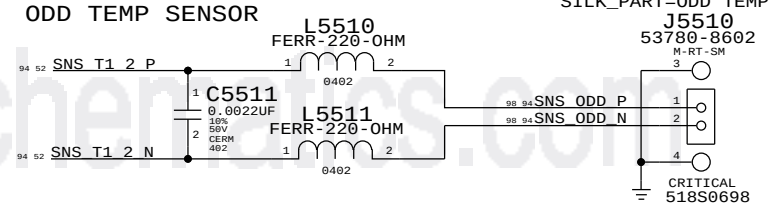


EMC1428-7: 6.8K PULL UP: I2C ADDRESS: WRITE: 0x92, READ: 0x93

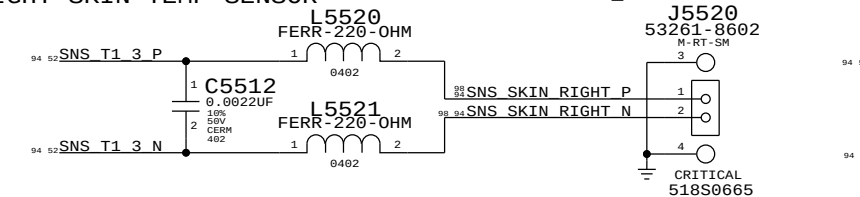
AMBIENT TEMP SENSOR



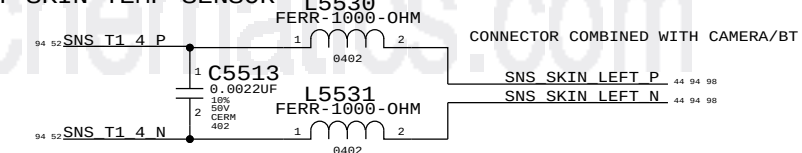
ODD TEMP SENSOR



RIGHT SKIN TEMP SENSOR

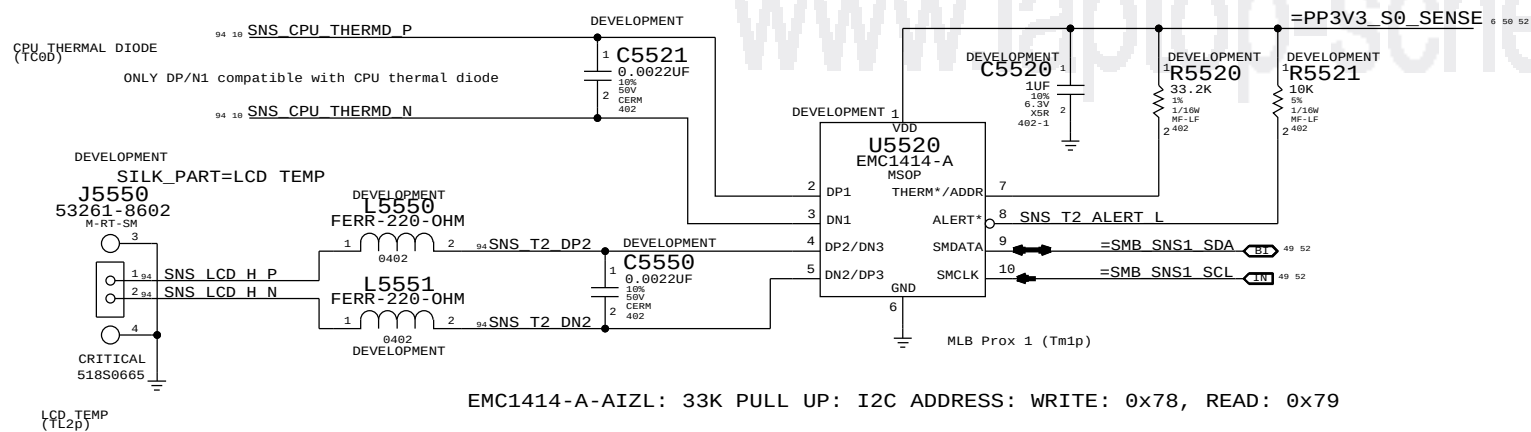


LEFT SKIN TEMP SENSOR



THIS PAGE DIFFERENT BETWEEN K60 AND K62.

SNS T2: DEVELOPMENT TEMP SENSOR IC



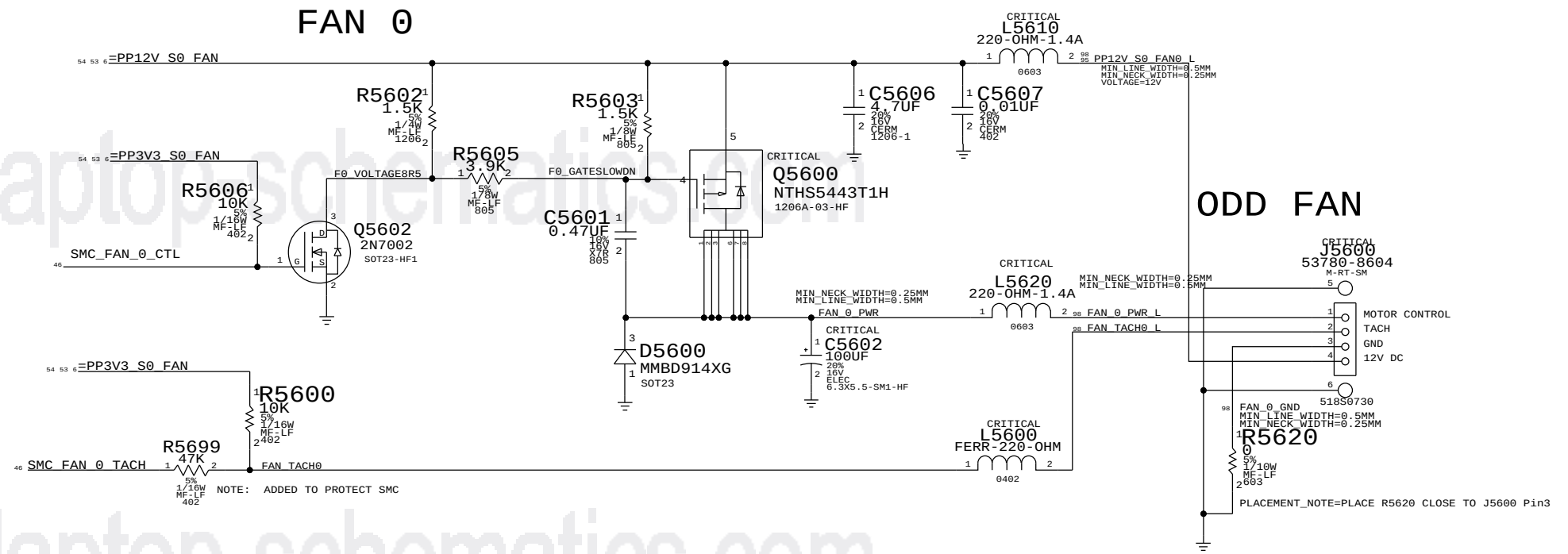
EMC1414-A-AIZL: 33K PULL UP: I2C ADDRESS: WRITE: 0x78, READ: 0x79

SYNC MASTER=K60 MARK		SYNC DATE=01/06/2011	
PAGE TITLE			
TEMP SENSORS			
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		BRANCH	
		PAGE	55 OF 110
		SHEET	52 OF 98

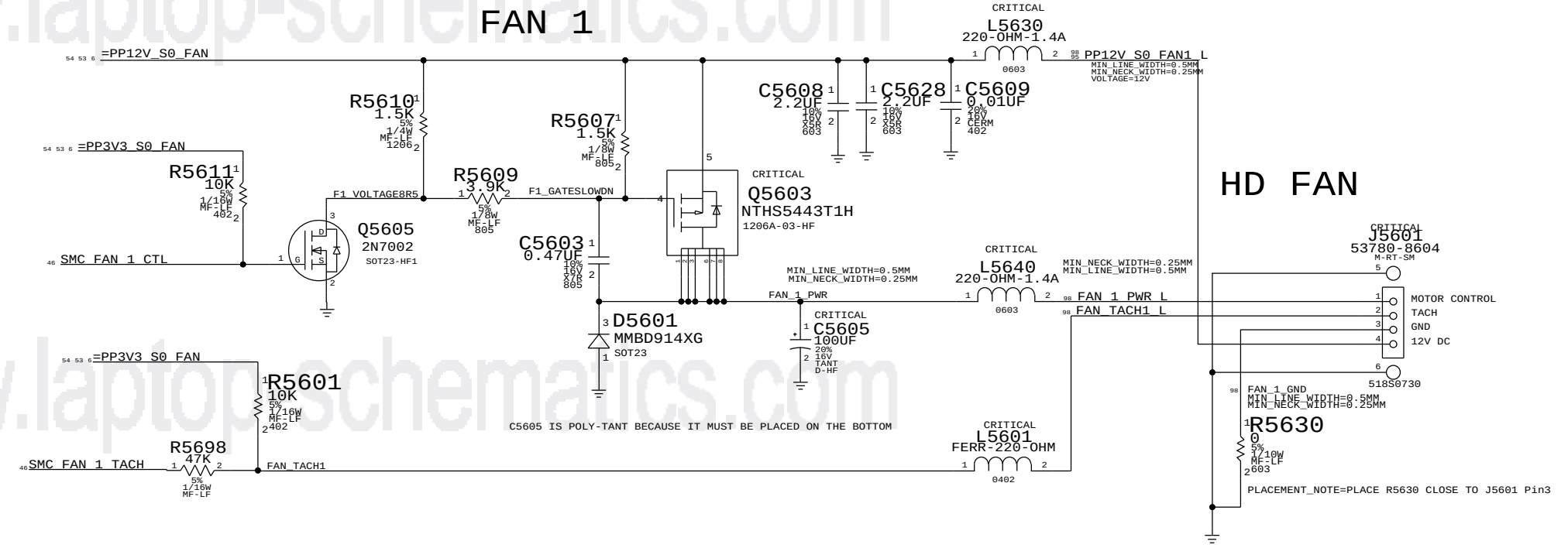


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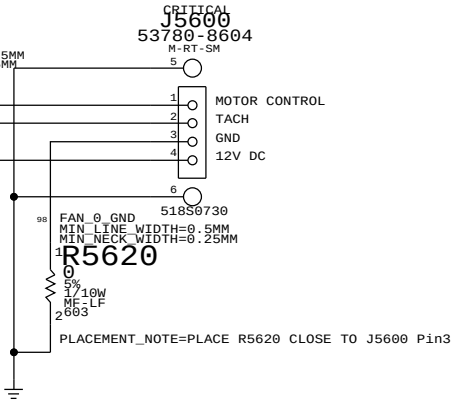
FAN 0



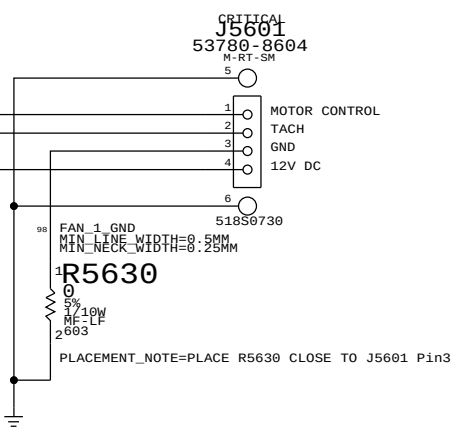
FAN 1



ODD FAN



HD FAN



SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		HD AND OD FAN	
Apple Inc.		DRAWING NUMBER	051-8115
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FAN 3 SMC CONTROL (UNUSED)



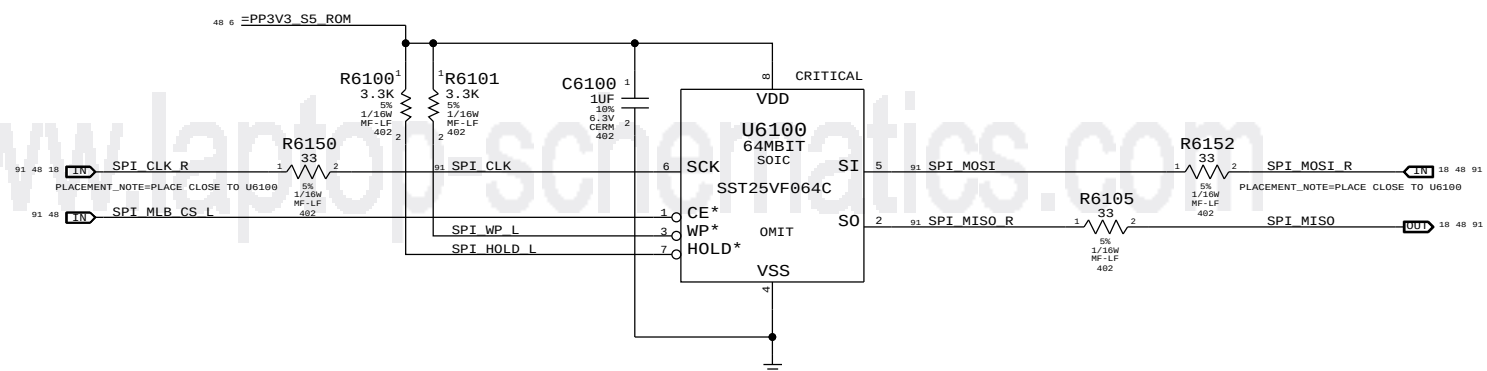
FAN 3 SMC CONTROL (UNUSED)

SYNC MASTER=K60 JERRY		SYNC DATE=01/06/2011	
PAGE TITLE		CPU FAN	
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		SIZE	D
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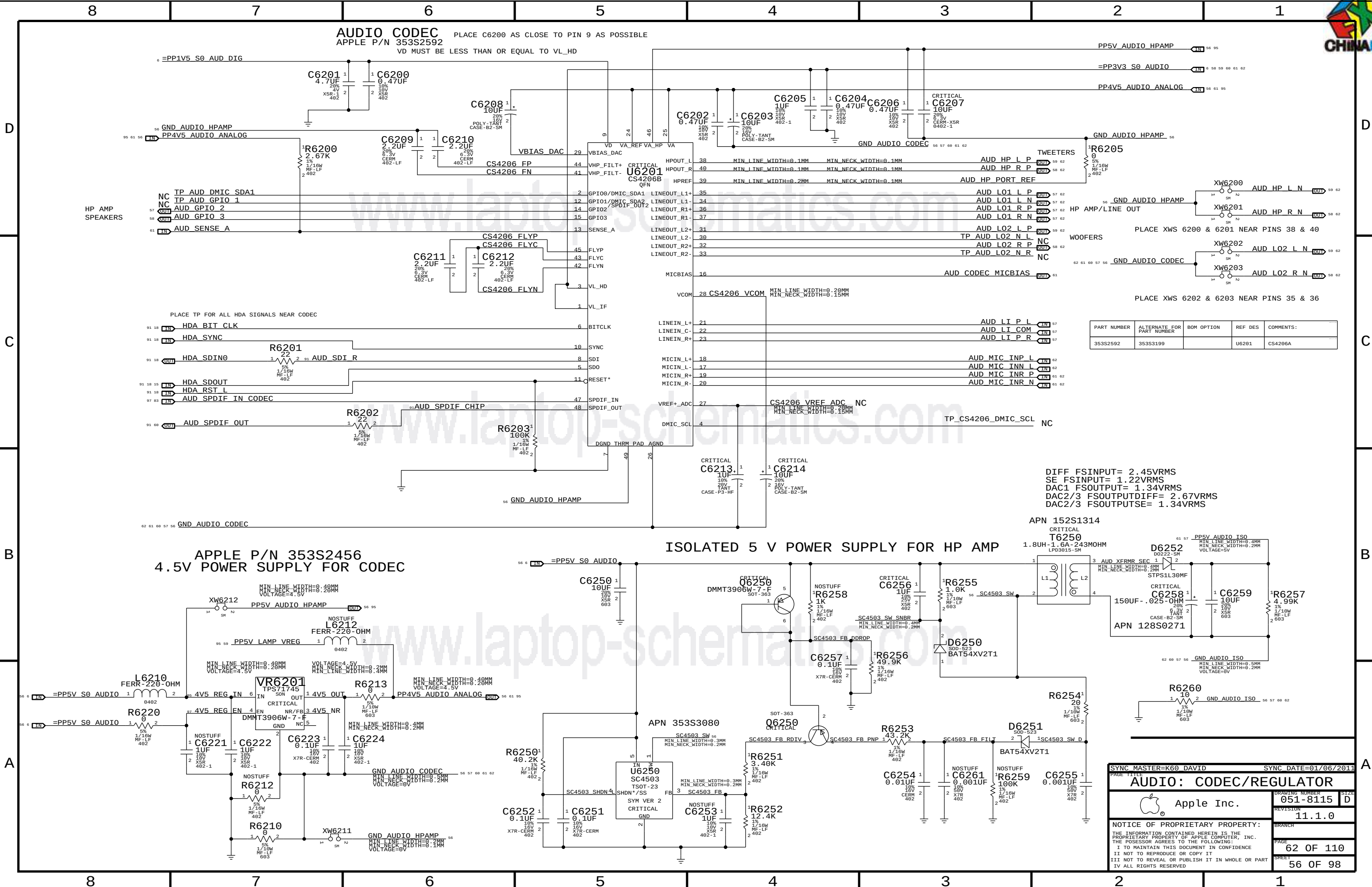
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SYNC MASTER=K62		SYNC DATE=01/06/2011	
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		DRAWING NUMBER	051-8115
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		PAGE	61 OF 110
		SHEET	55 OF 98



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S2592	353S3199		U6201	CS4206A

DIFF FSINPUT= 2.45VRMS
FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

APN 152S1314

CRITICAL
T6250

1.8UH-1.6A-243MOHM
LPD3015-SM

CRITICAL
D6252

DO222-SM

CRITICAL
C6258

150UF-025-0HM

APN 128S0271

CRITICAL
D6250

S00-523

BAT54XV2T1

CRITICAL
D6251

S00-523

BAT54XV2T1

CRITICAL
C6255

0.001UF

CRITICAL
C6254

0.01UF

CRITICAL
C6253

1UF

CRITICAL
C6252

0.1UF

CRITICAL
C6251

0.1UF

CRITICAL
C6250

10UF

CRITICAL
C6249

10UF

CRITICAL
C6248

10UF

CRITICAL
C6247

10UF

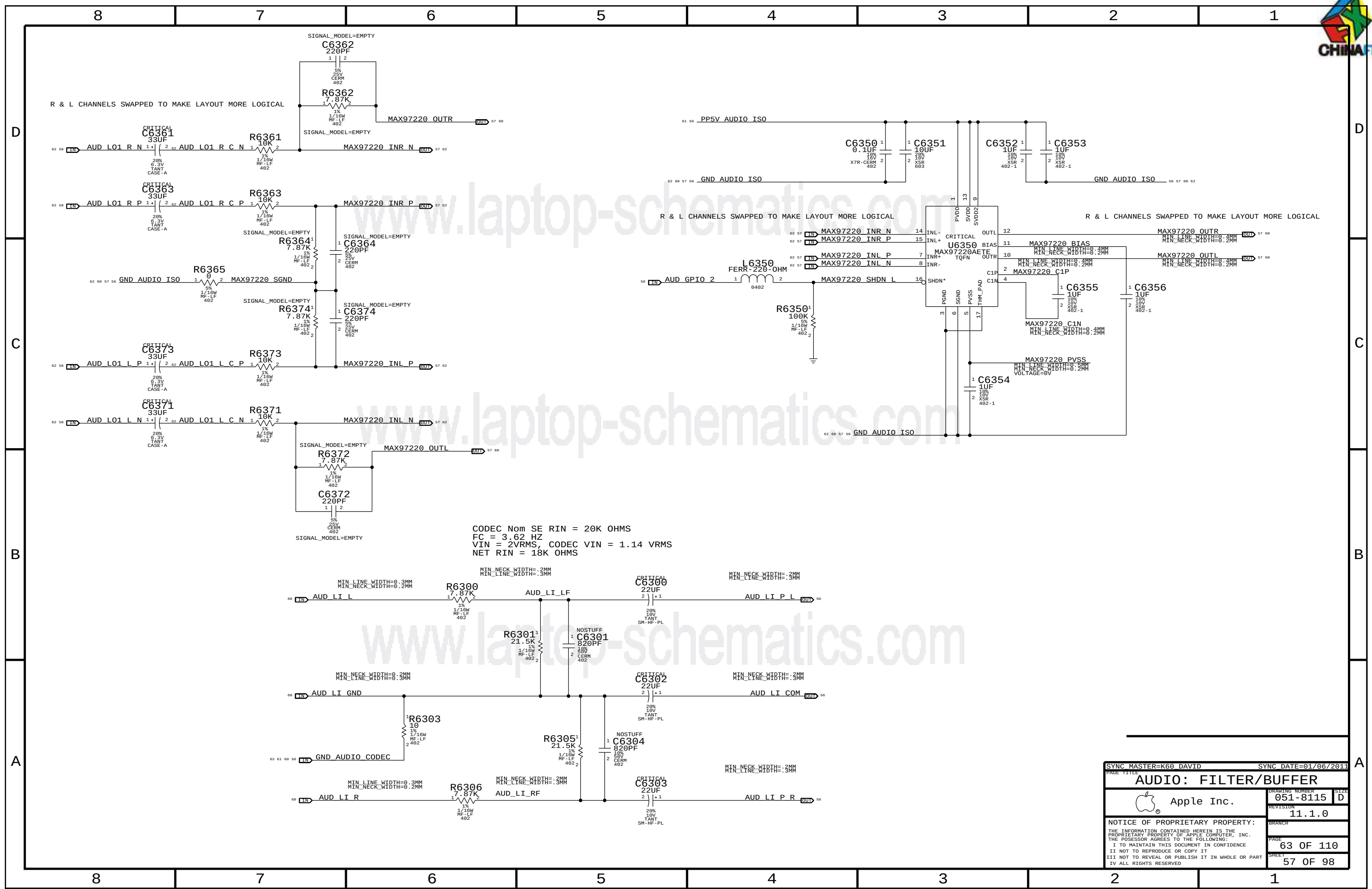
CRITICAL
C6246

10UF

CRITICAL
C6245

10UF

SYNC MASTER=K60 DAVID		SYNC DATE=01/06/2011	
PAGE TITLE		DRAWING NUMBER	
AUDIO: CODEC/REGULATOR		051-8115	
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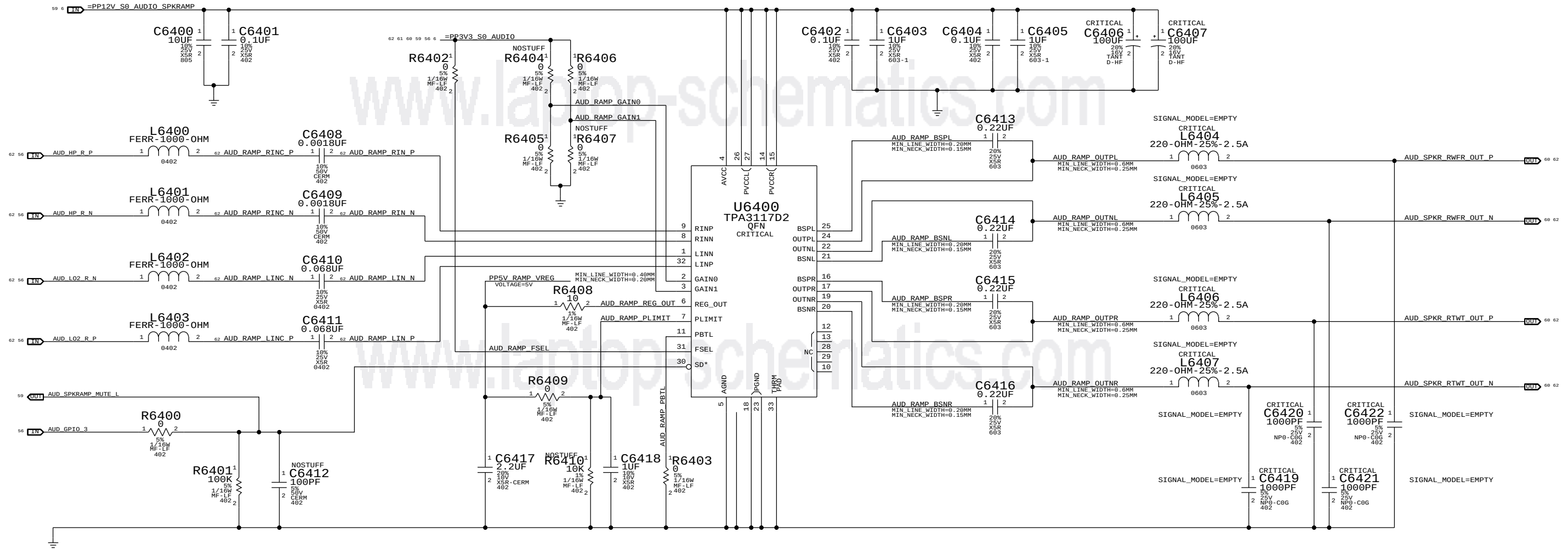


SYNC MASTER=K60 DAVID		SYNC DATE=01/06/2011	
PAGE TITLE		AUDIO: FILTER/BUFFER	
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RIGHT CH SPEAKER AMP
APPLE P/N 353S3069

R6402 = HIGH = 400 KHZ
SPEAKER AMP GAIN = +15.2 DB
SPEAKER AMP RIN = 104K NOMINAL W/ +15.2 DB GAIN
FC_HPF, TWEETERS = ~850 HZ (0.0018 UF)
FC_HPF, WOOFERS = ~22.5 HZ (0.068 UF)

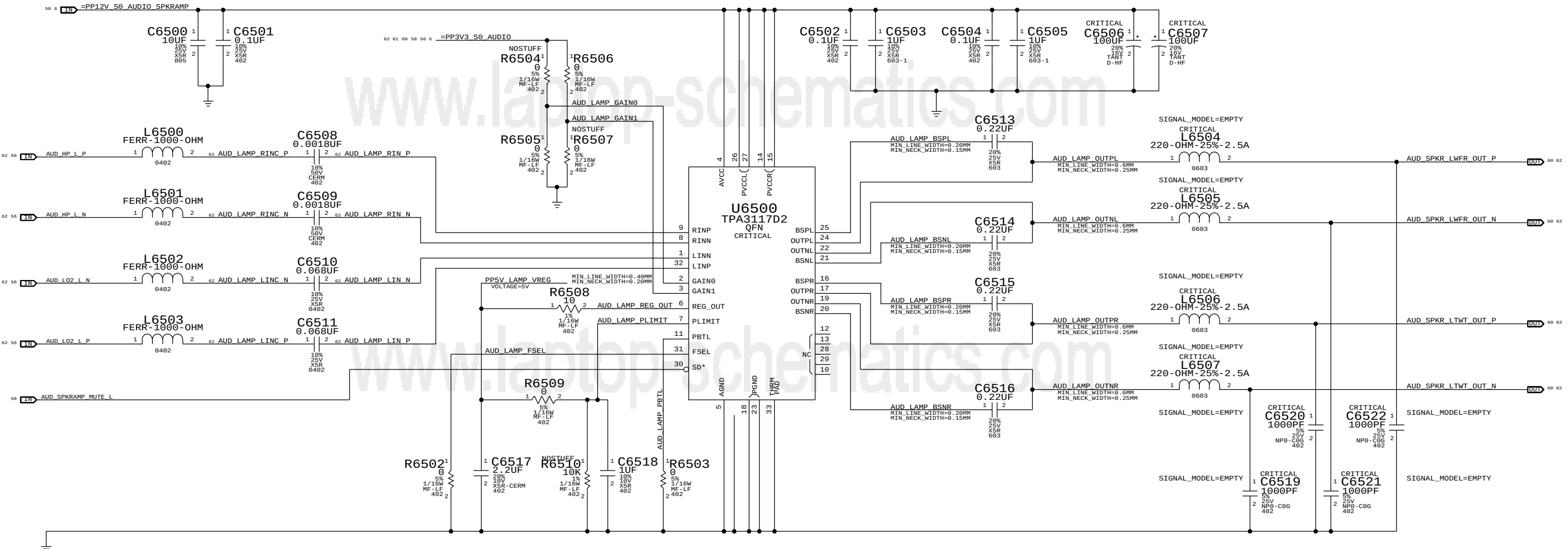


SYNC MASTER=K60 DAVID		SYNC DATE=01/06/2011	
PAGE TITLE			
AUDIO: SPEAKER AMP_1			
 Apple Inc.		DRAWING NUMBER	051-8115
		REVISION	11.1.0
		BRANCH	
		PAGE	64 OF 110
		SHEET	58 OF 98
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LEFT CH SPEAKER AMP
APPLE P/N 353S3069

R6502 = LOW = 300 KHZ
SPEAKER AMP GAIN = +15.2 DB
SPEAKER AMP RIN = 104K NOMINAL W/ +15.2 DB GAIN
FC HPF, TWEETERS = ~850 HZ (0.0018 UF)
FC HPF, WOOFERS = ~22.5 HZ (0.068 UF)



SYNC MASTER=K60 DAVID		SYNC DATE=01/06/2011	
PAGE TITLE			
AUDIO: SPEAKER AMP			
 Apple Inc.	DRAWING NUMBER	051-8115	SIZE
	REVISION	11.1.0	
	BRANCH		
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PAGE		65 OF 110	
SHEET		59 OF 98	



INTERNAL MIC CON
APPLE P/N 518S0677

SPEAKER CABLE CONNECTORS

APPLE P/N 518S0748
APPLE P/N 518S0656

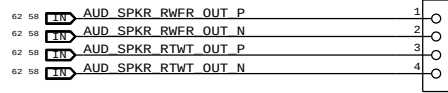
PROPERTIES FOR ALL SPKR NETS

CRITICAL

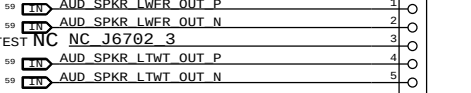
PROPERTIES FOR ALL SPKR NETS

CRITICAL

WOOFER (BR)
TWEETER (FR)



WOOFER (BL)
TWEETER (FL)



REMOTE I/O CONNECTOR
APPLE P/N 518S0723

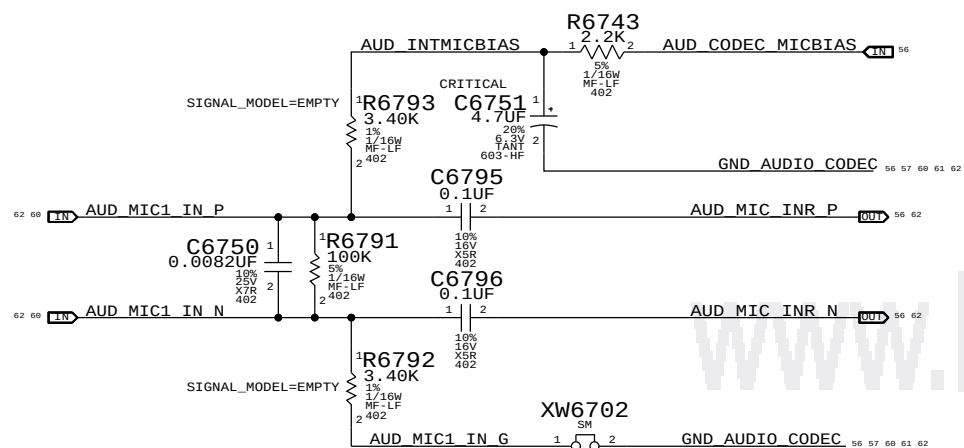
CRITICAL
J6600
20143-020E-20F
F-RT-SM

AUD SPDIFIN JACK
PP3V3 AUDIO SPDIF JACK
AUD LI_DET JACK
AUD LI R JACK
AUD LI GND JACK
AUD LI L JACK
AUD GND_DET JACK
HS MIC HI JACK
AUD HP GND JACK
AUD HP L JACK
AUD HP R JACK
AUD HP TYPEDET JACK
AUD IP PERPH JACK
AUD HP TIPDET JACK
PP3V3 AUDIO SPDIF JACK

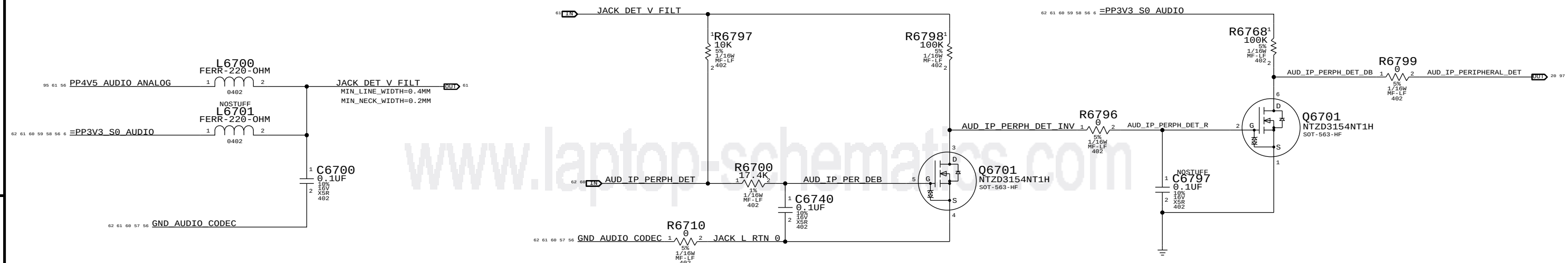
SYNC MASTER=K60 DAVID		SYNC DATE=01/06/2011	
PAGE TITLE		Audio: MLB to I/O Conn.	
Apple Inc.		DRAWING NUMBER	051-8115
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Internal Microphone Impedance Matching



IPHS HS Detect Debounce CKT

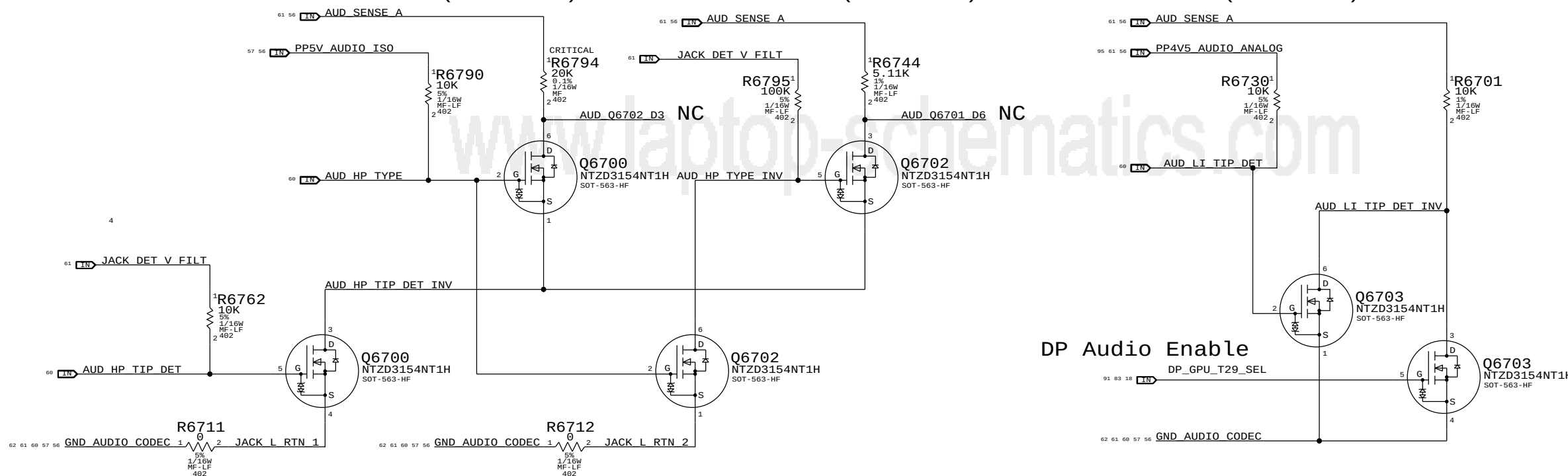


Digital Out (DETECT B)

Headphone Out (DETECT D)

LI Insert Detect (DETECT C)

DP Audio Enable

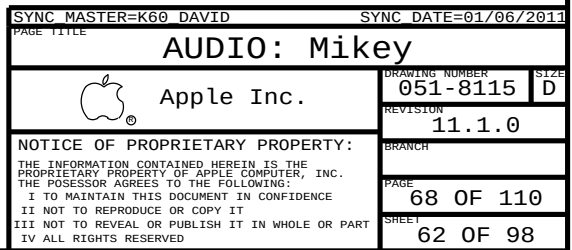


SYNC MASTER=K60 DAVID		SYNC DATE=11/24/2016	
PAGE TITLE		AUDIO: Detects/Grounding	
Apple Inc.		DRAWING NUMBER	051-8115
		REVISION	11.1.0
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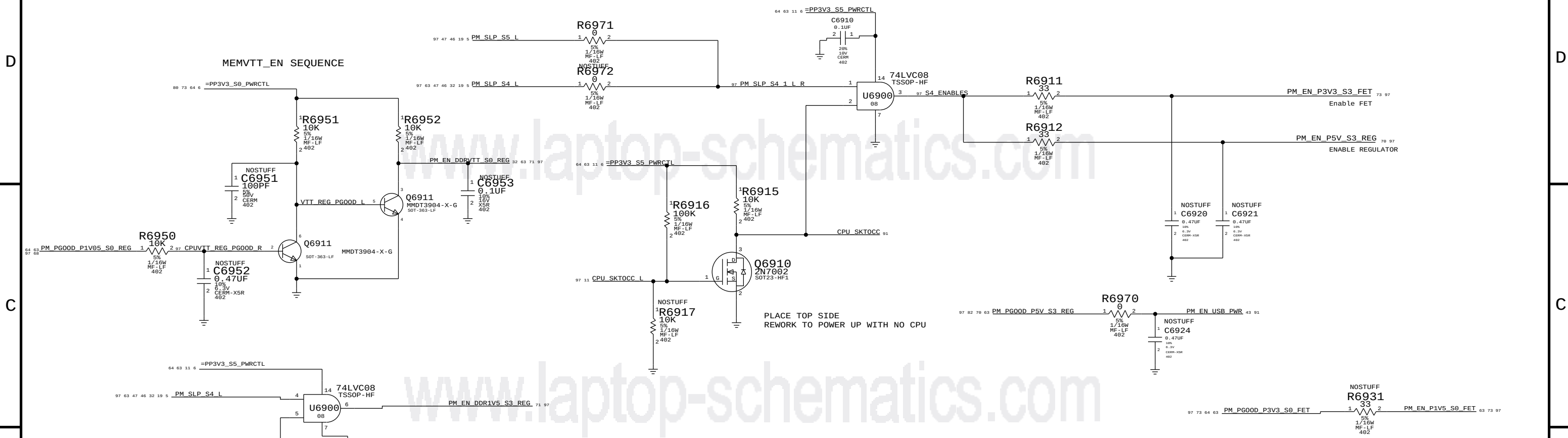
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUG1001FF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKR0U01FF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
1E001	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_HP_L_P	56 59
1E002	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_HP_L_N	56 59
1E003	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_HP_R_P	56 58
1E004	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_HP_R_N	56 58
1E005	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_L01_L_P	56 57
1E006	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_L01_L_N	56 57
1E007	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_L01_R_P	56 57
1E008	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_L01_R_N	56 57
1E009	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_L02_L_P	56 59
1E010	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_L02_L_N	56 59
1E011	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_L02_R_P	56 58
1E012	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_L02_R_N	56 58
1E013	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_RAMP_LINC_P	58
1E014	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_RAMP_LINC_N	58
1E015	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_RAMP_RINC_P	58
1E016	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_RAMP_RINC_N	58
1E017	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_RAMP_LIN_P	58
1E018	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_RAMP_LIN_N	58
1E019	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_RAMP_RIN_P	58
1E020	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_RAMP_RIN_N	58
1E021	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_LAMP_LINC_P	59
1E022	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_LAMP_LINC_N	59
1E023	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_LAMP_RINC_P	59
1E024	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_LAMP_RINC_N	59
1E025	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_LAMP_LIN_P	59
1E026	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_LAMP_LIN_N	59
1E027	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_LAMP_RIN_P	59
1E028	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_LAMP_RIN_N	59
1E029	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_L01_L_C_P	57
1E030	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_L01_L_C_N	57
1E031	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_L01_R_C_P	57
1E032	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_L01_R_C_N	57
1E033	AUDIO_DIFFPAIR	AUDIOTIEF	MAX97220_INL_P	57
1E034	AUDIO_DIFFPAIR	AUDIOTIEF	MAX97220_INL_N	57
1E035	AUDIO_DIFFPAIR	AUDIOTIEF	MAX97220_INR_P	57
1E036	AUDIO_DIFFPAIR	AUDIOTIEF	MAX97220_INR_N	57
1E037	SEKROUT_DIFFPAIR	SEKROUTTIEF	AUD_SPKR_RWFR_OUT_P	58 60
1E038	SEKROUT_DIFFPAIR	SEKROUTTIEF	AUD_SPKR_RWFR_OUT_N	58 60
1E039	SEKROUT_DIFFPAIR	SEKROUTTIEF	AUD_SPKR_RTWT_OUT_P	58 60
1E040	SEKROUT_DIFFPAIR	SEKROUTTIEF	AUD_SPKR_RTWT_OUT_N	58 60
1E041	SEKROUT_DIFFPAIR	SEKROUTTIEF	AUD_SPKR_LWFR_OUT_P	59 60
1E042	SEKROUT_DIFFPAIR	SEKROUTTIEF	AUD_SPKR_LWFR_OUT_N	59 60
1E043	SEKROUT_DIFFPAIR	SEKROUTTIEF	AUD_SPKR_LTWI_OUT_P	59 60
1E044	SEKROUT_DIFFPAIR	SEKROUTTIEF	AUD_SPKR_LTWI_OUT_N	59 60
1E045	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_MTC1_IN_P	60 61
1E046	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_MTC1_IN_N	60 61
1E047	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_MTC_INR_P	60 61
1E048	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_MTC_INR_N	60 61
1E049	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_MTC_IN1_CONN_P	60
1E050	AUDIO_DIFFPAIR	AUDIOTIEF	AUD_MTC_IN1_CONN_N	60

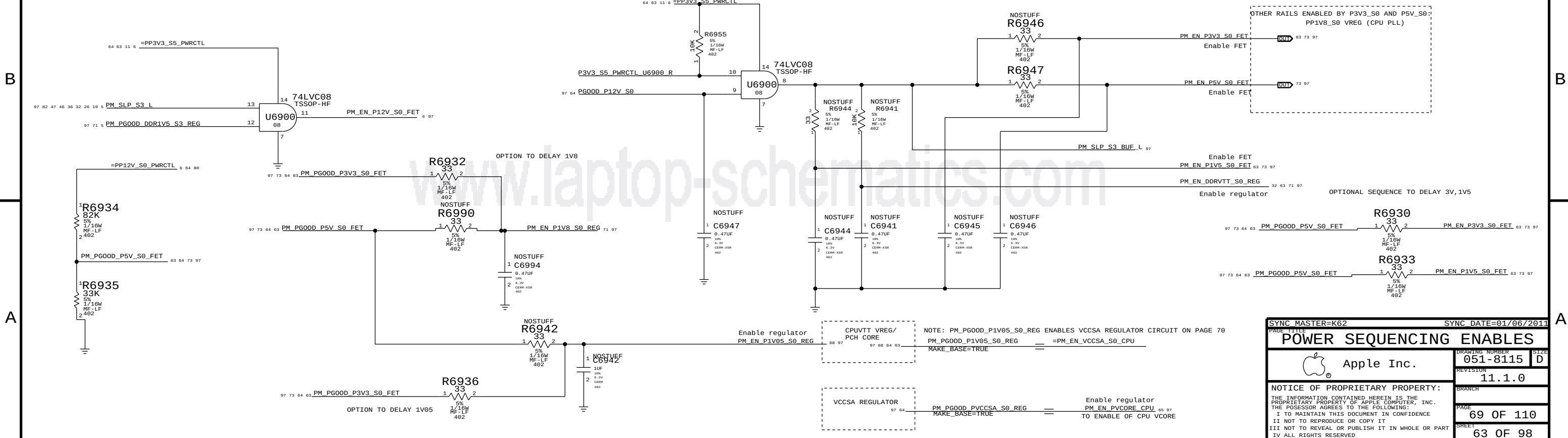
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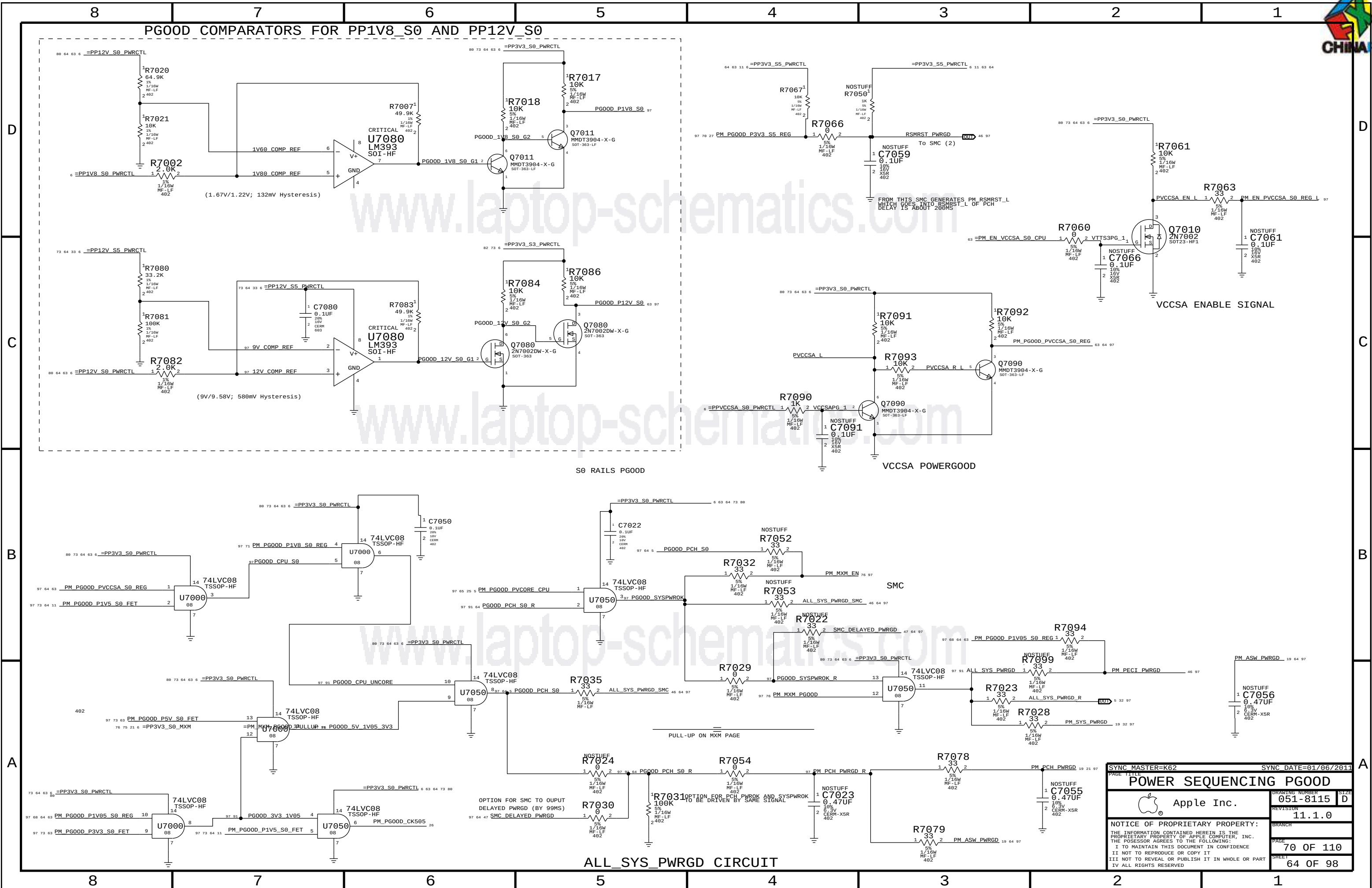
SLP_S4 ENABLES



SLP_S3 ENABLES



SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		POWER SEQUENCING ENABLES	
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		REVISION	11.1.0
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D

C

B

A

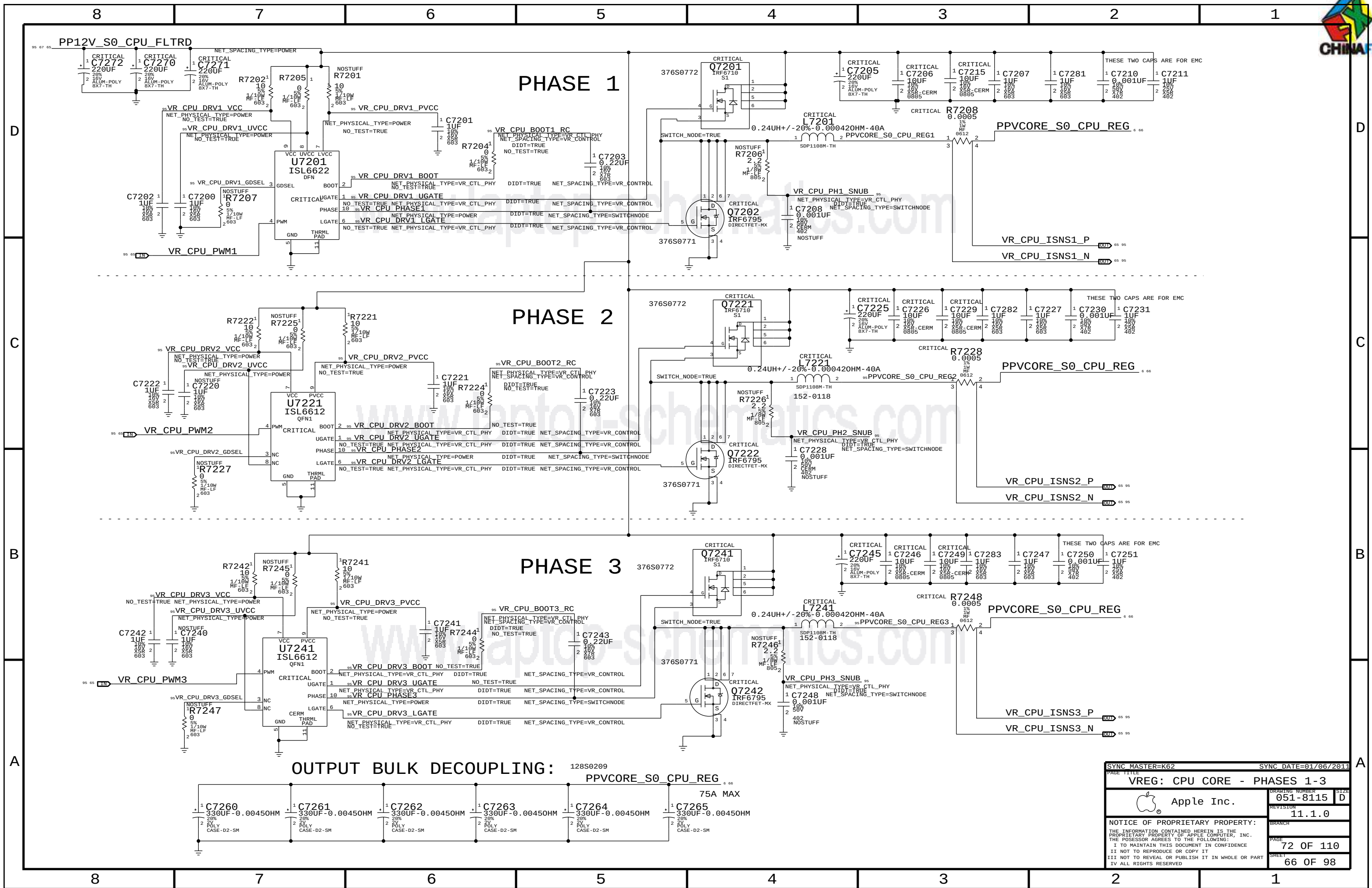
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C

3 |

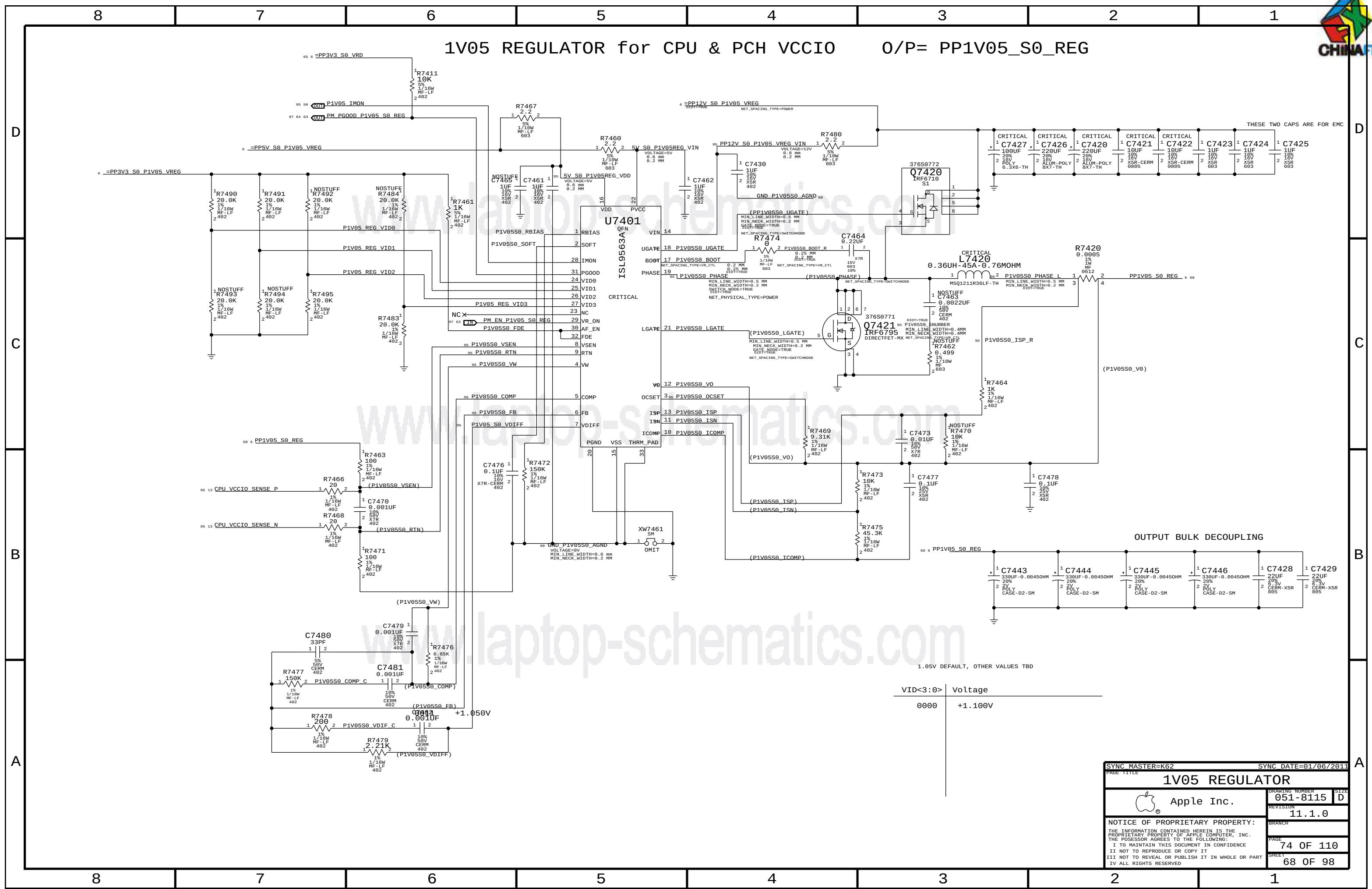
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SYNC MASTER=K60 AARON		SYNC DATE=N/A	
PAGE TITLE			
VREG: PPVCORE_S0_CPU			
 Apple Inc.		DRAWING NUMBER 051-8115	
		SIZE D	
		REVISION 11.1.0	
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BRANCH		PAGE	
		71 OF 110	
SHEET			
		65 OF 98	



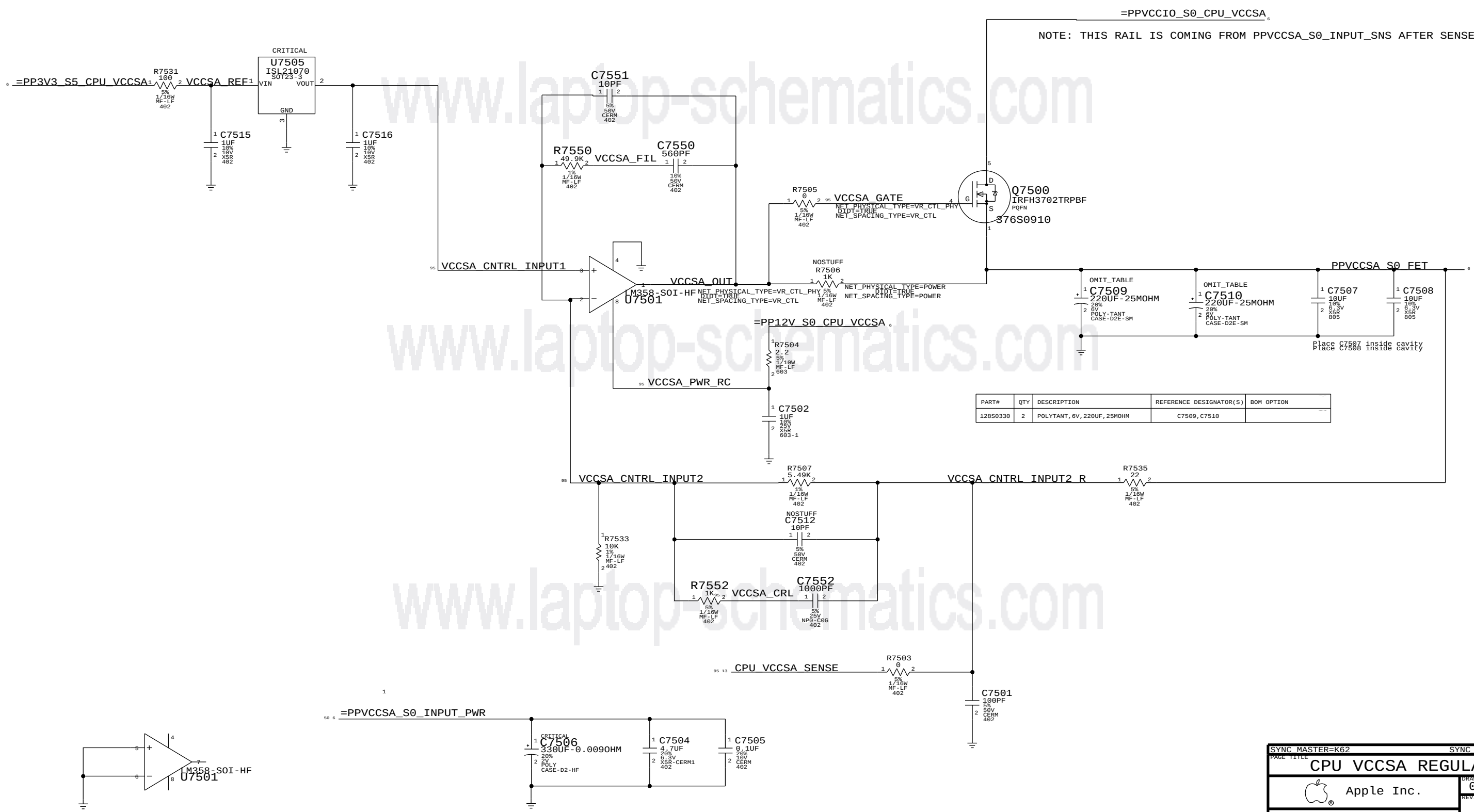
SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		VREG: CPU CORE - PHASES 1-3	
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		PAGE	72 OF 110
		SHEET	66 OF 98

SYNC MASTER-K60 AARON		SYNC DATE=N/A	
PAGE TITLE			
VREG:AXG PHASE/CORE - CAPS			
	Apple Inc.		DRAWING NUMBER
			051-8115
			SIZE
			D
		REVISION	
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		SHEET	67 OF 98





CPU VCCSA 0.925V (8.8A MAX)



NOTE: THIS RAIL IS COMING FROM PPVCCSA_S0_INPUT_SNS AFTER SENSE RES

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
128S0330	2	POLYTANT, 6V, 220UF, 25MOHM	C7509, C7510	

Place C7507 inside cavity
Place C7508 inside cavity

NOTE: THIS POWER RAIL IS BEFORE THE SENSE RES R5310

SYNC MASTER=K62

SYNC DATE=11/15/2016

CPU VCCSA REGULATOR

Apple Inc.

051-8115

11.1.0

75 OF 110

69 OF 98

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3V3 S5 REGULATOR

5V S3 REGULATOR

Power Rating ?

5V OUTPUT

3V3 OUTPUT

OUTPUT BULK DECOUPLING:

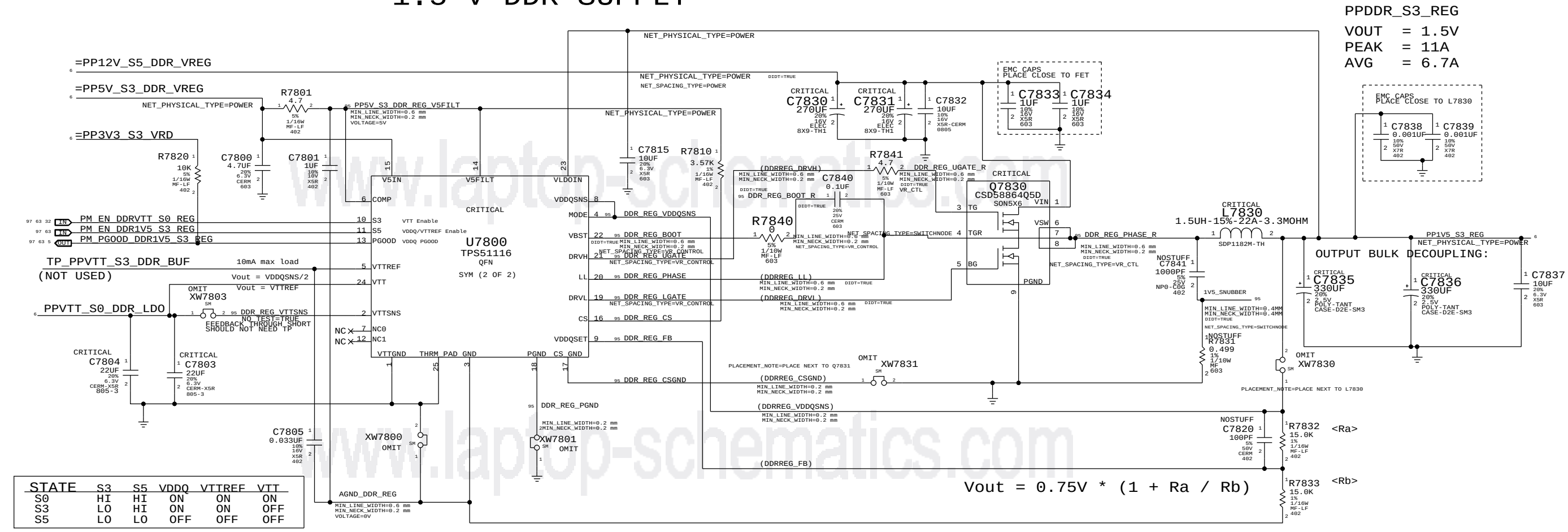
OUTPUT BULK DECOUPLING:

$$V_{out} = 0.6V * (1 + R_a / R_b)$$

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
5V_S3 / 3V3_S5 VREGS			
 Apple Inc.	DRAWING NUMBER	051-8115	SIZE D
	REVISION	11.1.0	
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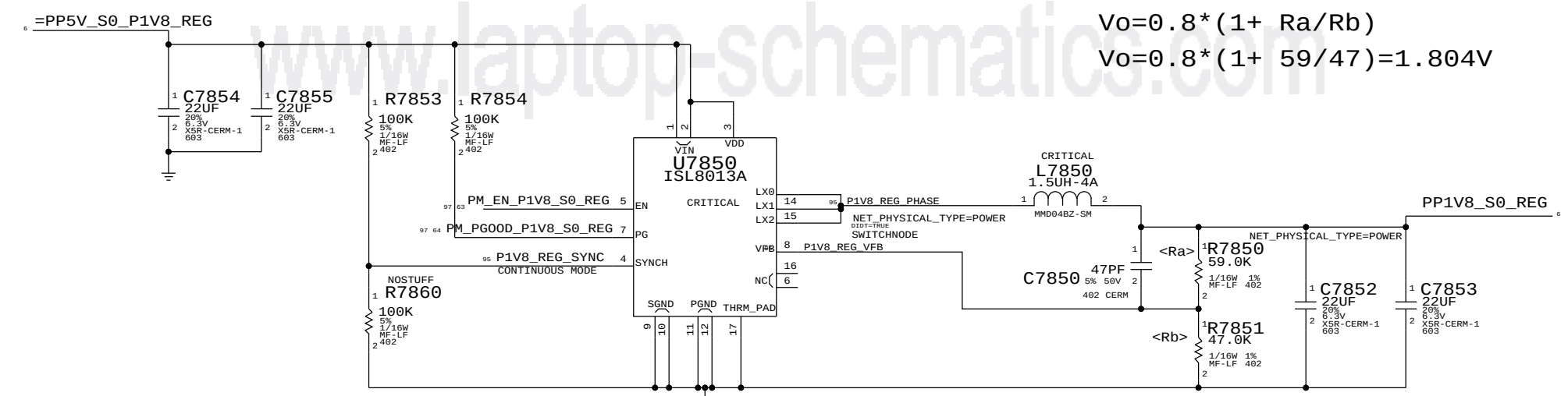
1.5 V DDR SUPPLY



1.8 V SUPPLY

1A Average current

$$V_o = 0.8 * (1 + R_a / R_b)$$
$$V_o = 0.8 * (1 + 59 / 47) = 1.804V$$



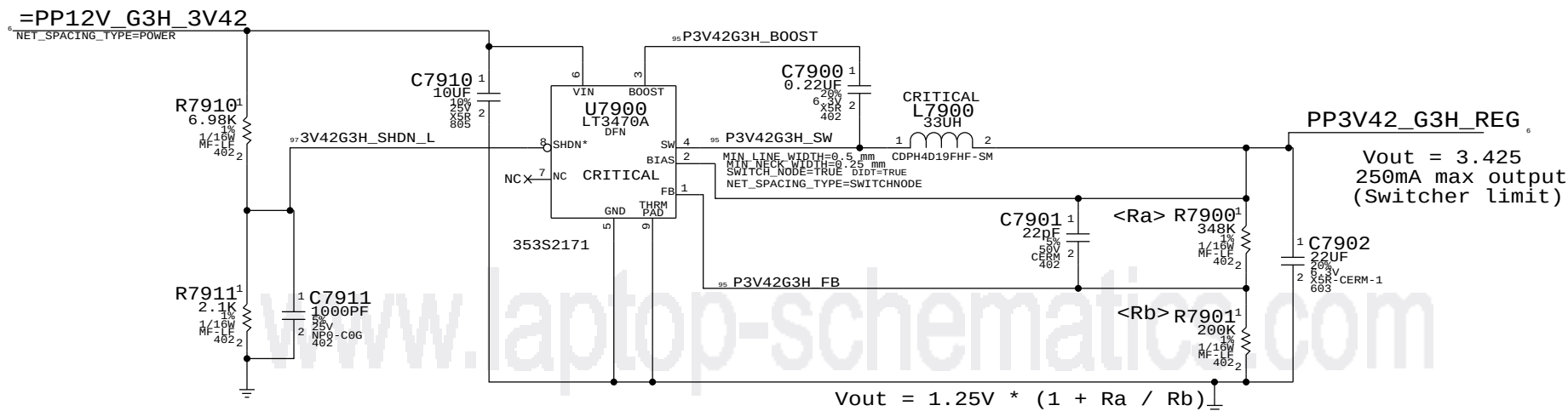
SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		1.5V / 1.8V VREGS	
Apple Inc.		DRAWING NUMBER	051-8115
		REVISION	11.1.0
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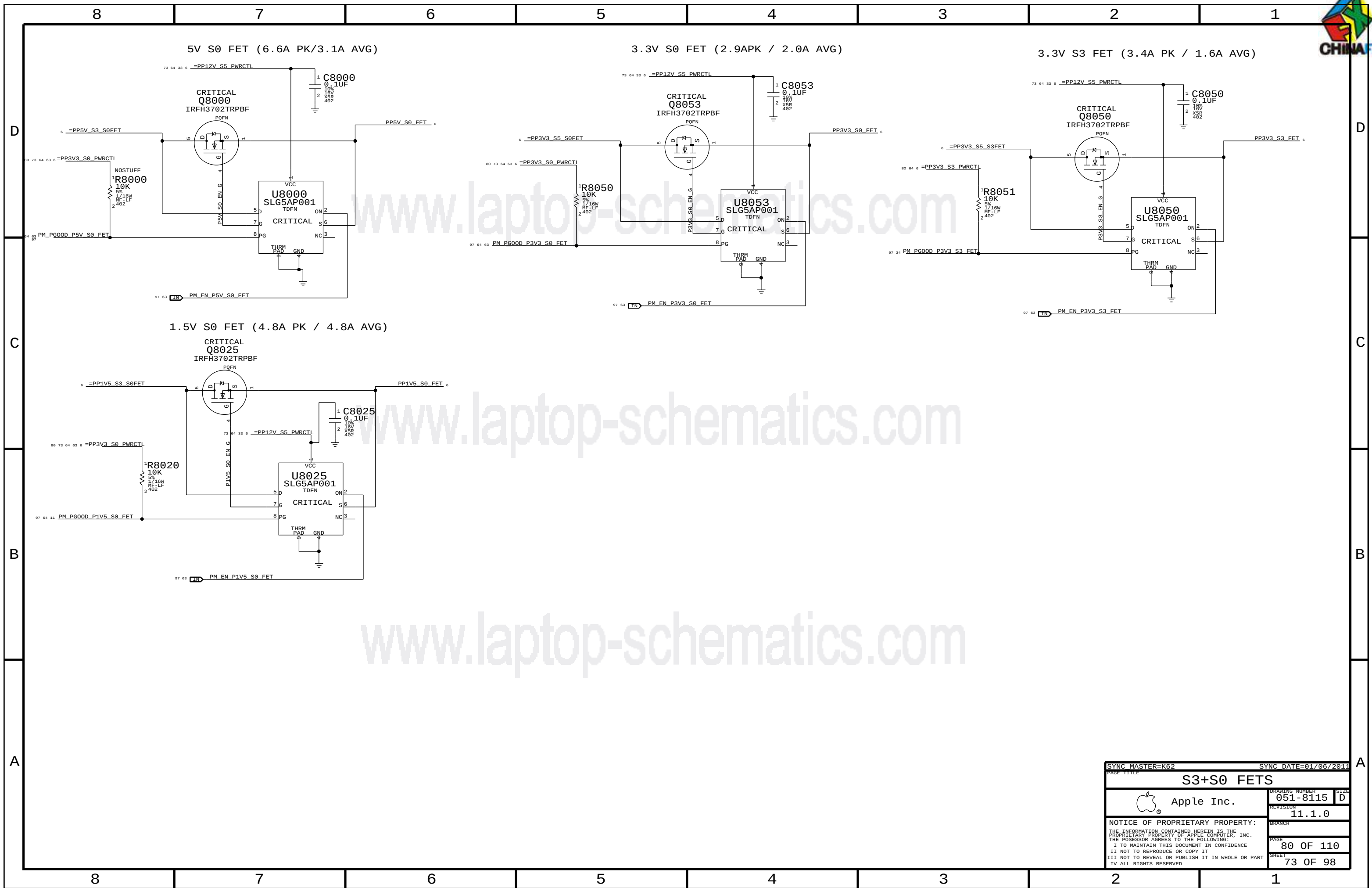
3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator



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SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
3.42		G3HOT SUPPLY	
	Apple Inc.		DRAWING NUMBER
			051-8115
		REVISION	D
		11.1.0	
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		BRANCH	
		PAGE	79 OF 110
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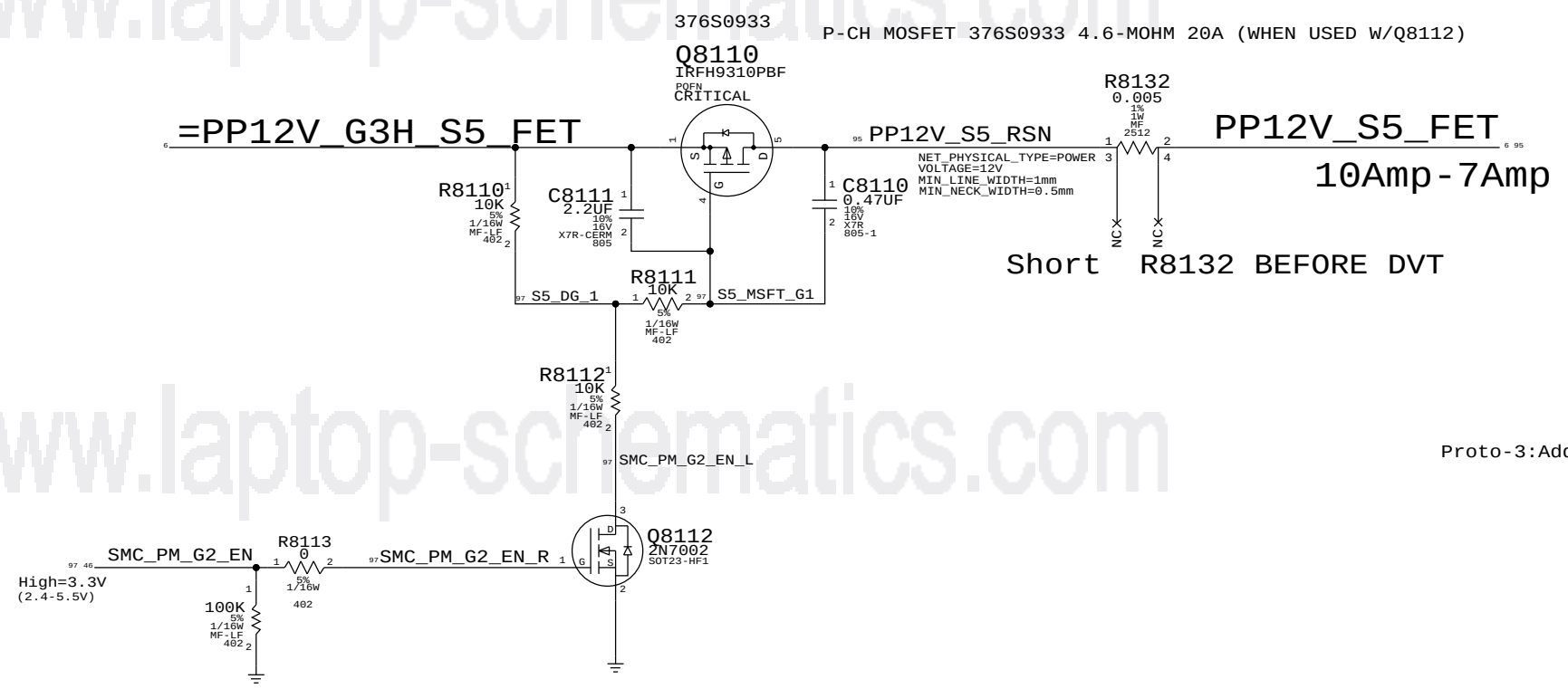




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Proto-3:Add C8111

SYNC MASTER=K60_JERRY		SYNC DATE=01/06/2011	
PAGE TITLE		12V_S0 & 12V_S5 switch	
DRAWING NUMBER		051-8115	SIZE D
REVISION		11.1.0	BRANCH
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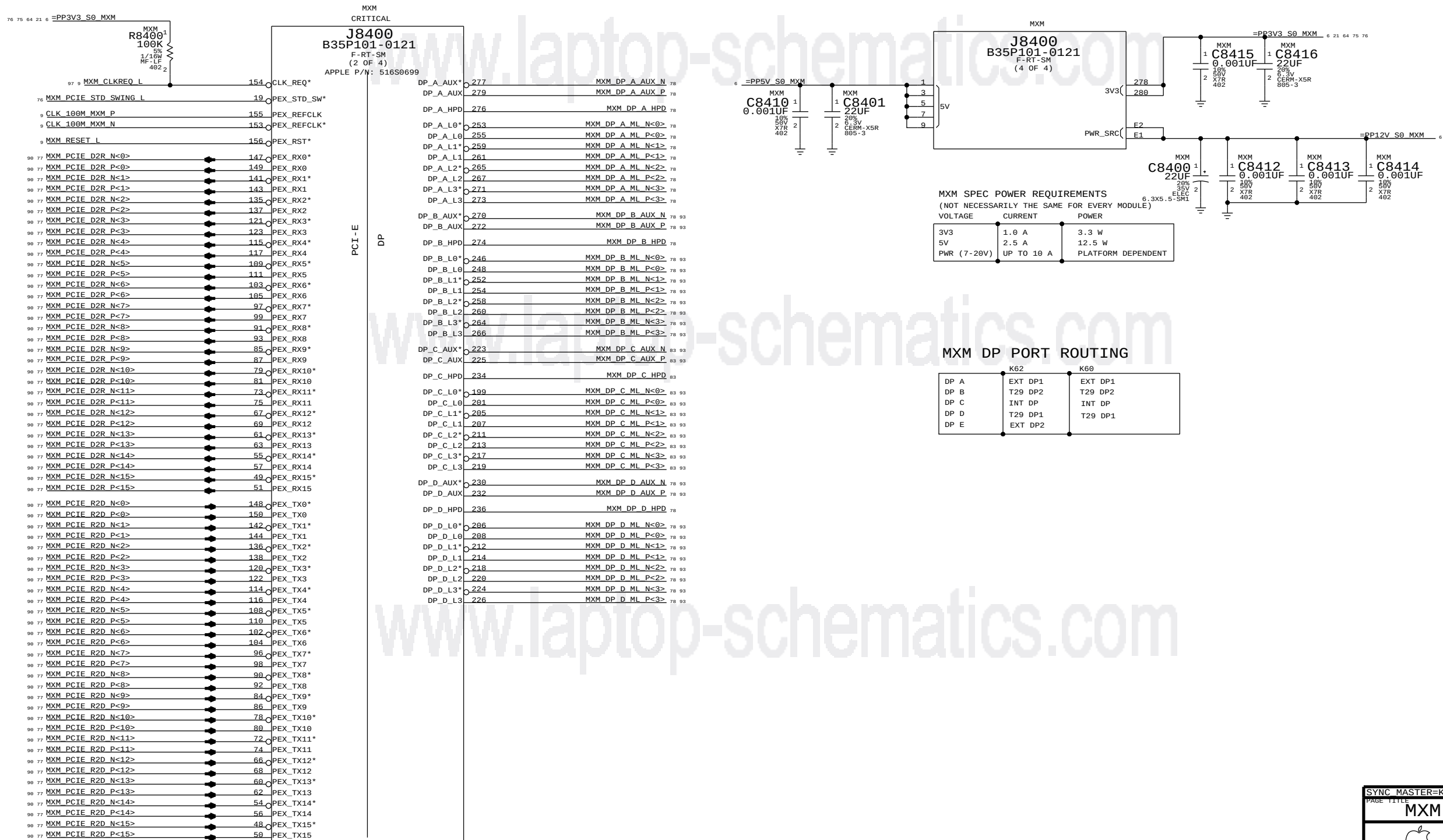
Power aliases required by this page:

- =PP3V3_S0_MXM
- =PP5V_S0_MXM
- =PPV_S0_MXM_PWRSRC

Signal aliases required by this page:
(NONE)

BOM options provided by this page:

- MXM



SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
MXM PCIe, DP & Power			
 Apple Inc.		DRAWING NUMBER	051-8115
		SIZE	D
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		PAGE	84 OF 110
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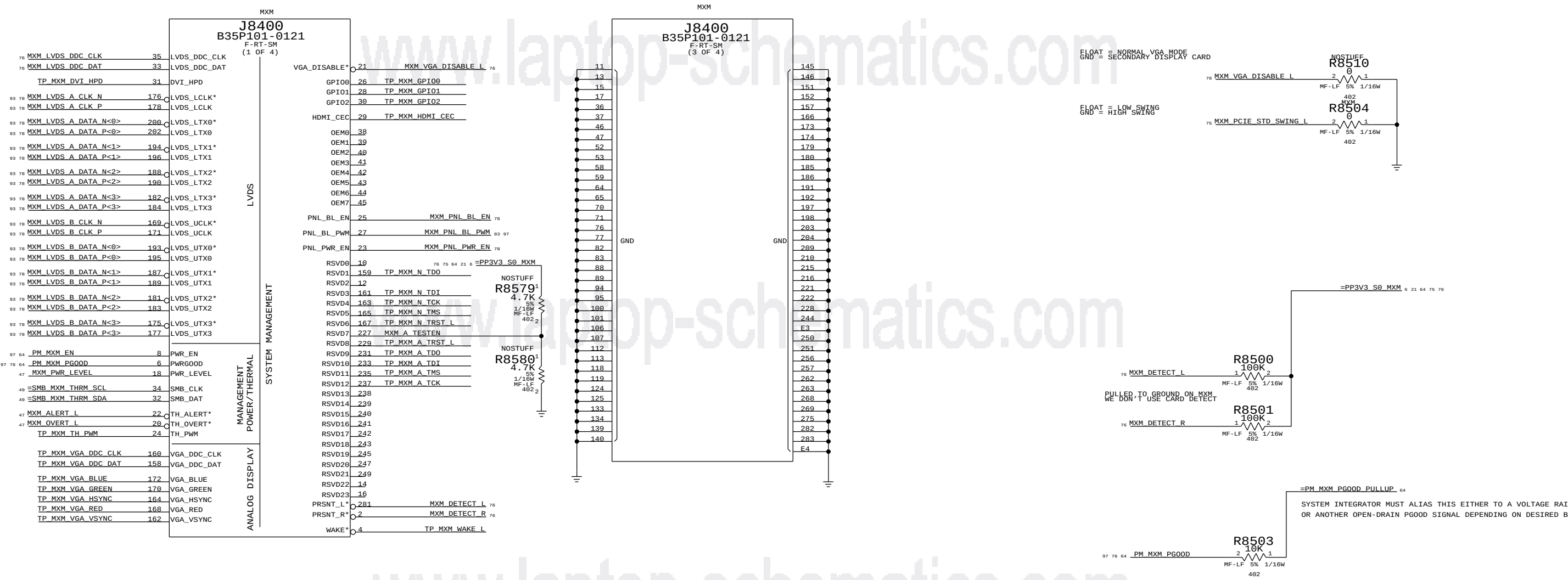
Page Notes

Power aliases required by this page:
- =PP3V3_S0_MXM

Signal aliases required by this page:
- =SMB_MXM_THRM_DATA - =PM_MXM_PG00D_PULLUP
- =SMB_MXM_THRM_CLK

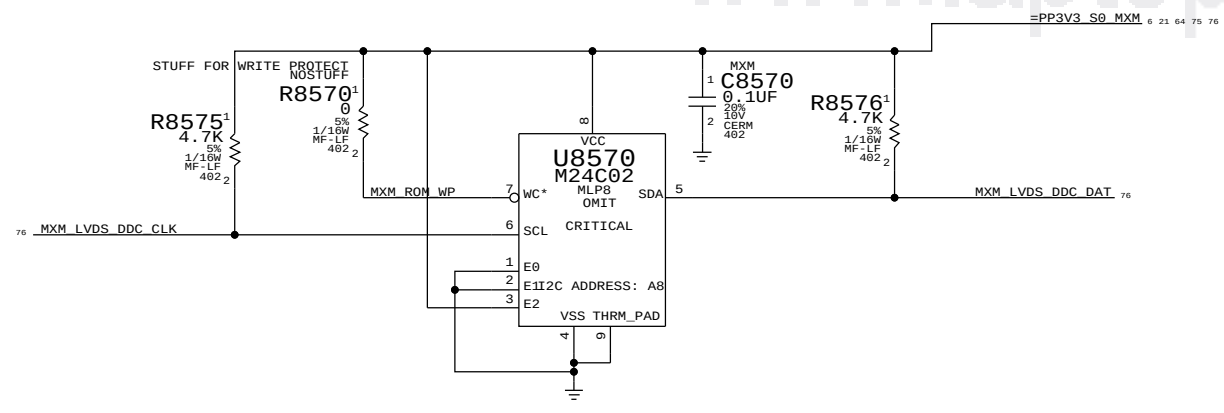
BOM options provided by this page:

PULLUPS & PULLDOWNS AT MXM CONNECTOR



MXM SYSTEM INFORMATION ROM

PLACE CLOSE TO J8400



PAGE TITLE		SYNC DATE=N/A	
MXM I/O		DRAWING NUMBER	SIZE
Apple Inc.		051-8115	D
REVISION		11.1.0	
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MXM TX CAPS

90 9	10	PEG_R2D_C_P<0>	MXM C8600 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_P<15>	75 90
90 9	10	PEG_R2D_C_N<0>	MXM C8601 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_N<15>	75 90
90 9	10	PEG_R2D_C_N<1>	MXM C8602 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_P<14>	75 90
90 9	10	PEG_R2D_C_P<1>	MXM C8603 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_N<14>	75 90
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90 9	10	PEG_R2D_C_N<4>	MXM C8608 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_P<11>	75 90
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90 9	10	PEG_R2D_C_N<5>	MXM C8610 0.1UF 1	2 10% 16V X5R 402	MXM_PCIE_R2D_P<10>	75 90
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MXM RX CAPS

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90 75	10	MXM_PCIE_D2R_P<12>	MXM C8638 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_N<3>	90 90
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90 75	10	MXM_PCIE_D2R_P<4>	MXM C8654 0.1UF 1	2 10% 16V X5R 402	PEG_D2R_N<11>	90 90
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SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
MXM PCIE CAPS			
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		REVISION	11.1.0
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Page Notes

Power aliases required by this page:

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

MXM ALIAS

75	MXM_DP_A_ML_P<0..3>	==	DP_EXT_A_ML_C_P<0..3>	84	93
		==	MAKE_BASE=TRUE NO_TEST=TRUE		
75	MXM_DP_A_ML_N<0..3>	==	DP_EXT_A_ML_C_N<0..3>	84	93
		==	MAKE_BASE=TRUE NO_TEST=TRUE		
75	MXM_DP_A_AUX_P	==	DP_EXT_A_AUXCH_C_P	78	84
		==	MAKE_BASE=TRUE NO_TEST=TRUE	84	93
75	MXM_DP_A_AUX_N	==	DP_EXT_A_AUXCH_C_N	78	84
		==	MAKE_BASE=TRUE NO_TEST=TRUE	84	93
75	MXM_DP_A_HPD	==	DP_EXT_A_HPD		
		==	MAKE_BASE=TRUE NO_TEST=TRUE	84	

T29 CONN POWER CONTROL ALIAS

95	84	PP3V3_SW_DPAPWR	==	=PP3V3_SW_DPAPWR	6
97	36	33	10	PCIE_WAKE_L	==
				=T29_WAKE_L	84

T29 MXM DP ALIAS

93	75	MXM_DP_D_ML_P<0..3>	==	DP_T29SNK0_ML_C_P<0..3>	86	96
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
93	75	MXM_DP_D_ML_N<0..3>	==	DP_T29SNK0_ML_C_N<0..3>	86	96
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
93	75	MXM_DP_D_AUX_P	==	DP_T29SNK0_AUXCH_C_P	86	96
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
93	75	MXM_DP_D_AUX_N	==	DP_T29SNK0_AUXCH_C_N	86	96
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
75		MXM_DP_D_HPD	==	DP_T29SNK0_HPD	86	
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
93	75	MXM_DP_B_ML_P<0..3>	==	DP_T29SNK1_ML_C_P<0..3>	86	96
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
93	75	MXM_DP_B_ML_N<0..3>	==	DP_T29SNK1_ML_C_N<0..3>	86	96
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93	75	MXM_DP_B_AUX_P	==	DP_T29SNK1_AUXCH_C_P	86	96
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
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			==	MAKE_BASE=TRUE NO_TEST=TRUE		
75		MXM_DP_B_HPD	==	DP_T29SNK1_HPD	86	
			==	MAKE_BASE=TRUE NO_TEST=TRUE		

Unused MXM Interfaces

93	76	MXM_LVDS_A_CLK_N	==	NC_MXM_LVDS_A_CLK_N		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
93	76	MXM_LVDS_A_CLK_P	==	NC_MXM_LVDS_A_CLK_P		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
93	76	MXM_LVDS_A_DATA_N<3..0>	==	NC_MXM_LVDS_A_DATA_N<3..0>		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
93	76	MXM_LVDS_A_DATA_P<3..0>	==	NC_MXM_LVDS_A_DATA_P<3..0>		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
93	76	MXM_LVDS_B_CLK_N	==	NC_MXM_LVDS_B_CLK_N		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
93	76	MXM_LVDS_B_CLK_P	==	NC_MXM_LVDS_B_CLK_P		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
93	76	MXM_LVDS_B_DATA_N<3..0>	==	NC_MXM_LVDS_B_DATA_N<3..0>		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
93	76	MXM_LVDS_B_DATA_P<3..0>	==	NC_MXM_LVDS_B_DATA_P<3..0>		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		

Unused T29 Interfaces

96	86	T29_D2R_N<2..3>	==	NC_T29_D2R_N<2..3>		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
96	86	T29_D2R_P<2..3>	==	NC_T29_D2R_P<2..3>		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
96	86	T29_R2D_C_N<2..3>	==	NC_T29_R2D_C_N<2..3>		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
96	86	T29_R2D_C_P<2..3>	==	NC_T29_R2D_C_P<2..3>		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
86		T29_LSE0<3>	==	T29_LSE0_LSE0E3	==	T29_LSE0<3>
			==	MAKE_BASE=TRUE NO_TEST=TRUE		86
86		T29_LSE0<2>	==	T29_LSE0_LSE0E2	==	T29_LSE0<2>
			==	MAKE_BASE=TRUE NO_TEST=TRUE		86

UNUSED MXM CONTROL SIGNALS

76		MXM_PNL_BL_EN	==	NC_MXM_PNL_BL_EN		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		
76		MXM_PNL_PWR_EN	==	NC_MXM_PNL_PWR_EN		
			==	MAKE_BASE=TRUE NO_TEST=TRUE		

DDC/AUX ALIAS

93	84	78	DP_EXT_A_AUXCH_C_P	==	DP_EXT_A_DDC_CLK	84
			MAKE_BASE=TRUE			
93	84	78	DP_EXT_A_AUXCH_C_N	==	DP_EXT_A_DDC_DATA	84
			MAKE_BASE=TRUE			

SYNC MASTER=K60 AARON		SYNC DATE=07/18/2016	
PAGE TITLE			
DP ALIAS AND CONTROL			
Apple Inc.		DRAWING NUMBER	051-8115
		REVISION	11.1.0
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GreenCLK Implementation Notes:

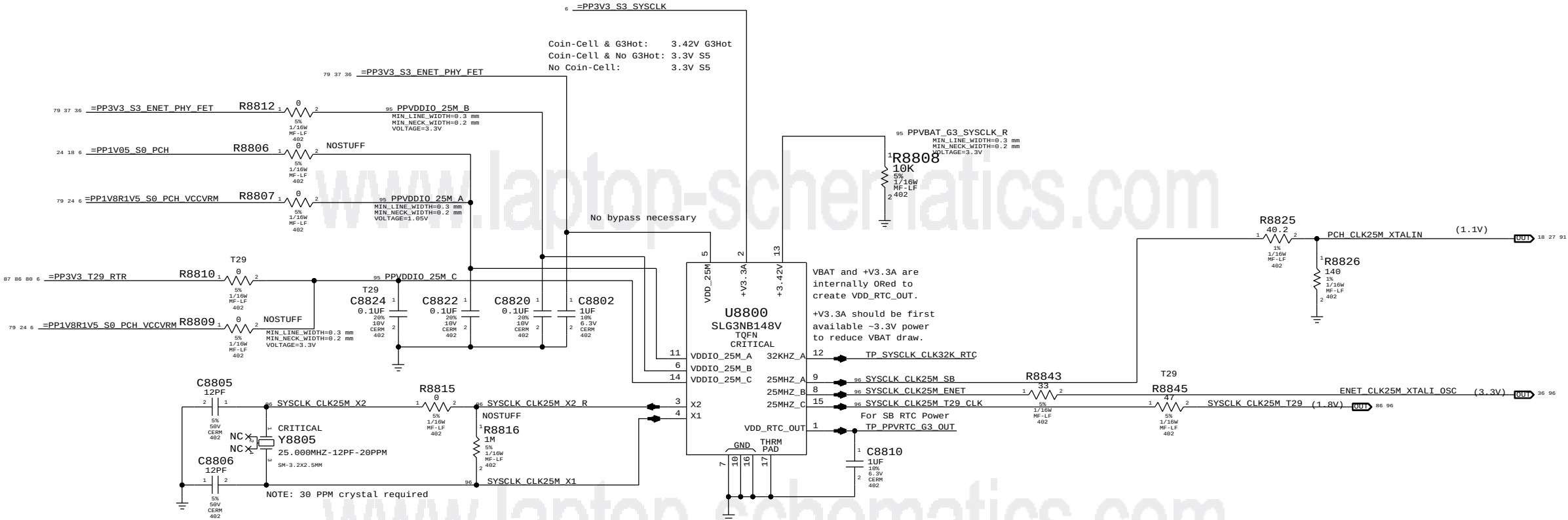
VBAT: Alias as appropriate (see note below & Desktop Example)
+V3.3A: Alias as appropriate (see note below)
VDD_25M: 3.3V matching 'highest' VDDIO power state (ENET)

VDDIO_25M_A: S8 power rail for XTAL circuit.
VDDIO_25M_B: Ethernet power rail for XTAL circuit.
VDDIO_25M_C: T29 power rail for XTAL circuit.

NOTE: VDD_25M must be powered if any VDDIO_25M_x is powered.

For Cougar Point Desktop: VDDIO = VCCVRM (1.8V), Vclk = 1.1V Max, Divider: 694 / 1099
For Cougar Point Mobile: VDDIO = VCCVRM (1.5V), Vclk = 1.1V Max, Divider: 332 / 1099
For Caesar-IV (BCM57765): VDDIO = XTALVDDH (3.3V), Vclk = 3.3V Max. No Divider Necessary

System RTC Power Source & 32kHz / 25MHz Clock Generator



SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
GREEN CLOCK			
 Apple Inc.	DRAWING NUMBER	051-8115	SIZE
	REVISION	11.1.0	
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Page Notes

Power aliases required by this page:

- PP3V3_T29_P3V3T29FET (3.3V FET Input)
- PP3V3_T29_FET (3.3V FET Output)
- PP3V3_S0_T29PWRCTL
- PP1V05_T29_P1V05T29FET (1.05V FET Input)
- PP1V05_T29_FET (1.05V FET Output)

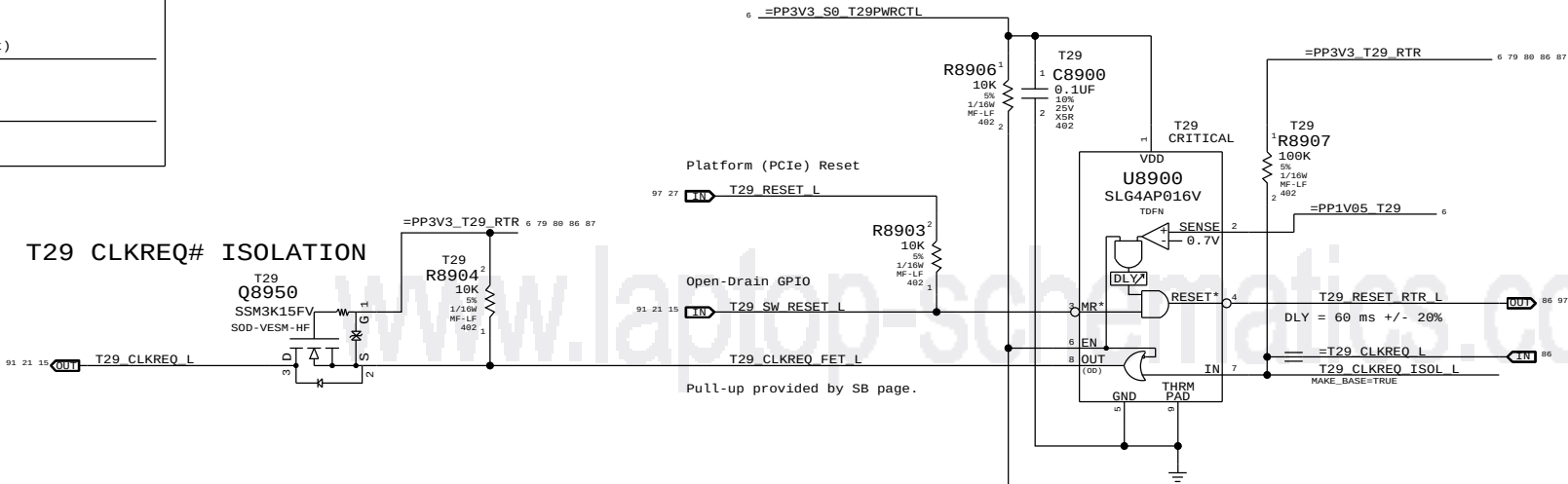
Signal aliases required by this page:

- T29_CLKREQ_L
- T29_RESET_L

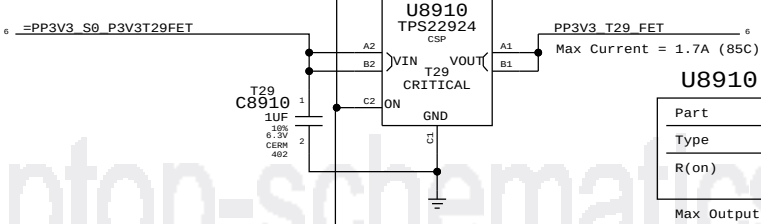
BOM options provided by this page:

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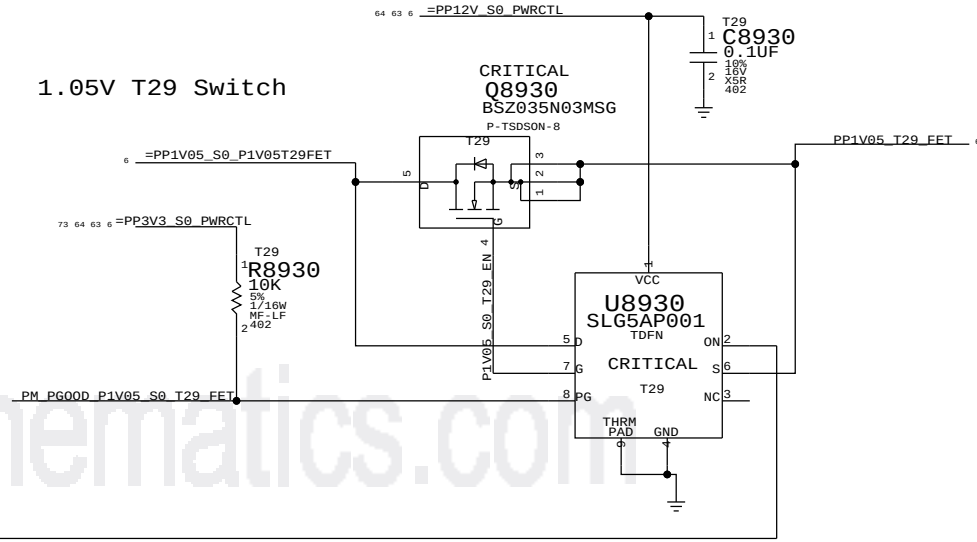
Supervisor & CLKREQ# Isolation



3.3V T29 Switch



1.05V T29 Switch



SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE		T29 POWER	
Apple Inc.		DRAWING NUMBER	051-8115
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Page Notes

Power aliases required by this page:

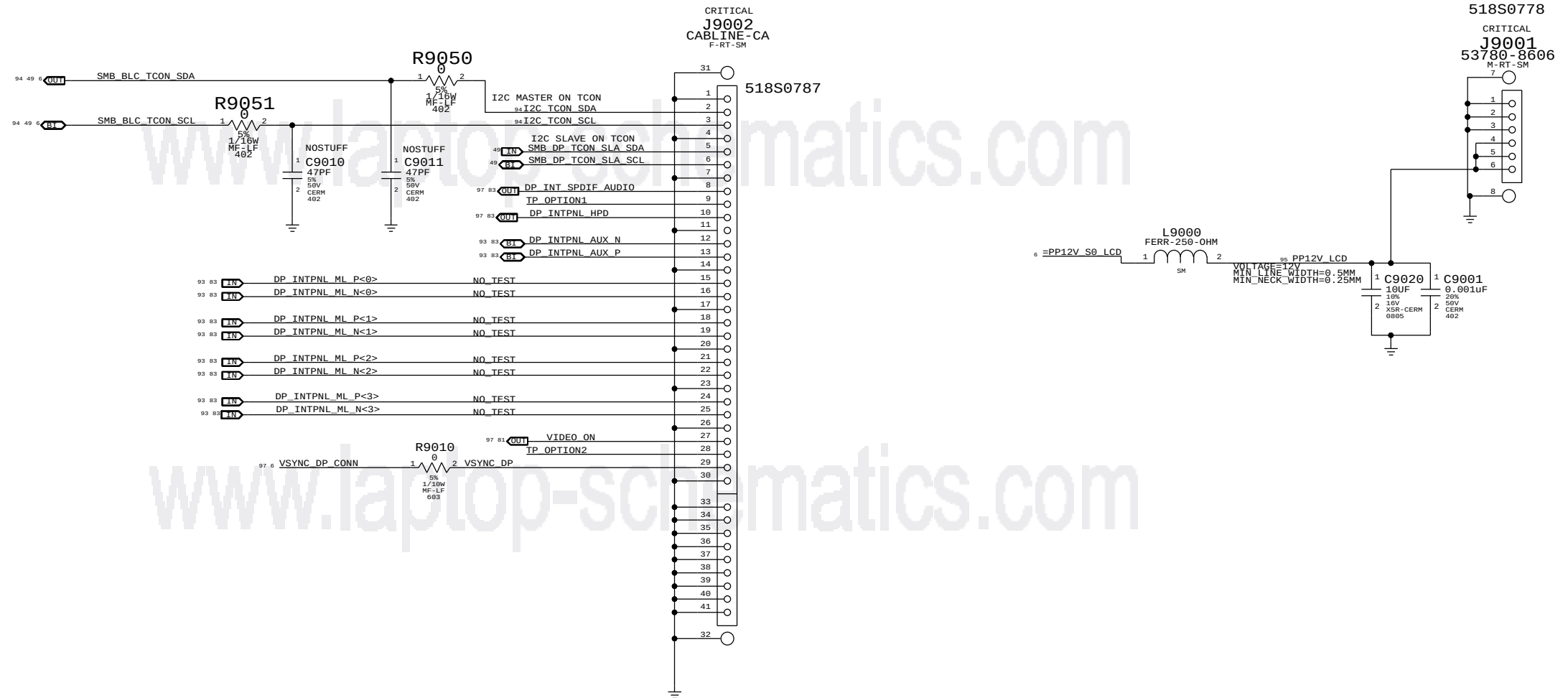
- =PP12V_S0_LCD
- =PP3V3_S0_VIDEO

Signal aliases required by this page:
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BOM options provided by this page:
IG, MXM, MLB_PNL_PWR, LCD_PNL_PWR

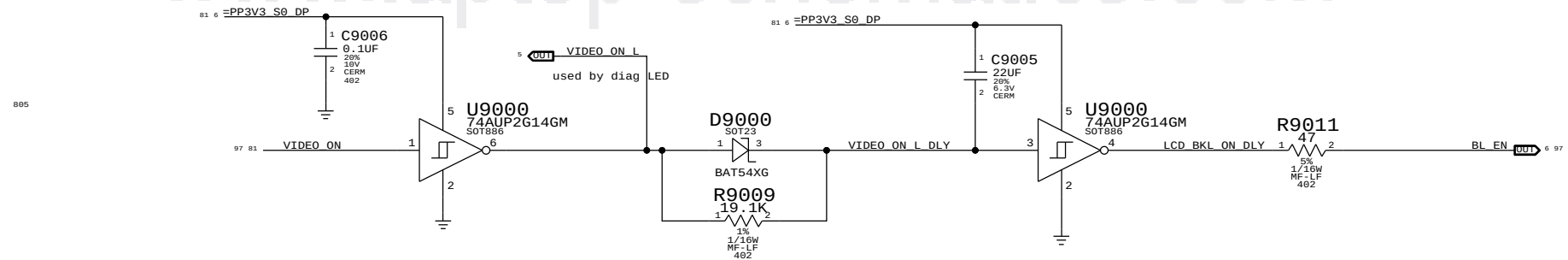
INTERNAL DP INTERFACE

INTERNAL DP POWER



BACKLIGHT CONTROL SUPPORT

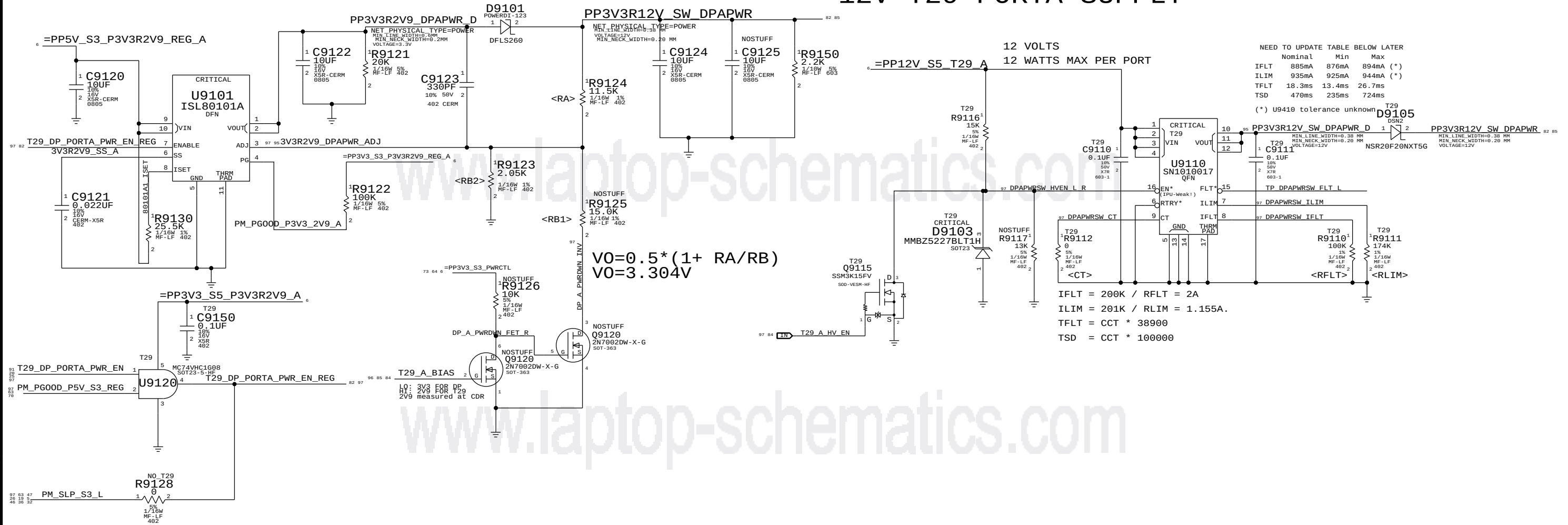
guarantee backlight is
only on when Panel has valid video

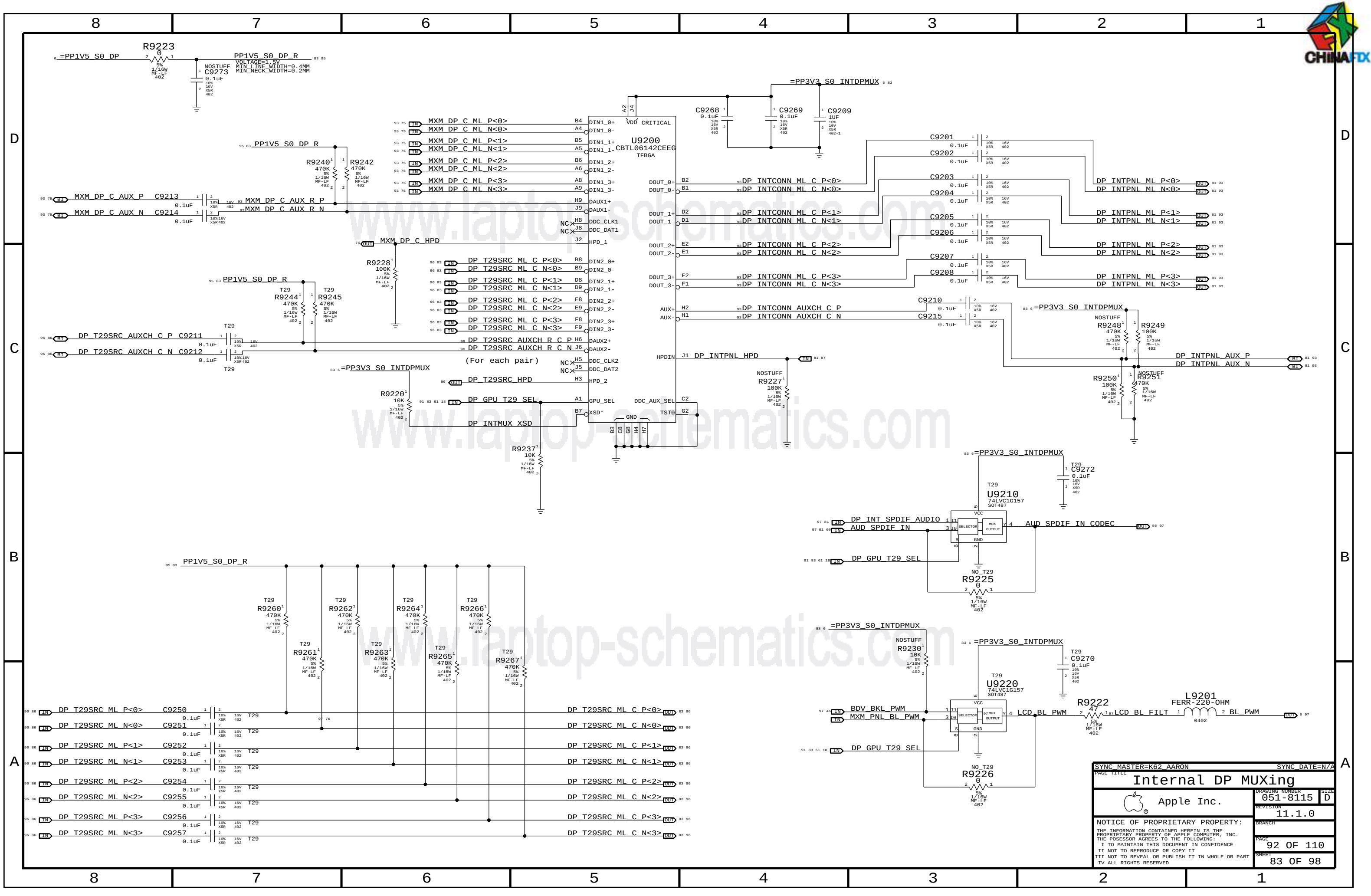


PAGE 1/TITLE		SYNC DATE=01/06/2011	
Display: Int DP Connector		DRAWING NUMBER	SIZE
Apple Inc.		051-8115	D
REVISION		11.1.0	
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3V3(DP)/2V9(T29) PORTA SUPPLY

12V T29 PORTA SUPPLY

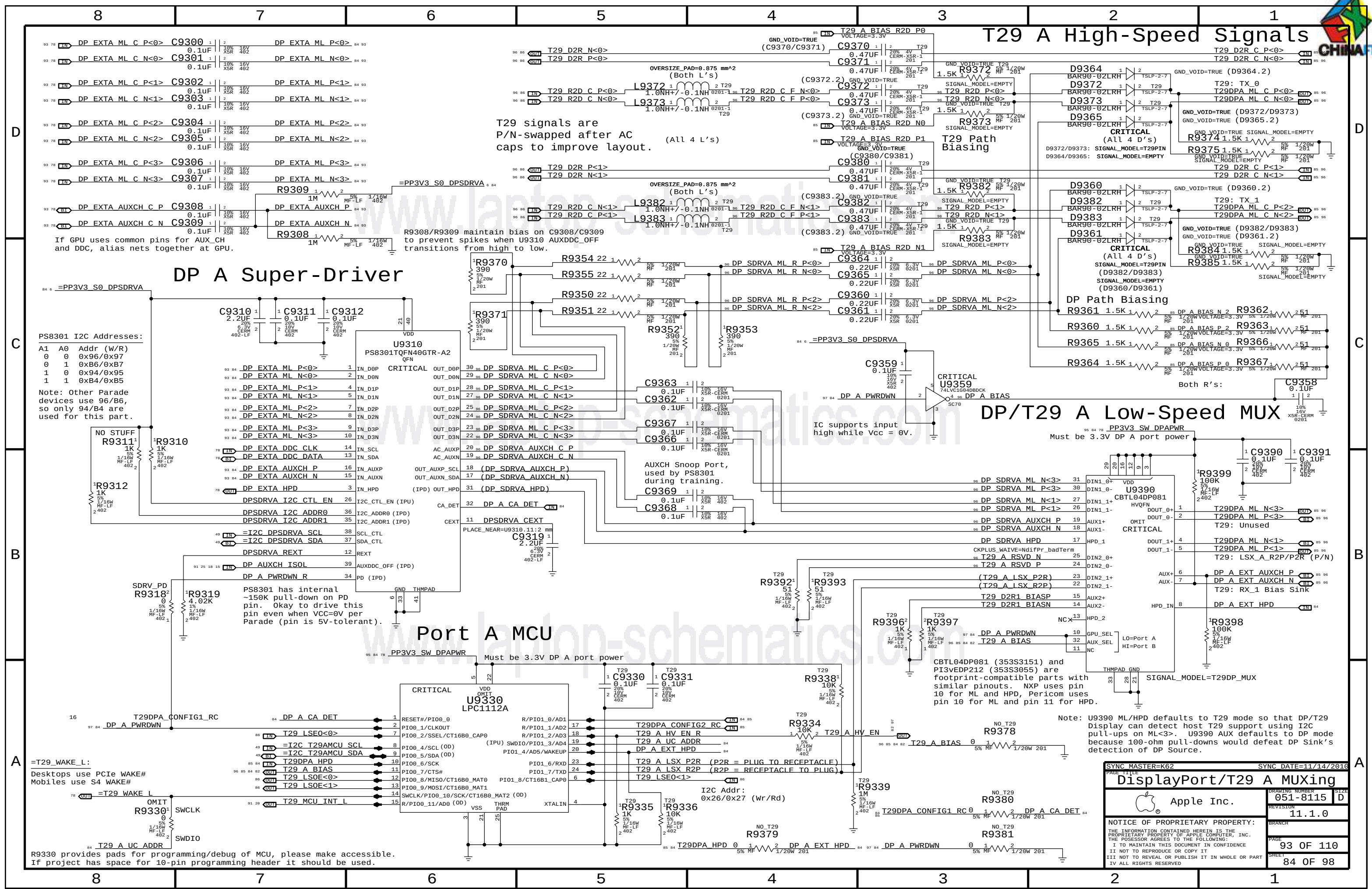




PAGE TITLE		SYNC DATE=N/A	
Internal DP MUXing		DRAWING NUMBER	
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T29 A High-Speed Signals

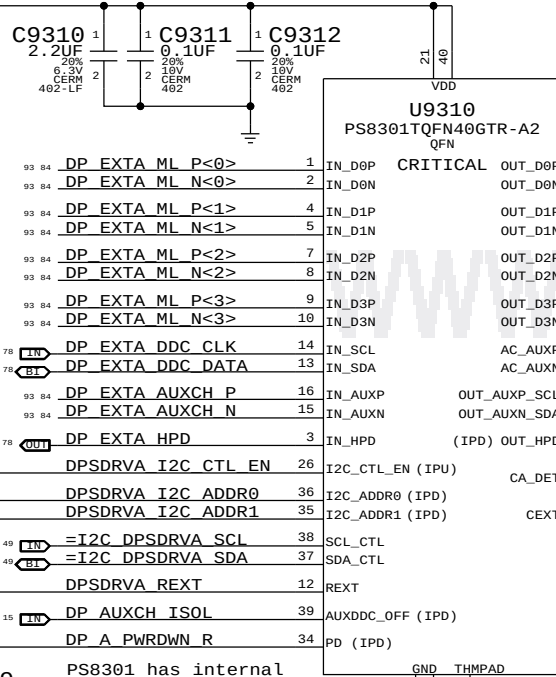


T29 signals are P/N-swapped after AC caps to improve layout. (All 4 L's)

DP A Super-Driver

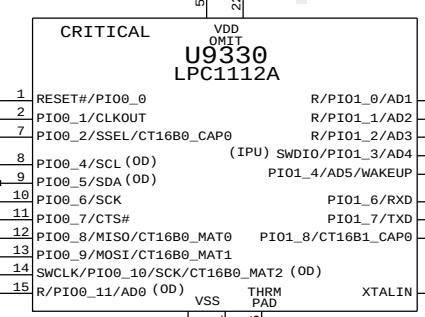
PS8301 I2C Addresses:
A1 A0 Addr (W/R)
0 0 0x96/0x97
0 1 0xB6/0xB7
1 0 0x94/0x95
1 1 0xB4/0xB5

Note: Other Parade devices use 96/B6, so only 94/B4 are used for this part.



PS8301 has internal ~150K pull-down on PD pin. Okay to drive this pin even when VCC=0V per Parade (pin is 5V-tolerant).

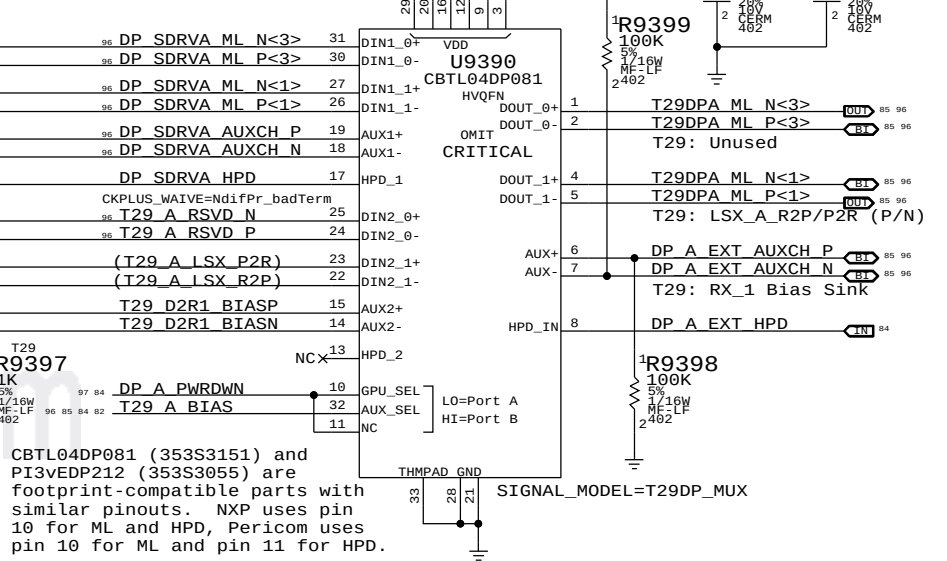
Port A MCU



=T29_WAKE_L:
Desktops use PCIe WAKE#
Mobiles use S4 WAKE#

R9330 provides pads for programming/debug of MCU, please make accessible. If project has space for 10-pin programming header it should be used.

DP/T29 A Low-Speed MUX

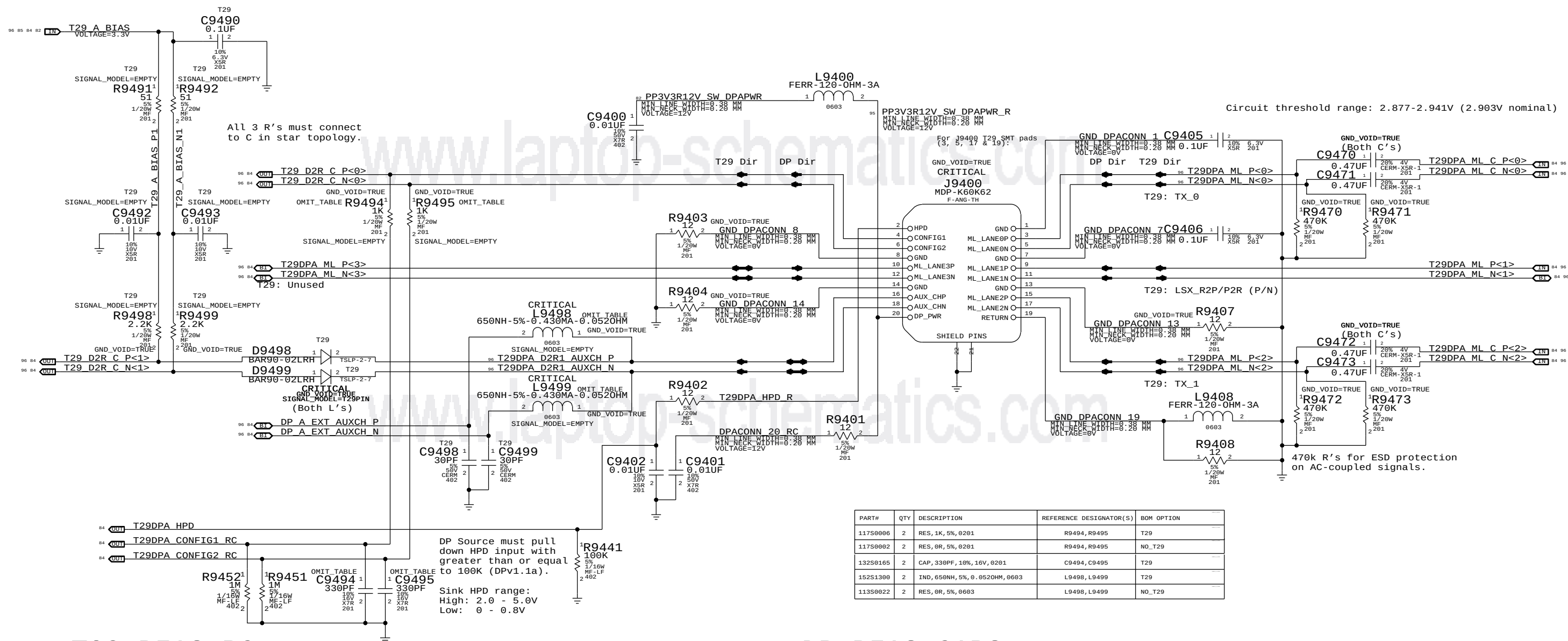


CBTL04DP081 (353S3151) and PI3VEDP212 (353S3055) are footprint-compatible parts with similar pinouts. NXP uses pin 10 for ML and HPD, Pericom uses pin 10 for ML and pin 11 for HPD.

Note: U9390 ML/HPD defaults to T29 mode so that DP/T29 Display can detect host T29 support using I2C pull-ups on ML<3>. U9390 AUX defaults to DP mode because 100-ohm pull-downs would defeat DP Sink's detection of DP Source.

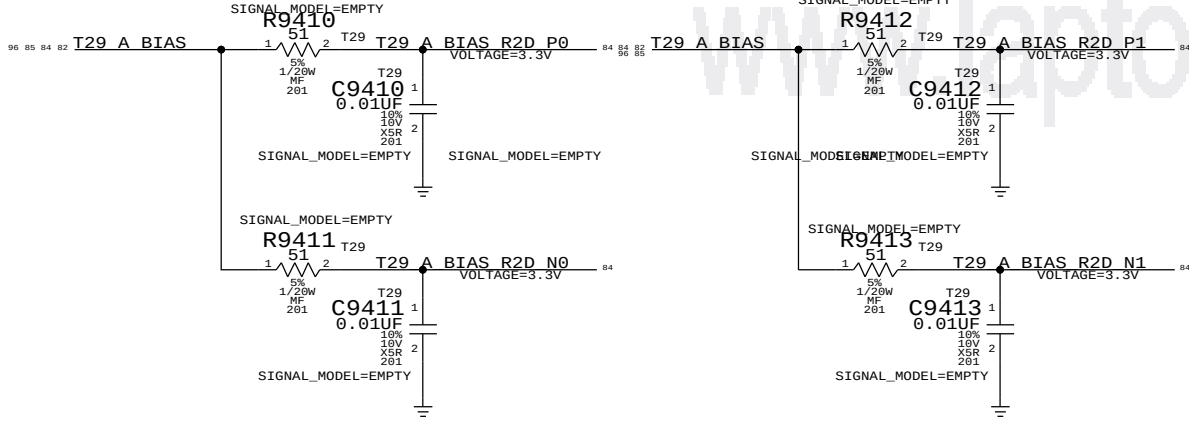
SYNC MASTER=K62		SYNC DATE=11/14/2016	
PAGE 11/16			
DisplayPort/T29 A MUXing		DRAWING NUMBER	051-8115
Apple Inc.		REVISION	11.1.0
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DisplayPort/T29 A Connector

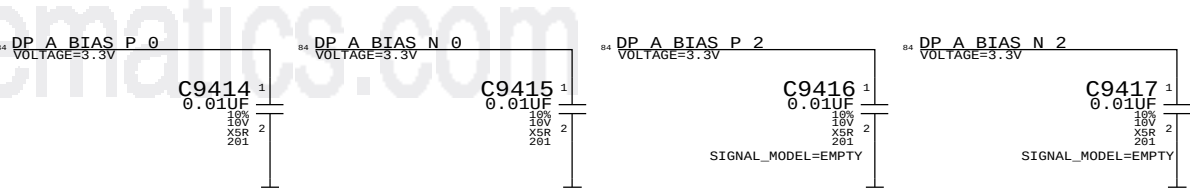


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
117S0006	2	RES, 1K, 5%, 0201	R9494, R9495	T29
117S0002	2	RES, 0R, 5%, 0201	R9494, R9495	NO_T29
132S0165	2	CAP, 330PF, 10%, 16V, 0201	C9494, C9495	T29
152S1300	2	IND, 650NH, 5%, 0.0520HM, 0603	L9498, L9499	T29
113S0022	2	RES, 0R, 5%, 0603	L9498, L9499	NO_T29

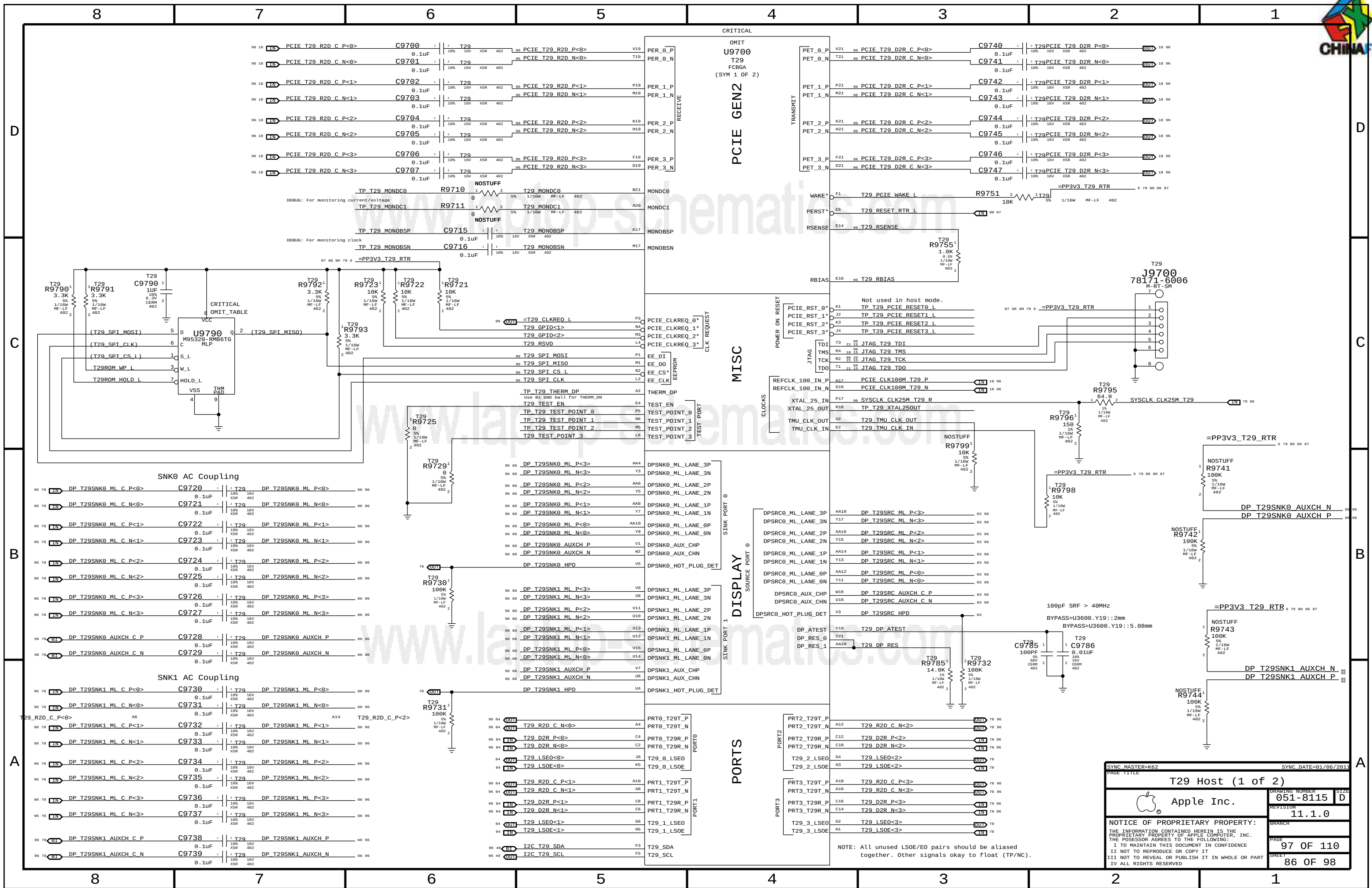
T29 BIAS RC

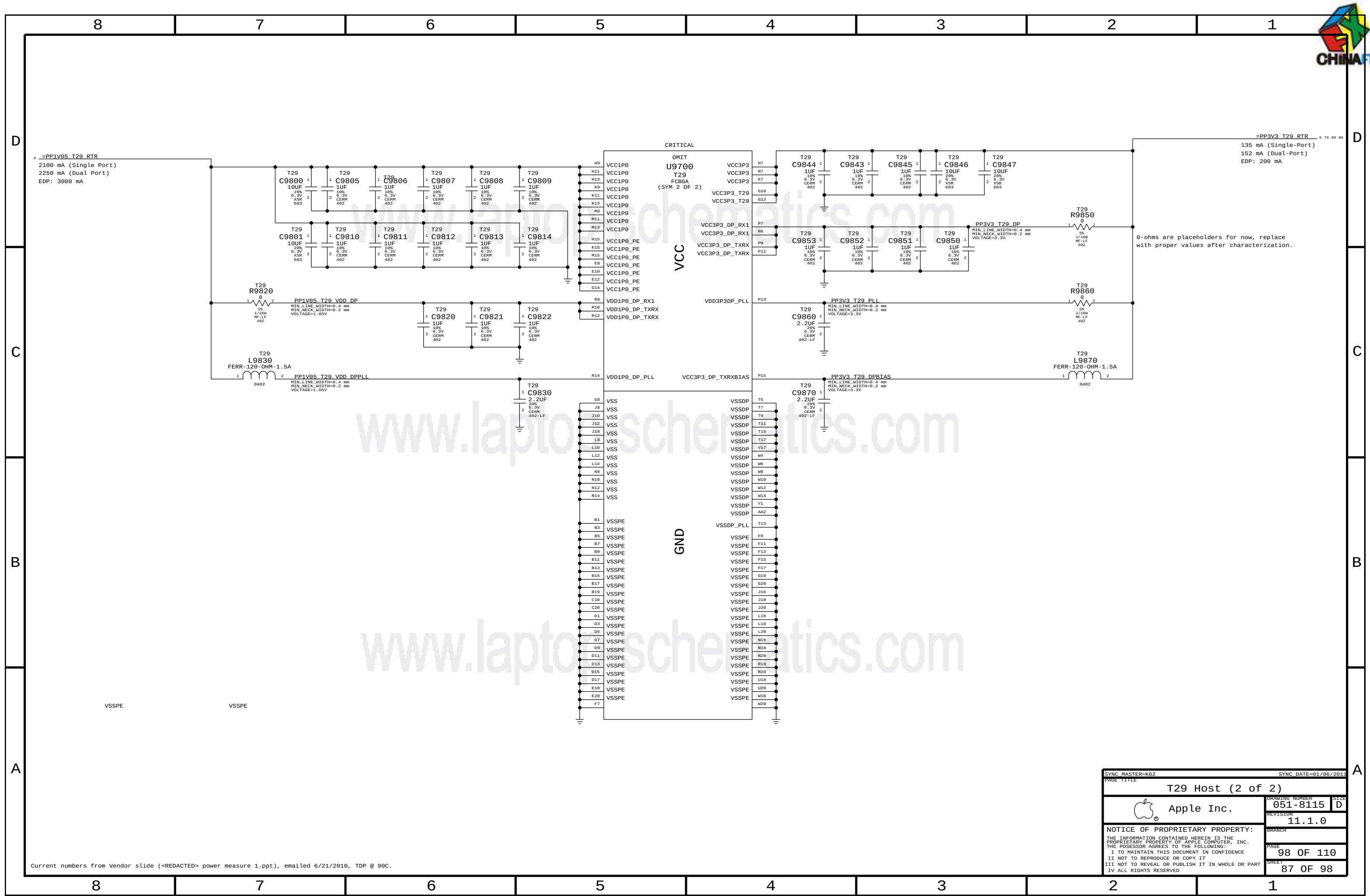


DP BIAS CAPS



SYNC MASTER=(MASTER)	SYNC DATE=(MASTER)
DisplayPort/T29 A Connector	
Apple Inc.	DRAWING NUMBER 051-8115
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Current numbers from Vendor slide (<REDACTED> power measure 1.ppt), emailed 6/21/2010, TDP @ 90C.

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
T29 Host (2 of 2)			
 Apple Inc.	DRAWING NUMBER		SIZE
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K60/62 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL OF MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, BOTTOM	NO_TYPE, BGA, BGA_P1MM	MM	15.5.1

PHYSICAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	100 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	TOP, BOTTOM	Y	0.21 MM	0.090 MM	=STANDARD		
34_OHM_SE	*	Y	0.19 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	TOP, BOTTOM	Y	0.175 MM	0.090 MM	=STANDARD		
39_OHM_SE	*	Y	0.150 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	TOP, BOTTOM	Y	0.151 MM	0.090 MM	=STANDARD		
42_OHM_SE	*	Y	0.135 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP, BOTTOM	Y	0.1 MM	0.085 MM	15 MM		
50_OHM_SE	*	Y	0.1 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP, BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD		
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
68_OHM_DIFF	ISL3, ISL6	Y	0.16 MM	0.090 MM	=STANDARD	0.13 MM	0.1 MM
68_OHM_DIFF	TOP, BOTTOM	Y	0.165 MM	0.090 MM	=STANDARD	0.13 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3, ISL6	Y	0.115 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM
85_OHM_DIFF	TOP, BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3, ISL6	Y	0.090 MM	0.085 MM	12 MM	0.2 MM	0.1 MM
90_OHM_DIFF	TOP, BOTTOM	Y	0.110 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3, ISL6	Y	0.081 MM	0.081 MM	=STANDARD	0.25 MM	0.1 MM
100_OHM_DIFF	TOP, BOTTOM	Y	0.091 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3, ISL6	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	TOP, BOTTOM	Y	0.075 MM	0.075 MM	=STANDARD	0.320 MM	0.15 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM
POWER_WIDTH	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_CTL	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

SPACING RULE SET

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
3.5:1_SPACING	*	0.35 MM	?
4:1_SPACING	*	0.4 MM	?
5:1_SPACING	*	0.5 MM	?
6:1_SPACING	*	0.6 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
GND_P2MM	*	0.2 MM	1000
PWR_P2MM	*	0.2 MM	1000
SWITCHNODE	*	0.8 MM	1000

CONSTRAINTS FOR BGA AREA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=DEFAULT	?
BGA_P1P5MM	*	0.15 MM	?
BGA_P2MM	*	0.2 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_P1MM	BGA_P1MM
*	*	BGA	BGA_P1MM
MEM_CLK	*	BGA	BGA_P1P5MM
CLK_PCIE	*	BGA	BGA_P1MM
CLK_LPC	*	BGA	BGA_P1MM
CLK_PCI	*	BGA	BGA_P1MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CONTROL	*	*	SWITCHNODE
VR_CONTROL	VR_CONTROL	*	STANDARD
VR_CONTROL	SWITCHNODE	*	STANDARD
VR_CONTROL	GND	*	STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER	BGA_P1MM	POWER_CTL
POWER	*	POWER_WIDTH
VR_CTL_PHY	BGA_P1MM	DEFAULT
VR_CTL_PHY	*	POWER_CTL

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.155 MM	?
2X_DIELECTRIC	TOP, BOTTOM	0.145 MM	?
3X_DIELECTRIC	*	0.230 MM	?
3X_DIELECTRIC	TOP, BOTTOM	0.215 MM	?
4X_DIELECTRIC	*	0.305 MM	?
4X_DIELECTRIC	TOP, BOTTOM	0.285 MM	?
5X_DIELECTRIC	*	0.380 MM	?
5X_DIELECTRIC	TOP, BOTTOM	0.355 MM	?
7X_DIELECTRIC	*	0.532 MM	?
7X_DIELECTRIC	TOP, BOTTOM	0.497 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	*	*	STANDARD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	STANDARD

BOARD STACK-UP

TOP	HALF OZ	SIGNAL
	0.071	PREPREG
2	TWO OZ	GND
	0.076	PREPREG
3	ONE OZ	SIGNAL
	0.370	PREPREG
4	TWO OZ	POWER
	0.101	CORE
5	TWO OZ	POWER
	0.370	PREPREG
6	ONE OZ	SIGNAL
	0.076	PREPREG
7	TWO OZ	GND
	0.071	PREPREG
BOTTOM	HALF OZ	SIGNAL

BOARD THICKNESS = 62 MIL (1.5748 mm)

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
K60/K62 RULE DEFINITIONS			
 Apple Inc.		DRAWING NUMBER	051-8115
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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
MEM_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
MEM_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
MEM_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF
MEM_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2.5:1_SPACING	?
MEM_CTRL2MEM	*	=3:1_SPACING	?
MEM_CMD2CMD	*	=2:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DQ_SAMEBYTE	*	=3:1_SPACING	?
MEM_DQ_DIFFBYTE	*	=5:1_SPACING	?
MEM_DATA2MEM	*	=4:1_SPACING	?
MEM_DQS2MEM	*	=4:1_SPACING	?
MEM_20THER	*	=5:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE0	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE1	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE2	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE3	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE4	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE5	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE6	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE7	*	MEM_DQS2MEM
MEM_DQS	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE5	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE5	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE5	MEM_DQ_BYTE5	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE5	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE5	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE5	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE6	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE6	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE6	MEM_DQ_BYTE6	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE6	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE6	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE7	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE7	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE7	MEM_DQ_BYTE7	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE7	*	*	MEM_20THER

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE0	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE0	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE0	MEM_DQ_BYTE0	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE1	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE2	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE3	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE1	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE1	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE1	MEM_DQ_BYTE1	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE2	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE3	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE2	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE2	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE2	MEM_DQ_BYTE2	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE3	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE3	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE3	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE3	MEM_DQ_BYTE3	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	**	*	

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE4	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE4	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE4	MEM_DQ_BYTE4	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE4	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE4	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE4	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE4	*	*	MEM_20THER

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
	MEM_68D	MEM_CLK	MEM A CLK P<3..0>	12 32
	MEM_68D	MEM_CLK	MEM A CLK N<3..0>	12 32
	MEM_39S	MEM_CTRL	MEM A CKE<3..0>	12 30
	MEM_39S	MEM_CTRL	MEM A CS L<3..0>	12 30
	MEM_39S	MEM_CTRL	MEM A ODT<3..0>	12 30
	MEM_34S	MEM_CMD	MEM A A<15..0>	12 30
	MEM_34S	MEM_CMD	MEM A BA<2..0>	12 30
	MEM_34S	MEM_CMD	MEM A RAS L	12 30
	MEM_34S	MEM_CMD	MEM A CAS L	12 30
	MEM_34S	MEM_CMD	MEM A WE L	12 30
	MEM_42S	MEM_DQ_BYTE0	MEM A DQ<7..0>	12 32
	MEM_42S	MEM_DQ_BYTE1	MEM A DQ<15..8>	12 32
	MEM_42S	MEM_DQ_BYTE2	MEM A DQ<23..16>	12 32
	MEM_42S	MEM_DQ_BYTE3	MEM A DQ<31..24>	12 32
	MEM_42S	MEM_DQ_BYTE4	MEM A DQ<39..32>	12 32
	MEM_42S	MEM_DQ_BYTE5	MEM A DQ<47..40>	12 32
	MEM_42S	MEM_DQ_BYTE6	MEM A DQ<55..48>	12 32
	MEM_42S	MEM_DQ_BYTE7	MEM A DQ<63..56>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS P<0>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS N<0>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS P<1>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS N<1>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS P<2>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS N<2>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS P<3>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS N<3>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS P<4>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS N<4>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS P<5>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS N<5>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS P<6>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS N<6>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS P<7>	12 32
	MEM_42S_D	MEM_DQS	MEM A DQS N<7>	12 32
	MEM_50S	PM	MEM RESET L	30 31 32 97

MEMORY MISC PROPERTIES

VOLTAGE	PHYSICAL	NET_TYPE	VOLTAGE
	MEM_POWER_PHY	MEM_POWER	CPU DIMM VREF A
	MEM_POWER_PHY	MEM_POWER	CPU DIMM VREF B
	MEM_POWER_PHY	MEM_POWER	CPU DDR VREF
	MEM_POWER_PHY	MEM_POWER	VREFMARGIN DIMMA DACOUT
	MEM_POWER_PHY	MEM_POWER	VREFMARGIN DIMMA OPFB
	MEM_POWER_PHY	MEM_POWER	VREFMARGIN DIMMA DQ
	MEM_POWER_PHY	MEM_POWER	CPU DIMM VREF A SW
	MEM_POWER_PHY	MEM_POWER	VREFMARGIN DIMMB DACOUT
	MEM_POWER_PHY	MEM_POWER	VREFMARGIN DIMMB OPFB
	MEM_POWER_PHY	MEM_POWER	VREFMARGIN DIMMB DQ
	MEM_POWER_PHY	MEM_POWER	CPU DIMM VREF B SW

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MEM_POWER_WIDTH	*	Y	0.500 MM	0.250 MM	=STANDARD	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_POWER_PHY	*	MEM_POWER_WIDTH

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MEM_POWER	*	=3:1_SPACING	?

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
	MEM_68D	MEM_CLK	MEM B CLK P<3..0>	12 32
	MEM_68D	MEM_CLK	MEM B CLK N<3..0>	12 32
	MEM_39S	MEM_CTRL	MEM B CKE<3..0>	12 31
	MEM_39S	MEM_CTRL	MEM B CS L<3..0>	12 31
	MEM_39S	MEM_CTRL	MEM B ODT<3..0>	12 31
	MEM_34S	MEM_CMD	MEM B A<15..0>	12 31
	MEM_34S	MEM_CMD	MEM B BA<2..0>	12 31
	MEM_34S	MEM_CMD	MEM B RAS L	12 31
	MEM_34S	MEM_CMD	MEM B CAS L	12 31
	MEM_34S	MEM_CMD	MEM B WE L	12 31
	MEM_42S	MEM_DQ_BYTE0	MEM B DQ<7..0>	12 32
	MEM_42S	MEM_DQ_BYTE1	MEM B DQ<15..8>	12 32
	MEM_42S	MEM_DQ_BYTE2	MEM B DQ<23..16>	12 32
	MEM_42S	MEM_DQ_BYTE3	MEM B DQ<31..24>	12 32
	MEM_42S	MEM_DQ_BYTE4	MEM B DQ<39..32>	12 32
	MEM_42S	MEM_DQ_BYTE5	MEM B DQ<47..40>	12 32
	MEM_42S	MEM_DQ_BYTE6	MEM B DQ<55..48>	12 32
	MEM_42S	MEM_DQ_BYTE7	MEM B DQ<63..56>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS P<0>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS N<0>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS P<1>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS N<1>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS P<2>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS N<2>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS P<3>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS N<3>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS P<4>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS N<4>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS P<5>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS N<5>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS P<6>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS N<6>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS P<7>	12 32
	MEM_42S_D	MEM_DQS	MEM B DQS N<7>	12 32

SYNC MASTER=K60 ROSITA		SYNC DATE=01/06/2011	
PAGE TITLE		Memory Constraints	
Apple Inc.		DRAWING NUMBER	051-8115
		REVISION	11.1.0
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PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCIE_850	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_900	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCIE	*	=4X_DIELECTRIC	?	PCIE	TOP,BOTTOM	=4:1_SPACING	?
CLK_PCIE	*	0.5 MM	?				

CPU

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_RCOMP_PHY	*	Y	0.254 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CPU_ITP	*	0.2 MM	?
CPU_RCOMP	*	0.2 MM	?

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SATA_900	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SATA	*	=6:1_SPACING	?	SATA	TOP,BOTTOM	=6:1_SPACING	?

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
PCIE GRAPHICS	PCIE_850	PCIE	PEG_R2D_C_P<15..0>	9 77
	PCIE_850	PCIE	PEG_R2D_C_N<15..0>	9 77
	PCIE_850	PCIE	PEG_D2R_P<15..0>	9 77
	PCIE_850	PCIE	PEG_D2R_N<15..0>	9 77
	PCIE_850	PCIE	MXM_PCIE_R2D_P<7..0>	75 77
	PCIE_850	PCIE	MXM_PCIE_R2D_N<7..0>	75 77
	PCIE_850	PCIE	MXM_PCIE_D2R_P<7..0>	75 77
	PCIE_850	PCIE	MXM_PCIE_D2R_N<7..0>	75 77
	PCIE_850	PCIE		
	PCIE_850	PCIE		
PCIE I/O	PCIE_850	PCIE	PCIE_MINI_R2D_P	33
	PCIE_850	PCIE	PCIE_MINI_R2D_N	33
	PCIE_850	PCIE	PCIE_MINI_R2D_C_P	18 33
	PCIE_850	PCIE	PCIE_MINI_R2D_C_N	18 33
	PCIE_850	PCIE	PCIE_MINI_D2R_P	18 33
	PCIE_850	PCIE	PCIE_MINI_D2R_N	18 33
	PCIE_850	PCIE		
	PCIE_850	PCIE		
	PCIE_850	PCIE		
	PCIE_850	PCIE		
DMI	CLK_PCIE_900	CLK_PCIE	DMI_MIDBUS_CLK100M_N	
	CLK_PCIE_900	CLK_PCIE	DMI_MIDBUS_CLK100M_P	
	CLK_PCIE_900	CLK_PCIE	DMI_CLK100M_CPU_N	11 18
	CLK_PCIE_900	CLK_PCIE	DMI_CLK100M_CPU_P	11 18
	PCIE_850	PCIE	DMI_S2N_P<3..0>	10 19
	PCIE_850	PCIE	DMI_S2N_N<3..0>	10 19
	PCIE_850	PCIE	DMI_N2S_P<3..0>	10 19
	PCIE_850	PCIE	DMI_N2S_N<3..0>	10 19
	PCIE_850	PCIE		
	PCIE_850	PCIE		
PCIE REF CLOCKS	CLK_PCIE_900	CLK_PCIE	GPU_CLK100M_PCIE_P	9
	CLK_PCIE_900	CLK_PCIE	GPU_CLK100M_PCIE_N	9
	CLK_PCIE_900	CLK_PCIE	PCIE_CLK100M_MINI_P	18 33
	CLK_PCIE_900	CLK_PCIE	PCIE_CLK100M_MINI_N	18 33
	CLK_PCIE_900	CLK_PCIE	PCIE_CLK100M_FW_P	18 39
	CLK_PCIE_900	CLK_PCIE	PCIE_CLK100M_FW_N	18 39
	ENET_1000	ENET_MII	PCIE_CLK100M_ENET_P	18 37
	ENET_1000	ENET_MII	PCIE_CLK100M_ENET_N	18 37
	ENET_1000	ENET_MII		
	ENET_1000	ENET_MII		
SATA	SATA_900	SATA	SATA_HDD_R2D_C_P	18 42
	SATA_900	SATA	SATA_HDD_R2D_C_N	18 42
	SATA_900	SATA	SATA_HDD_R2D_P	42
	SATA_900	SATA	SATA_HDD_R2D_N	42
	SATA_900	SATA	SATA_HDD_D2R_P	18 42
	SATA_900	SATA	SATA_HDD_D2R_N	18 42
	SATA_900	SATA	SATA_HDD_D2R_C_P	42
	SATA_900	SATA	SATA_HDD_D2R_C_N	42
	SATA_900	SATA	SATA_ODD_R2D_C_P	18 42
	SATA_900	SATA	SATA_ODD_R2D_C_N	18 42
CLOCKS	CLK_PCIE_900	CLK_PCIE	PCH_CLK100M_DMI_P	18 26
	CLK_PCIE_900	CLK_PCIE	PCH_CLK100M_DMI_N	18 26
	CLK_PCIE_900	CLK_PCIE	PCH_CLK96M_D0T_P	18 26
	CLK_PCIE_900	CLK_PCIE	PCH_CLK96M_D0T_N	18 26
	CLK_PCIE_900	CLK_PCIE	PCH_CLK100M_SATA_P	18 26
	CLK_PCIE_900	CLK_PCIE	PCH_CLK100M_SATA_N	18 26
	CLK_PCIE_900	CLK_PCIE	ITP_XDP_CLK100M_N	18 25
	CLK_PCIE_900	CLK_PCIE	ITP_XDP_CLK100M_P	18 25
	CLK_PCIE_900	CLK_PCIE	ITPCPU_CLK100M_N	11 18
	CLK_PCIE_900	CLK_PCIE	ITPCPU_CLK100M_P	11 18
UNUSED CLOCKS	CLK_PCIE_900	CLK_PCIE	XDP_CPU_CLK100M_P	25
	CLK_PCIE_900	CLK_PCIE	XDP_CPU_CLK100M_N	25
UNUSED PCIE	PCIE_850	PCIE	MXM_PCIE_R2D_P<8..15>	75 77
	PCIE_850	PCIE	MXM_PCIE_R2D_N<8..15>	75 77
	PCIE_850	PCIE	MXM_PCIE_D2R_P<8..15>	75 77
	PCIE_850	PCIE	MXM_PCIE_D2R_N<8..15>	75 77

ANY OTHER LYNNFIELD CONSTRAINTS NOT COVERED ON PAGES 101 AND 107 SHOULD GO ON THIS PAGE TOO

SYNC_MASTER=K62		SYNC_DATE=01/06/2011	
PAGE TITLE		PCIE/DMI/FDI/SATA CONSTRAINTS	
		DRAWING NUMBER	051-8115
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PCH CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	0.2 MM	?
ITP_PCH	*	0.2 MM	?

PCI Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCI	*	=STANDARD	?
CLK_PCI	*	0.2 MM	?

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
LPC	*	0.15 MM	?
CLK_LPC	*	0.2 MM	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SMB	*	=2X_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
HDA	*	=2X_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE		

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SPI	*	0.2 MM	?

XTAL Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

PHYSICAL	NET_TYPE	SPACING
18320	PM	T29_CLKREQ0_L
18320	PM	FW_MINI_CLKREQ0_L
18320	PM	BLC_GPIO
18320	PM	T29_SW_RESET_L
18320	PM	ENET_CLKREQ0_L
18320	PM	DP_GPU_T29_SEL
18320	PM	T29_MCU_INT_L
18320	PM	T29_DP_PORTA_PWR_EN
18320	PM	T29_DP_PORTB_PWR_EN
18320	PM	DP_AUXCH_ISOL
18320	PM	PLT_RST_BUF_L
18320	PM	XDPCPU_PLTRST_L
18320	PM	PCH_PEG_CLKREQ0_L
18320	PM	ENET_SW_RESET_L
18320	PM	CPU_SKTOCC
18320	PM	PM_EN_USB_PWR

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING
18320	PCI_55S	PCI	PCI_REQ0_L
18320	PCI_55S	PCI	PCI_REQ1_L
18320	PCI_55S	PCI	PCI_REQ2_L
18320	PCI_55S	PCI	PCI_REQ3_L
18320	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIOUT
18320	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIIIN
18320	LPC_55S	LPC	LPC_AD<3..0>
18320	LPC_55S	LPC	LPC_FRAME_L
18320	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R
18320	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC
18320	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS
18320	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK_R
18320	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK
18320	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R
18320	LPC_55S	LPC	LPC_R_AD<3..0>
18320	LPC_55S	LPC	LPC_FRAME_R_L
18320	SPI_55S	SPI	SPI_CLK_1_R
18320	SPI_55S	SPI	SPI_MOSI_1_R
18320	CLK_XTAL	XTAL	USB_HUB2_XTAL1
18320	CLK_XTAL	XTAL	USB_HUB2_XTAL2
18320	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1_R
18320	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R
18320	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1
18320	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2
18320	CLK_XTAL	XTAL	CK505_XTAL_IN
18320	CLK_XTAL	XTAL	CK505_XTAL_OUT
18320	CLK_XTAL	XTAL	CK505_XTAL_OUT_R
18320	CLK_PCH_55S	CLK_PCH	PCH_CLK14P3M_REFCLK

PHYSICAL	NET_TYPE	SPACING
18320	PM	ENET_RESET_LOGIC_L
18320	PM	ENET_RESET_FET_L
18320	PM	ENET_CLKREQ_FET_L
18320	PM	PGOOD_5V_1V05_3V3
18320	PM	PGOOD_CPU_UNCORE
18320	PM	ALL_SYS_PMRGD
18320	PM	PGOOD_3V3_1V05
18320	PM	PGOOD_PCH_S0_R
18320	PM	AUD_IPHS_SWITCH_EN_PCH

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING
18320	SPT_55S	SPT	SPI_CLK_R
18320	SPT_55S	SPT	SPI_CLK
18320	SPT_55S	SPT	SPI_MOSI_R
18320	SPT_55S	SPT	SPI_MOSI
18320	SPT_55S	SPT	SPI_MISO
18320	SPT_55S	SPT	SPI_MISO_R
18320	SPT_55S	SPT	SPI_CS0_R_L
18320	SPT_55S	SPT	SPI_CS0_L
18320	SPT_55S	SPT	SPI_MLB_CS_L
18320	SPT_55S	SPT	SPI_ALT_CS_L
18320	SPT_55S	SPT	SPIROM_USE_MLB
18320	SPT_55S	SPT	SPI_ALT_MOSI
18320	SPT_55S	SPT	SPI_ALT_MISO
18320	SPT_55S	SPT	SPI_ALT_CLK
18320	HDA_55S	HDA	HDA_BIT_CLK
18320	HDA_55S	HDA	HDA_BIT_CLK_R
18320	HDA_55S	HDA	HDA_RST_L
18320	HDA_55S	HDA	HDA_RST_R_L
18320	HDA_55S	HDA	HDA_SDOOUT
18320	HDA_55S	HDA	HDA_SDOOUT_R
18320	HDA_55S	HDA	HDA_SYNC
18320	HDA_55S	HDA	HDA_SYNC_R
18320	HDA_55S	HDA	HDA_SDIIN0
18320	HDA_55S	HDA	AUD_SDI_R
18320	PM	PM	AUD_SPDIF_IN
18320	HDA	HDA	AUD_SPDIF_OUT
18320	HDA	HDA	AUD_SPDIF_CHIP
18320	HDA_55S	HDA	AUD_SPKR_OUTL01L_NOUT
18320	HDA_55S	HDA	AUD_SPKR_OUTL01L_POUT
18320	HDA_55S	HDA	AUD_SPKR_OUTL01R_NOUT
18320	HDA_55S	HDA	AUD_SPKR_OUTL01R_POUT
18320	HDA_55S	HDA	AUD_SPKR_OUTL02L_NOUT
18320	HDA_55S	HDA	AUD_SPKR_OUTL02L_POUT
18320	HDA_55S	HDA	AUD_SPKR_OUTL02R_NOUT
18320	HDA_55S	HDA	AUD_SPKR_OUTL02R_POUT
18320	CLK_XTAL	XTAL	PCH_CLK25M_XTALOUT_R
18320	CLK_XTAL	XTAL	PCH_CLK25M_XTALIN_R
18320	CLK_XTAL	XTAL	PCH_CLK25M_XTALOUT
18320	CLK_XTAL	XTAL	PCH_CLK25M_XTALIN
18320	PCH_55S	COMP_PCH	PCH_USB_RB1AS
18320	PCH_55S	COMP_PCH	PCH_SATA3COMP
18320	PCH_55S	COMP_PCH	PCH_XCLK_RCOMP
18320	PCH_55S	COMP_PCH	PCH_DMI_COMP
18320	PCH_55S	COMP_PCH	PCH_SATA1COMP
18320	CLK_XTAL	XTAL	USB_HUB1_XTAL1
18320	CLK_XTAL	XTAL	USB_HUB1_XTAL2
18320	PCH_55S	COMP_PCH	USB_HUB1_RB1AS
18320	PCH_55S	ITP_PCH	XDP_PCH_TCK
18320	PCH_55S	ITP_PCH	XDP_PCH_TMS
18320	PCH_55S	ITP_PCH	XDP_PCH_TDI
18320	PCH_55S	ITP_PCH	XDP_PCH_TDO
18320	PCH_55S	COMP_PCH	PCH_DMI2RB1AS
18320	PCH_55S	COMP_PCH	PCH_SATA3RB1AS
18320	PCH_55S	COMP_PCH	USB_HUB2_RB1AS

SYNC MASTER=K62		SYNC DATE=01/06/2011	
PAGE TITLE			
IBEX PEAK CONSTRAINTS			
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CAESAR II (ETHERNET) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
ENET_MII	*	0.3 MM	?
ENET_SE	*	=STANDARD	?

SOURCE:BROADCOM 5764M-DS04-RDS. PAGE 38

CAESAR II (ETHERNET) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
ENET_DIFF	*	0.6 MM	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_20THER	*	=50MIL_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
50MIL_SPACING	*	1.27 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_DIFF_T	*	*	ENET_20THER

CAESAR IV (SD) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SD	*	=3:1_SPACING	?

FireWire Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
FW_110D	*	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB	*	=3:1_SPACING	?	USB	TOP,BOTTOM	=4X_DIELECTRIC	?

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
	ENET_50S	ENET_SE	ENET RDAC 37
	CLK_PCH_55S	XTAL	ENET_CLK25M_XTAL1 36 37
	CLK_PCH_55S	XTAL	ENET_CLK25M_XTAL0 36 37
	CLK_PCH_55S	XTAL	ENET_CLK25M_XTAL0_R 36
	ENET_100D	ENET_DIFF	ENETCONN MDI P<3..0> 37 38
	ENET_100D	ENET_DIFF	ENETCONN MDI N<3..0> 37 38
	ENET_100D	ENET_DIFF_T	ENETCONN MDI T P<3..0> 38
	ENET_100D	ENET_DIFF_T	ENETCONN MDI T N<3..0> 38
	PCIE_85D	ENET_MII	PCIE ENET R2D P 37
	PCIE_85D	ENET_MII	PCIE ENET R2D N 37
	PCIE_85D	ENET_MII	PCIE ENET D2R P 18 37
	PCIE_85D	ENET_MII	PCIE ENET D2R N 18 37
	PCIE_85D	ENET_MII	PCIE ENET R2D C P 18 37
	PCIE_85D	ENET_MII	PCIE ENET R2D C N 18 37
	PCIE_85D	ENET_MII	PCIE ENET D2R C P 37
	PCIE_85D	ENET_MII	PCIE ENET D2R C N 37
	SD_50S	SD	ENET SD CMD 37
	SD_50S	SD	SDCONN CMD 37 45
	SD_50S	SD	SDCONN CLK 37 45
	SD_50S	SD	ENET SD CLK 37
	SD_50S	SD	SDCONN DATA<7..0> 37 45
	SD_50S	SD	ENET_CR DATA<7..0> 37
	SD_50S	FW	ENET MEDIA SENSE 15 18 37
	SD_50S	SD	ENET MEDIA SENSE R 37
	SD_50S	SD	ENET SD DETECT L 37
	SD_50S	FW	SDCONN DETECT BUF L 97

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
USB_90D	USB	USB	USB_EXTN_P 34 43
USB_90D	USB	USB	USB_EXTN_N 34 43
USB_90D	USB	USB	USB_PORT0_P 43
USB_90D	USB	USB	USB_PORT0_N 43
USB_90D	USB	USB	USB_EXTB_P 35 43
USB_90D	USB	USB	USB_EXTB_N 35 43
USB_90D	USB	USB	USB_PORT1_P 43
USB_90D	USB	USB	USB_PORT1_N 43
USB_90D	USB	USB	USB_EXTC_P 34 43
USB_90D	USB	USB	USB_EXTC_N 34 43
USB_90D	USB	USB	USB_PORT2_P 43
USB_90D	USB	USB	USB_PORT2_N 43
USB_90D	USB	USB	USB_EXTD_P 35 43
USB_90D	USB	USB	USB_EXTD_N 35 43
USB_90D	USB	USB	USB_D_MUXED_P 43
USB_90D	USB	USB	USB_D_MUXED_N 43
USB_90D	USB	USB	USB_PORT3_P 43
USB_90D	USB	USB	USB_PORT3_N 43
USB_90D	USB	USB	USB_CAMERA_P 20 44
USB_90D	USB	USB	USB_CAMERA_N 20 44
USB_90D	USB	USB	USB_CAMERA_L_P 44 98
USB_90D	USB	USB	USB_CAMERA_L_N 44 98
USB_90D	USB	USB	USB_BT_P 35 44
USB_90D	USB	USB	USB_BT_N 35 44
USB_90D	USB	USB	USB_BT_L_P 44 98
USB_90D	USB	USB	USB_BT_L_N 44 98
USB_90D	USB	USB	USB_IR_P 34 44
USB_90D	USB	USB	USB_IR_N 34 44
USB_90D	USB	USB	USB_IR_L_P 44 98
USB_90D	USB	USB	USB_IR_L_N 44 98
USB_90D	USB	USB	USB_SDCARD_P 34 44
USB_90D	USB	USB	USB_SDCARD_N 34 44
USB_90D	USB	USB	USB_SDCARD_L_P 44
USB_90D	USB	USB	USB_SDCARD_L_N 44
USB_90D	USB	USB	USB_HUB1_UP_P 20 34
USB_90D	USB	USB	USB_HUB1_UP_N 20 34
USB_90D	USB	USB	USB_HUB2_UP_P 20 35
USB_90D	USB	USB	USB_HUB2_UP_N 20 35
USB_90D	USB	USB	USB_HUB2UNUSED_P 35
USB_90D	USB	USB	USB_HUB2UNUSED_N 35

FireWire Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
CLK_PCH_55S	XTAL		FW_CLK24P576M_X0 39
CLK_PCH_55S	XTAL		FW_CLK24P576M_X0_R 39
CLK_PCH_55S	XTAL		FW_CLK24P576M_XI 39
FW_110D	FW_TP		FW_PORT0_TPA_P 40 41
FW_110D	FW_TP		FW_PORT0_TPA_N 40 41
FW_110D	FW_TP		FW_PORT0_TPB_P 40 41
FW_110D	FW_TP		FW_PORT0_TPB_N 40 41
UNUSED FW NETS PHYSICAL PROPERTIES			
FW_110D	FW_TP		FW_P1_TPA_P 39 40
FW_110D	FW_TP		FW_P1_TPA_N 39 40
FW_110D	FW_TP		FW_P2_TPA_P 39 40
FW_110D	FW_TP		FW_P2_TPA_N 39 40
FW_110D	FW_TP		FW_P1_TPB_P 39 40
FW_110D	FW_TP		FW_P1_TPB_N 39 40
FW_110D	FW_TP		FW_P2_TPB_P 39 40
FW_110D	FW_TP		FW_P2_TPB_N 39 40

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PAGE_TITLE			
USB/ENET/SD/FW/AUD		CONSTRAINTS	
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GRAPHICS CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
DP_850	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	*	=3:1_SPACING	?

USE 5X_DIELECTRIC IN K62

PAIRS SHOULD BE WITHIN 100 MILS OF CLOCK LENGTH.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.

ELECTRICAL_CONSTRAINT_SET ASSIGNED IN CONT. MGR.				NET_TYPE
		PHYSICAL	SPACING	
DP_850	DISPLAYPORT	DP_INTCONN ML C P<3..0>	83	
DP_850	DISPLAYPORT	DP_INTCONN ML C N<3..0>	83	
DP_850	DISPLAYPORT	DP_INTCONN AUXCH C P	83	
DP_850	DISPLAYPORT	DP_INTCONN AUXCH C N	83	
DP_850	DISPLAYPORT	DP_INTPNL ML P<3..0>	81 83	
DP_850	DISPLAYPORT	DP_INTPNL ML N<3..0>	81 83	
DP_850	DISPLAYPORT	DP_INTPNL AUX P	81 83	
DP_850	DISPLAYPORT	DP_INTPNL AUX N	81 83	
DP_850	DISPLAYPORT	DP_EXTB ML P<3..0>	84	
DP_850	DISPLAYPORT	DP_EXTB ML N<3..0>	84	
DP_850	DISPLAYPORT	DP_EXTB AUXCH P	84	
DP_850	DISPLAYPORT	DP_EXTB AUXCH N	84	
DP_850	DISPLAYPORT	DP_EXTB ML C P<3..0>	78 84	
DP_850	DISPLAYPORT	DP_EXTB ML C N<3..0>	78 84	
DP_850	DISPLAYPORT	DP_EXTB AUXCH C P	78 84	
DP_850	DISPLAYPORT	DP_EXTB AUXCH C N	78 84	
DP_850	DISPLAYPORT	DP_EXTB ML P<3..0>	75 78	
DP_850	DISPLAYPORT	DP_EXTB ML N<3..0>	75 78	
DP_850	DISPLAYPORT	DP_EXTB AUXCH P	75 78	
DP_850	DISPLAYPORT	DP_EXTB AUXCH N	75 78	
DP_850	DISPLAYPORT	DP_EXTB ML C P<3..0>	75 83	
DP_850	DISPLAYPORT	DP_EXTB ML C N<3..0>	75 83	
DP_850	DISPLAYPORT	DP_EXTB AUXCH C P	75 83	
DP_850	DISPLAYPORT	DP_EXTB AUXCH C N	75 83	
DP_850	DISPLAYPORT	DP_EXTB ML P<3..0>	75 78	
DP_850	DISPLAYPORT	DP_EXTB ML N<3..0>	75 78	
DP_850	DISPLAYPORT	DP_EXTB AUXCH P	75 78	
DP_850	DISPLAYPORT	DP_EXTB AUXCH N	75 78	
DP_850	DISPLAYPORT	MXM_DP_B ML P<3..0>	75 78	
DP_850	DISPLAYPORT	MXM_DP_B ML N<3..0>	75 78	
DP_850	DISPLAYPORT	MXM_DP_B AUX P	75 78	
DP_850	DISPLAYPORT	MXM_DP_B AUX N	75 78	
DP_850	DISPLAYPORT	MXM_DP_C ML P<3..0>	75 83	
DP_850	DISPLAYPORT	MXM_DP_C ML N<3..0>	75 83	
DP_850	DISPLAYPORT	MXM_DP_C AUX P	75 83	
DP_850	DISPLAYPORT	MXM_DP_C AUX N	75 83	
DP_850	DISPLAYPORT	MXM_DP_C AUX R P	83	
DP_850	DISPLAYPORT	MXM_DP_C AUX R N	83	
DP_850	DISPLAYPORT	MXM_DP_D ML P<3..0>	75 78	
DP_850	DISPLAYPORT	MXM_DP_D ML N<3..0>	75 78	
DP_850	DISPLAYPORT	MXM_DP_D AUX P	75 78	
DP_850	DISPLAYPORT	MXM_DP_D AUX N	75 78	

UNUSED VIDEO NET PHYSICAL CONSTRAINTS

DP_850	DISPLAYPORT	MXM_LVDS_A_CLK_P	76 78
DP_850	DISPLAYPORT	MXM_LVDS_A_CLK_N	76 78
DP_850	DISPLAYPORT	MXM_LVDS_B_CLK_P	76 78
DP_850	DISPLAYPORT	MXM_LVDS_B_CLK_N	76 78
DP_850	DISPLAYPORT	MXM_LVDS_A_DATA_P<3..0>	76 78
DP_850	DISPLAYPORT	MXM_LVDS_A_DATA_N<3..0>	76 78
DP_850	DISPLAYPORT	MXM_LVDS_B_DATA_P<3..0>	76 78
DP_850	DISPLAYPORT	MXM_LVDS_B_DATA_N<3..0>	76 78

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SYNC_DATE=01/06/2011

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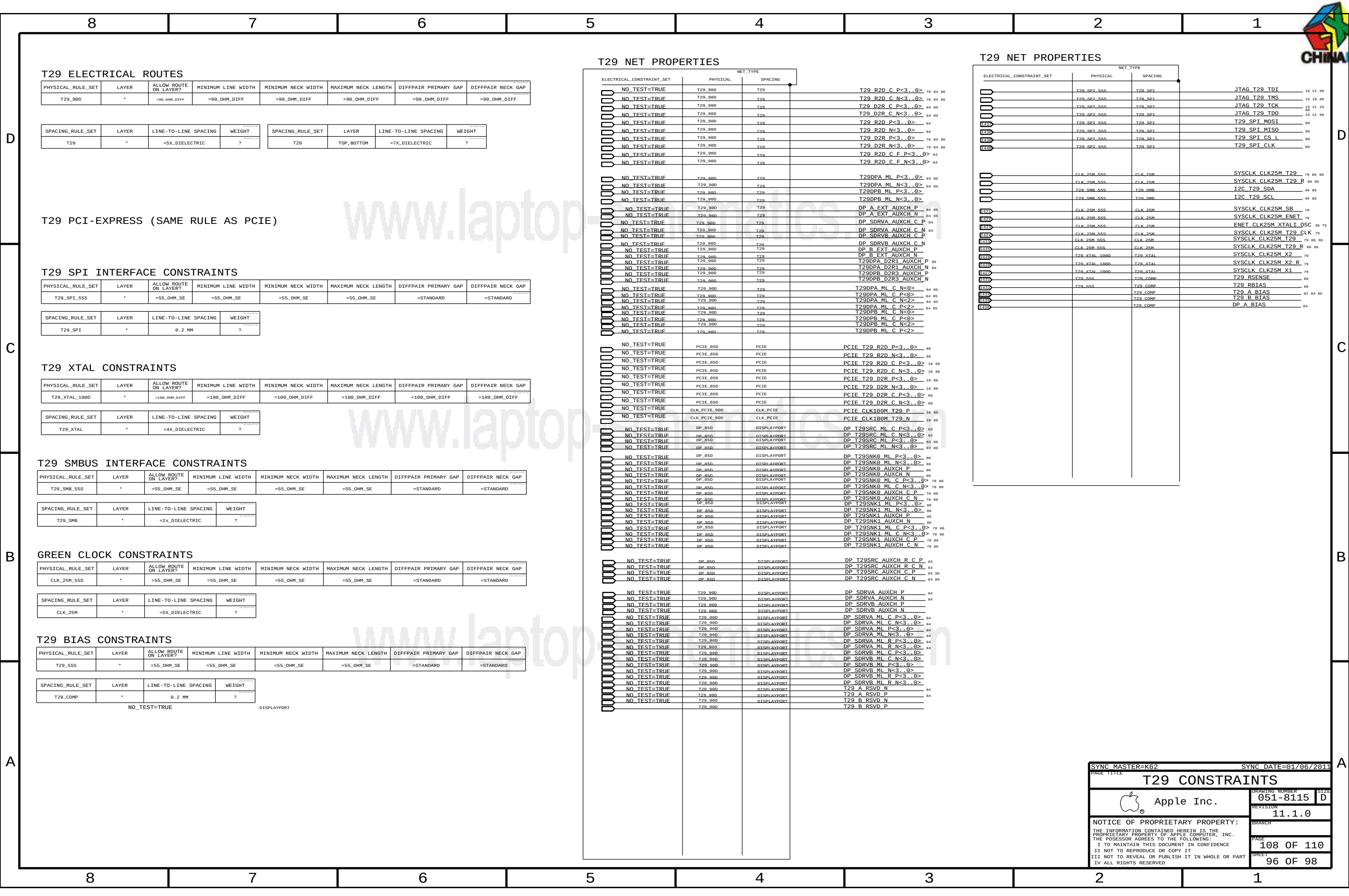
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SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL	*	0.35MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_VIDS	*	=4X_DIELECTRIC	?

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PM NET PROPERTIES
(PM, RESET, EN, PG00D)

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PM	*	*	2:1_SPACING
PM_VTT	PM_VTT	*	2:1_SPACING
PM_VTT	*	*	3:1_SPACING
PM_VTT	GND	*	DEFAULT
PM	GND	*	DEFAULT

NET_TYPE		PHYSICAL	SPACING	
PHYSICAL	SPACING			
PM	4V5_REG_EN	56		
PM	3V42G3H_SHDN_L	72		
PM	ALL_SYS_PWRGD_R	5	32 64	
PM	ALL_SYS_PWRGD_SMC	46	64	
PM	AP_PWR_EN	20	25 33	
PM	AP_MINI_RESET_L	33		
PM	AUD_I2C_INT_L	20	62	
PM	AUD_IP_PERIPHERAL_DET	20	61	
PM	AUD_IPHS_SWITCH_EN	21	62	
PM	AUD_SPDIF_IN	60	83 91	
PM	AUD_SPDIF_IN_CODEC	56	83	
PM	BDV_BKL_PWM	46	83 97	
PM	BL_PWM	6	83	
PM	BL_EN	6	83	
PM	BDV_BKL_PWM	46	83 97	
PM	CK505_27MHZ_EN	26		
PM	CPUVTT_REG_EN			
PM_VTT	CPUVTT_REG_PG00D_R	63		
PM	CPU_MEM_RESET_L	11	32	
PM_VTT	CPU_PECI_R	46		
PM	CPU_PWRGD	11	21 25	
PM	CPU_RESET_L	11	27	
PM	CPU_SKTOCC_L	11	63	
PM	CPU_CATERR_L	11		
PM	CPU_PECI	11	21 46	
PM	CPU_PROCHOT_L	11	47 65	
PM	CPU_THRMTRIP_L	11	47	
PM	CPU_PROC_SEL	11	19	
PM	DEBUG_RESET_L	27	48	
PM	DDRVTT_EN	81	83	
PM	DP_INT_SPDIF_AUDIO	81	83	
PM	DP_INTPNL_HPD	81	83	
PM	3V3R2V9_DPAPWR_ADJ	82	95	
PM	DP_A_PWRDWN	82		
PM	DP_A_PWRDWN_FET_R	82		
PM	DP_A_PWRDWN_INV	82		
PM	DPAPWRSW_HVEN_L_R	82		
PM	DPAPWRSW_CT	82		
PM	DPAPWRSW_ILIM	82		
PM	DPAPWRSW_IFLT	82		
PM	T29_A_HV_EN	82	84	
PM	3V3R2V9_DPBWPR_ADJ			
PM	DP_B_PWRDWN			
PM	DP_B_PWRDWN_FET_R			
PM	DP_B_PWRDWN_INV			
PM	DPBPWRSW_HVEN_L_R			
PM	DPBPWRSW_CT			
PM	DPBPWRSW_ILIM			
PM	DPBPWRSW_IFLT			
PM	T29_B_HV_EN			
PM	T29_PWR_EN	18	80 97	
PM	T29_RESET_RTR_L	80	86	
PM	LCD_BL_FILT	83		
PM	LCD_BLK_ON_DLY	83		
PM	LCD_BL_PWM	83		
PM	MXM_PNL_BL_PWM	76	83	

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NET_TYPE		PHYSICAL	SPACING	
PHYSICAL	SPACING			
PM	ENET_PWR_EN	20	25 36	
PM	ENET_LOW_PWR	15	21 37	
PM	FW_RESET_L	27	39	
PM	ENET_RESET_L	27	36	
PM	FW_PME_L	15	21 39	
PM	FW_PWR_EN	15	21	
PM	FW_CLKREQ_L	15	39	
PM	ISOLATE_CPU_MEM_L	21	25 32	
PM	LPC_PWRDWN_L	19	46 48	
PM	MEM_RESET_L	30	31 32 89	
PM	MINI_CLKREQ_L	15	33	
PM	MINI_RESET_L	27	33	
PM	MXM_CLKREQ_L	9	75	
PM	MXM_G00D	5	21 25	
PM	ODD_PWR_EN_L	15	21 42	
PM	RTC_RESET_L	18	27 97	
PM	RSMRST_PWRGD	46	64	
PM	RTC_RESET_L	18	27 97	
PM	S4_ENABLES	63		
PM	SDCONN_STATE_RST_L			
PM	SDCONN_DETECT_BUF_L	92		
PM	SDCONN_STATE_CHANGE	20	25 45	
PM	SDCARD_RESET	15	21 44 98	
PM	SDCARD_RESET_L	44		
PM	SDCARD_PLT_RST_L	27	44	
PM	SDCARD_PLT_RST_L_R			
PM	SMC_PM_G2_EN	46	74	
PM	SMC_PM_G2_EN_R	74		
PM	SMC_PM_G2_EN_L	74		
PM	SS_DG_1	74		
PM	SS_MSFT_G1	74		
PM	USE_HDD_O0B_L	20	51	
PM	HDD_O0B_1V00_REF	51		
PM	SMC_ADAPTER_EN	19	46 47	
PM	SMC_RUNTIME_SCI_L	21	46 47	
PM	SMC_WAKE_SCI_L	15	18 21 46	
PM	SMC_DELAYED_PWRGD	47	64	
PM	SMC_LRESET_L	27	46	
PM	SMC_RESET_L	46	47 48	
PM	SMC_PROCHOT	46	47	
PM	SMC_PROCHOT_3_3_L	46	47	
PM	SMC_ONOFF_L	46	47	
PM	SMC_MANUAL_RST_L	47		
PM	SPI_DESCRIPTOR_OVERRIDE_L	18	46	
PM	T29_PWR_EN	18	80 97	
PM	T29_RESET_L	27	80	
PM	T29_DP_PORTA_PWR_EN	20	25 82 91	
PM	T29_DP_PORTA_PWR_EN_REG	82		
PM_VTT	XDP_CPUPWRGD			
PM_VTT	XDP_DBRESET_L	11	25	
PM_VTT	XDP_PWRGD			
PM	XDPPCH_PLTRST_L	25	27	
PM	USB_HUB_SOFT_RESET_L	20	25 34	
PM	VSYNC_DP_CONN	6	81	
PM	VSYNC_DP	61		
PM	VIDEO_ON	61		
PM	VTT_REG_PG00D_L	63		

NET_TYPE		PHYSICAL	SPACING	
PHYSICAL	SPACING			
PM	PLT_RESET_L	20	27	
PM_VTT	PLT_RESET_LS1V05_L	11		
PM	PM_BATLOW_L	15	19 40	
PM	PM_CLK32K_SUSCLK	9	46 91	
PM	PM_CLK32K_SUSCLK_R	9	19 91	
PM	PM_CLKRUN_L	15	19 46 48	
PM	PM_PWRBTN_L	19	25 46	
PM	PM_RSMRST_L	27	46	
PM	PM_RSMRST_PCH_L	19	27	
PM	PCH_SRTCST_L	18		
PM	PCH_INTVRMEN_L	18		
PM	PCH_DSWVRMEN	19		
PM	PCH_DF_TVS	19		
PM	PCH_PROCPWRGD	21		
PM	PCIE_WAKE_L	19	33 36 78	
PM	PM_DSW_PWRGD	19		
PM	PM_ASW_PWRGD	19	64	
PM	PM_MEM_PWRGD_R	11		
PM	PM_EN_DDR1V5_S3_REG	63	71	
PM	PM_EN_DDRVTT_S0_REG	32	63 71	
PM	PM_EN_P12V_S0_FET	6	63	
PM	PM_EN_P1V05_S0_REG	63	68	
PM	PM_EN_P1V05_S3_REG			
PM	PM_EN_P1V5_S0_FET	63	73	
PM	PM_EN_P1V8_S0_REG	63	71	
PM	PM_EN_P3V3_S0_FET	63	73	
PM	PM_EN_P3V3_S3_FET	63	73	
PM	PM_EN_P3V3_S5_REG	70		
PM	PM_EN_P5V_S0_FET	63	73	
PM	PM_EN_P5V_S3_REG	63	70	
PM	PM_EN_PVCCSA_S0_REG_L	64		
PM	PM_EN_VCCSA_S0_CPU			
PM	PM_EN_PVCORE_CPU	63	65	
PM_VTT	PM_MEM_PWRGD	11	19 97	
PM	PM_MXM_EN	64	76	
PM	PM_PCH_PWRGD_R	64		
PM	PM_PECI_PWRGD	46	64	
PM	PM_PECI_PWRGD_R	46		
PM	PM_PG00D_DDR1V5_S3_REG	5	63 71	
PM	PM_PG00D_P1V05_S0_REG	63	64 68	
PM	PM_PG00D_P1V5_S0_FET	11	64 73	
PM	PM_PG00D_P1V8_S0_REG	64	71	
PM	PM_PG00D_P3V3_S0_FET	63	64 73	
PM	PM_PG00D_P3V3_S3_FET	34	73	
PM	PM_PG00D_P3V3_S5_REG	27	64 70	
PM	PM_PG00D_P5V_S0_FET	63	64 73	
PM	PM_PG00D_MINI	33		
PM	PM_PG00D_PVCORE_CPU			
PM	PM_PG00D_PVCCSA_S0_REG	5	25 64 65	
PM	PM_PG00D_P5V_S3_REG	63	64	
PM	PM_PG00D_PVAXG	63	70 82	
PM	PM_PG00D_PVAXG	5	65	
PM_VTT	PM_MEM_PWRGD	11	19 97	
PM	PM_MEM_PWRGD_L	11		
PM	PM_MXM_PG00D	64	76	
PM	PM_PCH_PWRGD	19	21 64	
PM	PM_SLP_S3_5V	32		
PM	PM_SLP_S3_5V_L	32		
PM	PM_SLP_S3_5V_R2	32		
PM	PM_SLP_S3_L	5	19 26 32 36 46 47 63 82	
PM	PM_SLP_S4_L	5	19 32 46 47 63 97	
PM	PM_SLP_S5_L	5	19 46 47 63	
PM_VTT	PM_SYNC	11	19	
PM	PM_SYSRST_L	19	25 27 46	
PM	PM_SYS_PWRGD	19	32 64	
PM_VTT	PM_THRMTRIP_L	21	47	
PM	PM_SLP_S3_BUF_L	63		
PM	PM_SLP_S4_1_L_R	63		
PM	PM_SLP_S4_D_L	32		
PM	PM_SLP_S4_L	5	19 32 46 47 63 97	
PM	PG00D_P1V5_S0_DLY	11		
PM	PG00D_1V8_S0_G1	64		
PM	PG00D_1V8_S0_G2	64		
PM	PG00D_P12V_S0	63	64	
PM	PG00D_P1V8_S0	64		
PM	PG00D_PCH_S0	5	64	
PM	PG00D_PCH_S0_R	64	91	
PM	PG00D_SYSPWR0K	64		
PM	PG00D_SYSPWR0K_R	64		
PM	POWER_BUTTON_L	47		
PM	PEG_RESET_L	9	27	
PM	PG00D_CPU_S0	64		
PM	PG00D_CPU_UNCORE	64	91	
PM	PG00D_5V_1V05_3V3	64	91	
PM	PG00D_3V3_1V05	64	91	
PM	PG00D_12V_S0_G1	64		
PM	PG00D_12V_S0_G2	64		
PM	9V_COMP_REF	64		
PM	12V_COMP_REF	64		
PM	ALL_SYS_PWRGD	64	91	

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PM RESETS ENABLES PG00D CONST

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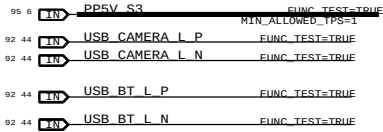
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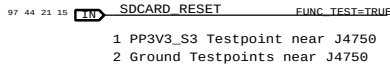
FUNCTIONAL TESTPOINTS FOR MAC-1 & ICT

J4700 USB CAMERA

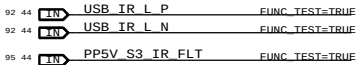


1 PP5V_S3_REG Testpoint near J4700
1 PP3V3_S3 TESTPOINT NEAR J4700
6 GROUND TESTPOINTS NEAR J4700

J4750 USB CARD READER

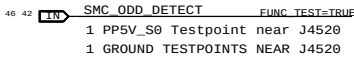


J4780 IR BOARD

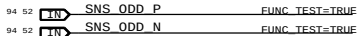


1 GROUND TESTPOINT NEAR J4780

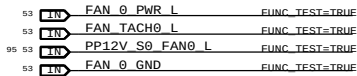
J4520 SATA ODD (HIGH SPEED)



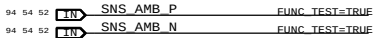
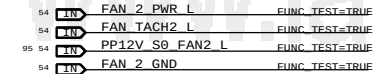
J5551 ODD TEMP SENSOR



J5600 ODD FAN

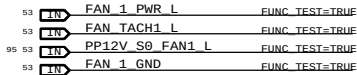


J5700 CPU FAN



1 GROUND TESTPOINT NEAR J5700

J5601 HD FAN

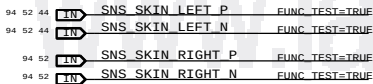


J5400 HDD TEMP SENSOR

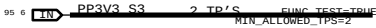
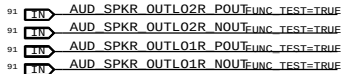


1 GROUND TESTPOINTS NEAR J5400

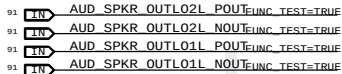
J5560 SKIN TEMP SENSOR



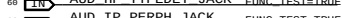
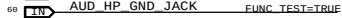
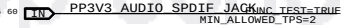
J6602 AUDIO RIGHT SPEAKER



J6603 AUDIO LEFT SPEAKER



J6600 AUDIO AUXILIARY CONNECTOR



4 GROUND TESTPOINTS NEAR J6600

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